UNITED STATES
SECURITIES AND EXCHANGE COMMISSION

Washington, D.C. 20549

FORM 10-K

☑ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2023

or

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from to

Commission File Number: 1-33100

Owens Corning

(Exact name of registrant as specified in its charter)

Delaware	**43-2109021**
(State or other jurisdiction of incorporation or organization)	*(I.R.S. Employer Identification No.)*
One Owens Corning Parkway, Toledo, OH	**43659**
(Address of principal executive offices)	*(Zip Code)*

(419) 248-8000

(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol	Name of each exchange on which registered
Common Stock, par value $0.01 per share	**OC**	**New York Stock Exchange**

Securities registered pursuant to Section 12(g) of the Act:

None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☑ No ❒

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or 15(d) of the Act. Yes ❒ No ☑

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or Section 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☑ No ❒

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☑ No ❒

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer ☑ Accelerated filer ❒ Non-accelerated filer ❒ Smaller reporting company ☐ Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ❒

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☑

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ❒

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ❒

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☑

On June 30, 2023, the last business day of the registrant's most recently completed second fiscal quarter, the aggregate market value of $0.01 par value common stock (the voting stock of the registrant) held by non-affiliates (assuming for purposes of this computation only that the registrant had no affiliates) was approximately $11,723,204,804.

As of February 9, 2024, 87,006,138 shares of the registrant's common stock, par value $0.01 per share, were outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of Owens Corning's proxy statement to be delivered to stockholders in connection with the Annual Meeting of Stockholders to be held on or about April 18, 2024 (the "2024 Proxy Statement") are incorporated by reference into Part III hereof.

		Page
PART I		
ITEM 1.	Business	1
	Overview	1
	Segment overview	1
	General	3
	Availability of information	6
ITEM 1A.	Risk factors	7
ITEM 1B.	Unresolved staff comments	18
ITEM 1C.	Cybersecurity	19
ITEM 2.	Properties	21
ITEM 3.	Legal proceedings	22
ITEM 4.	Mine safety disclosures	22
	Information about our Executive Officers	23
PART II		
ITEM 5.	Market for Owens Corning's common equity, related stockholder matters and issuer purchases of equity securities	24
ITEM 6.	Reserved	25
ITEM 7.	Management's discussion and analysis of financial condition and results of operations	26
ITEM 7A.	Quantitative and qualitative disclosures about market risk	44
ITEM 8.	Financial statements and supplementary data	46
ITEM 9.	Changes in and disagreements with accountants on accounting and financial disclosure	46
ITEM 9A.	Controls and procedures	46
ITEM 9B.	Other information	46
ITEM 9C.	Disclosure regarding foreign jurisdictions that prevent inspections	46
PART III		
ITEM 10.	Directors, executive officers and corporate governance	47
ITEM 11.	Executive compensation	47
ITEM 12.	Security ownership of certain beneficial owners and management and related stockholder matters	47
ITEM 13.	Certain relationships and related transactions, and director independence	47
ITEM 14.	Principal accountant fees and services	47
PART IV		
ITEM 15.	Exhibits and financial statement schedules	48
ITEM 16.	Form 10-K Summary	53
	Signatures	54
	Index to Consolidated Financial Statements	57
	Management's Report on Internal Control Over Financial Reporting	58
	Report of Independent Registered Public Accounting Firm	59
	Consolidated Financial Statements	61
	Notes to Consolidated Financial Statements	66
	Index to Consolidated Financial Statement Schedule	116
	Schedule II	117

PART I

ITEM 1. BUSINESS

OVERVIEW

Owens Corning is a global building and construction materials leader committed to building a sustainable future through material innovation. Its roofing products and systems enhance curb appeal of people's homes and protect residential and commercial buildings. Its insulation products conserve energy and improve acoustics, fire resistance and air quality in the spaces where people live, work and play. Its fiberglass composites make thousands of products lighter, stronger and more durable.

The business is global in scope, with operations in 30 countries, and human in scale, with approximately 18,000 employees and longstanding, local relationships with its customers. Founded in 1938 and based in Toledo, Ohio, Owens Corning recorded net sales in 2023 of $9.7 billion.

Unless the context indicates otherwise, the terms "Owens Corning," "Company," "we" and "our" in this report refer to Owens Corning and its subsidiaries. References to a particular year mean the Company's year commencing on January 1 and ending on December 31 of that year.

SEGMENT OVERVIEW

The Company has an integrated business model with three reportable segments: Roofing, Insulation and Composites. Our Roofing, Insulation and Composites reportable segments accounted for approximately 40%, 37% and 23% of our total reportable segment net sales, respectively, in 2023.

Roofing

Our primary products in the Roofing segment are laminate and strip asphalt roofing shingles. Other products include roofing components and oxidized asphalt. We have been able to meet the growing demand for longer lasting, aesthetically attractive laminate products with modest capital investment.

We sell shingles and roofing components primarily through distributors, home centers and lumberyards in the United States. Oxidized asphalt is a significant input used in the production of our asphalt roofing shingles. We are vertically integrated and have manufacturing facilities that process asphalt for use in our roofing shingles manufacturing process. In addition, we sell processed asphalt to other shingle manufacturers, to roofing contractors for built-up roofing asphalt systems and to manufacturers in a variety of other industries, including automotive, chemical, rubber and construction. Asphalt input costs and third-party asphalt sales prices are correlated to crude oil prices.

Demand for products in our Roofing segment is generally driven by both residential repair and remodeling activity and by new residential construction. Roofing damage from major storms can significantly increase demand in this segment. As a result, sales in this segment do not always follow seasonal home improvement, remodeling and new construction industry patterns as closely as our Insulation segment.

Our Roofing segment competes primarily with asphalt shingle manufacturers in the United States.According to various industry reports and Company estimates, Owens Corning's Roofing segment is the second largest producer of asphalt roofing shingles in the United States. Principal methods of competition include innovation and product design, proximity to customers, quality and price.

Our manufacturing operations are generally continuous in nature, and we warehouse much of our production prior to sale since we operate with relatively short delivery cycles. One of the raw materials important to this segment is sourced from a sole supplier. We have a long-term supply contract for this material and have no reason to believe that any availability issues will exist. If this supply was to become unavailable, our production could be interrupted until such time as the supplies again became available or the Company reformulated its products. Additionally, the supply of asphalt, another significant raw material in this segment, has been constricted at times. Although this has not caused a significant interruption of our production in the past, prolonged asphalt shortages would restrict our ability to produce products in this segment.

Insulation

Our insulating products provide a variety of benefits such as energy conservation, thermal functionality, improved acoustical performance and convenience of installation and use. Our Insulation segment includes a diverse portfolio of high, mid and low-temperature products with a geographic mix of United States, Canada, Europe, Asia-Pacific and Latin America, a market mix of residential, commercial, industrial and other markets, and a channel mix of retail, contractor and distribution.

Our products in the North American residential market include thermal and acoustical batts, loosefill insulation, spray foam, foam sheathing and accessories, and are sold under well-recognized brand names and trademarks, such as Owens Corning PINK® Next Gen™ FIBERGLAS™ Insulation. Our products in the commercial and industrial markets include glass fiber pipe insulation, energy efficient flexible duct media, bonded and granulated stone wool insulation, cellular glass insulation and foam insulation used in above- and below-grade construction applications, and are sold under well-recognized brand names and trademarks, such as FOAMULAR®, FOAMGLAS® and Paroc®. We sell our insulation products primarily to insulation installers, home centers, lumberyards, retailers and distributors in the United States, Canada, Europe, Asia-Pacific and Latin America.

Demand for Owens Corning's insulating products is driven by North American new residential construction, repair and remodeling activity, commercial and industrial construction activity in the United States, Canada, Europe, Asia-Pacific and Latin America, and increasingly stringent building codes and the growing need for energy efficiency. Demand in the segment typically follows seasonal home improvement, remodeling and renovation and residential, commercial and industrial construction industry patterns. Demand for residential insulation in North America typically follows housing starts on a three-month lagged basis, although the new residential construction cycle can elongate due to labor availability and other factors beyond our control. The peak season for home construction and remodeling in our geographic markets generally corresponds with the second and third calendar quarters. Demand for commercial and industrial applications is more heavily tied to industrial production growth, commercial construction activity, and overall economic conditions in the global markets we serve.

Our Insulation segment competes primarily with fiberglass insulation manufacturers in the United States, with an international presence in Canada, Europe, Asia-Pacific and Latin America. According to industry reports and Company estimates, Owens Corning is North America's largest producer of residential, commercial and industrial insulation. Principal methods of competition include innovation and product design, service, location, quality, price and compatibility of systems solutions.

Composites

Owens Corning glass fiber materials can be found in over 40,000 end-use applications primarily within three markets: building and construction, renewable energy and infrastructure. Such end-use applications include building structures, roofing shingles, tubs and showers, pools, decking, flooring, pipes and tanks, poles, electrical equipment and wind-energy turbine blades. Our products are manufactured and sold worldwide. We primarily sell our products directly to parts molders and fabricators. Within the building and construction market, our Composites segment sells glass fiber and/or glass mat directly to a small number of major shingle manufacturers, including our own Roofing segment.

Our Composites segment includes vertically integrated material solutions. The Company manufactures, fabricates and sells glass reinforcements in the form of fiber. Glass reinforcement materials are also used downstream by the Composites segment to manufacture and sell glass fiber products in the form of non-wovens, fabrics and composite lumber.

Demand for composites is driven by general global economic activity and, more specifically, by the increasing replacement of traditional materials, such as aluminum, wood, paper and steel with composites that offer lighter weight, improved strength, lack of conductivity and corrosion resistance.

We compete with glass fiber and building material manufacturers worldwide. According to various industry reports and Company estimates, our Composites segment is a world leader in the production of glass fiber reinforcement and other building materials. Primary methods of competition include innovation, quality, customer service, global geographic reach, sustainability and product customization.

GENERAL

Intellectual Property

The Company relies on a combination of intellectual property laws, as well as confidentiality procedures and contractual provisions, to protect our intellectual property, proprietary technology and our brands. Through continuous and extensive use of the color PINK® since 1956, Owens Corning became the first owner of a single color trademark registration. In addition to our Owens Corning and PINK® brands, the Company has registered, and applied for the registration of, U.S. and international trademarks, service marks, and domain names. Additionally, the Company owns numerous U.S. and international patents and patent applications, covering certain of our proprietary technology resulting from research and development efforts. Over time, the Company has assembled a portfolio of intellectual property rights including patents, trademarks, service marks, copyrights, domain names, know-how and trade secrets covering our products, services and manufacturing processes. Our proprietary technology is not dependent on any single or group of intellectual property rights and the Company does not expect the expiration of existing intellectual property to have a material adverse effect on the business as a whole. The Company believes the duration of our patents is adequate relative to the expected lives of our products. Although the Company protects its intellectual property and proprietary technology, any significant impairment of, or third-party claim against, our intellectual property rights could harm our business or our ability to compete.

Environmental Control

Owens Corning has established policies and procedures to ensure that its operations are conducted in compliance with all relevant laws and regulations and that enable the Company to meet its high standards for corporate sustainability and environmental stewardship. Our manufacturing facilities are subject to numerous foreign, federal, state and local laws and regulations relating to the presence of hazardous materials, pollution and protection of the environment, including emissions to air, reductions of greenhouse gases, discharges to water, management of hazardous materials, handling and disposal of solid wastes, use of chemicals in our manufacturing processes, and remediation of contaminated sites. All Company manufacturing facilities are either ISO 14001 certified or deploy environmental management systems based on ISO 14001 principles. The Company's 2030 Sustainability Goals include targets related to significant global reductions in energy use, water consumption, waste to landfill, and emissions of greenhouse gases, fine particulate matter, and volatile organic air emissions, and protection of biodiversity. The Company is dedicated to continuous improvement in its environmental, health and safety performance and to achieving its 2030 Sustainability Goals.

The Company has not experienced a material adverse effect upon its capital expenditures or competitive position as a result of environmental control legislation and regulations. Operating costs associated with environmental compliance were approximately $49 million in 2023. The Company continues to invest in equipment and process modifications to remain in compliance with applicable environmental laws and regulations worldwide.

Our manufacturing facilities are subject to numerous national, state and local environmental protection laws and regulations. Regulatory activities of particular importance to our operations include those addressing air pollution, water pollution, waste disposal and chemical control. It is possible that new laws and regulations will specifically address climate change, volatile organic compounds, ozone forming emissions and fine particulate matter. New environmental and chemical regulations could impact our ability to expand production or construct new facilities in geographic regions in which we operate. However, based on information known to the Company, including the nature of our manufacturing operations and associated air emissions, at this time we do not expect any of these new laws, regulations or activities to have a material adverse effect on our results of current operations, financial condition or long-term liquidity.

Owens Corning is involved in remedial response activities and is responsible for environmental remediation at a number of sites, including certain of its currently owned or formerly owned plants. These responsibilities arise under a number of laws, including, but not limited to, the Federal Resource Conservation and Recovery Act, and similar state or local laws pertaining to the management and remediation of hazardous materials and petroleum. The Company has also been named a potentially responsible party under the United States Federal Superfund law, or state equivalents, at a number of disposal sites. The Company became involved in these sites as a result of government action or in connection with business acquisitions. At the end of 2023, the Company was involved with a total of 22 sites worldwide, including 10 Superfund and state or country equivalent sites and 12 owned or formerly owned sites. None of the liabilities for these sites are individually significant to the Company.

Remediation activities generally involve a potential range of activities and costs related to soil, groundwater and sediment contamination. This can include pre-cleanup activities such as fact finding and investigation, risk assessment, feasibility studies, remedial action design and implementation (where actions may range from monitoring to removal of contaminants, to installation of longer-term remediation systems). A number of factors affect the cost of environmental remediation, including the number of parties involved in a particular site, the determination of the extent of contamination, the length of time the remediation may require, the complexity of environmental regulations, variability in clean-up standards, the need for legal action, and changes in remediation technology. Taking these factors into account, Owens Corning estimates the costs of remediation to be paid over a period of years. The Company accrues an amount on an undiscounted basis, when a liability is probable and reasonably estimable. Actual cost may differ from these estimates for the reasons mentioned above.

At December 31, 2023, the Company had an accrual totaling $4 million for its environmental liabilities, of which the current portion is $1 million. Changes in required remediation procedures or timing of those procedures at existing legacy sites, or discovery of contamination at additional sites, could result in material increases to the Company's environmental obligations.

Additional Government Laws and Regulations

In addition to environmental laws and regulations, we are subject to various laws and regulations around the world. For example, trade regulations, including tariffs or other import or export restrictions, may increase the cost of some of our raw materials or cross-border shipments, and limit our ability to do business in certain countries or with certain individuals. Our business is also subject to competition laws in the various jurisdictions where we operate, including the Sherman Antitrust Act and related federal and state antitrust laws in the United States, as well as similar foreign laws and regulations. These laws and regulations generally prohibit competitors from fixing prices, boycotting competitors, or engaging in other conduct that unreasonably restrains competition, and such laws and regulations may impact potential business relationships or transactions with third parties in the future. In addition, health and safety regulations have necessitated, and may continue to necessitate, increased operating costs or capital investments to promote a safe working environment. The Company is also required to comply with increasingly complex and changing laws and regulations enacted to protect business and personal data in the United States and other jurisdictions regarding privacy, data protection and data security, including those related to the collection, storage, use, transmission and protection of personal information and other consumer, customer, vendor or employee data. Further, an increasing number of laws and regulations focused on product and chemical hazards, including regulations concerning the impact of product manufacturing and use on climate change, and resulting preferential product selection could also impact our ability to manufacture and sell certain products or require significant research and development investment and capital expenditures to meet regulatory requirements. With respect to the laws and regulations noted above, as well as other applicable laws and regulations, the Company's compliance programs, may under certain circumstances, involve material investments in the form of additional processes, training, personnel, information technology and capital. For a discussion of the risks associated with certain applicable laws and regulations, see Item 1A, "Risk Factors."

Sustainability

As a worldwide leader in our industry, our goal is to be at the forefront of corporate sustainability efforts. It is our ambition to be a net-positive company, that is, one whose positive impact of our people and products, is greater than the negative impact of manufacturing our products. We work to continually increase the good our people and products do while we concurrently strive to reduce the negative environmental impact of our operations.

Our climate-related sustainability efforts have led Owens Corning to develop a range of strategies and tactics that have had a significant impact on the way we conduct our business. We strive to reduce the greenhouse gas emissions released throughout the entire life cycle of our products by improving the use-phase impacts of our products, making our manufacturing processes more energy-efficient, sourcing more renewable electricity, improving our supply chain logistics, increasing recycled content, and developing end-of-life recycling solutions. Together, this work helps to reduce the environmental impact of our operations and lowers the embodied carbon in our products – an attribute of growing importance to our customers.

Many of Owens Corning's products are made using heavy, industrialized manufacturing processes. While we strive to continue our progress to reduce our impact, our factories produce various emissions, including greenhouse gases. Owens Corning is subject to or has chosen to voluntarily participate in Emissions Trading Schemes around the world. Broad and gradual tightening of national, regional, and state government limits on emissions could disrupt our access to energy sources or specific raw materials, which in turn could disrupt the manufacturing of products dependent upon them. Owens Corning invests in research and development on climate-related risks and opportunities.

Human Capital Resources

The Company's long-term success is dependent upon its access to and development of management and primary employees who are sufficiently skilled and capable of the work necessary to achieve the Company's short-and long-term business objectives. To maintain employee engagement, Owens Corning strives to ensure its people feel valued, included, and engaged – from recruitment to retirement. That is why Owens Corning is dedicated to fostering an environment of learning and growth within a supportive, caring culture. We are committed to providing a safe, healthy workplace and a meaningful, engaging employee experience.

As of December 31, 2023, Owens Corning had approximately 18,000 employees, of which approximately 9,000 were located outside the United States. Approximately 8,200 (46%) of hourly employees are subject to collective bargaining agreements. The Company regularly engages its salaried, non-represented and represented primary employees to collect feedback and based on that feedback believes employee engagement and relations are good. In 2023, the Company also engaged in an Employee Value Proposition survey and considered employee feedback in formulating its value proposition, including changes to compensation and benefits offerings such as sick leave enhancements for primary employees and improvements to our facilities, including the roll-out of personal dignity spaces.

Safety and Well-Being

One of our primary objectives is the safety and well-being of our employees. Working safely is an unconditional, organization-wide expectation at Owens Corning, which we believe directly benefits employees' lives, improves our manufacturing processes and reduces our costs. The Company maintains comprehensive safety programs focused on identifying hazards and eliminating risks that can lead to severe injuries. In the fall of 2023, the Company kicked off its employee-developed "Safer Together" initiative, intended to increase employee focus and collective engagement on safety. One of our primary safety measures is the Recordable Incident Rate ("RIR") as defined by the United States Bureau of Labor Statistics. For the year ended December 31, 2023, our RIR was 0.60, compared to 0.65 as reported in the same period for the prior year.

Additionally, with our Healthy Living platform, we provide a multifaceted well-being program designed to drive sustainable, long-term change, improve the health and lives of employees, and strengthen the culture and work experience.

Employee Performance and Related Objectives

We also focus on evaluating and managing employee performance, development, succession planning, and turnover. Our goal is to create a high-performance culture and teams that are diverse, capable and engaged. We strive to have clear objectives, effective performance management, and a structure that includes regular feedback, talent reviews, succession planning, development, and compensation analysis.

Corporate Culture

Another objective we pursue is maintaining a corporate culture focused on inclusion and diversity, ethics and compliance, training and positive employee relations and engagement. The Company believes its success and sustainability are enhanced by an inclusive and diverse workforce. We believe that inclusion and diversity add value to the business by fostering an environment that leads to high engagement and innovative thinking in the workplace. Five years following the Company's pledge to diversity & inclusion, there is more work to be done, and the Company continues to pursue diversity in its workforce through diverse candidate slates, diversity on hiring committees, and development programs, and the continued focus on development of management skills needed to sustain progress in this area through the roll-out of inclusive leadership training across the organization. Owens Corning operates programs that foster gender and ethnic diversity as well as equality within its workforce, including supporting various employee-led affinity groups, so its employees feel valued and appreciated for the distinct voices they bring to the team.

As of December 31, 2023, the composition of our Board of Directors was 60% demographically diverse, which includes gender, race, ethnicity, nationality, national origin or other elements of one's identity. Leadership positions were comprised of approximately 29% women globally and 18% people of color in the United States. Our 2030 diversity goals set targets for our leadership positions of 35% women globally and 22% people of color in the United States.

The Company has a robust pay equity system, which includes multiple processes and controls to prevent pay equity gaps from occurring. We perform a biennial pay equity review with the assistance of a third-party vendor who utilizes a strong, statistical analysis of pay equity across our global salaried workforce. We promptly remediate all identified and substantiated pay gaps through pay increases.

Ethics and compliance efforts include our support of the Owens Corning Code of Conduct ("Code of Conduct"), which is dedicated to encouraging compliance with a range of legal guidelines and our corporate values. Our training efforts encompass the Code of Conduct and other areas of compliance and development as relevant to employees. We also seek to foster positive and productive relations with the labor organizations representing them.

Owens Corning employees contribute service hours to boards, special causes and nonprofit organizations in the communities where they live and operate. These programs aim to enable the Company's employees to connect with the community, further improve its reputation locally and globally, and instill a sense of pride in the workforce.

Owens Corning is a recognized leader on advancing social issues, including issues related to diversity, equity and inclusion and human rights.Select awards and honors earned by the Company include:

- Earned a Top 50 ranking on the Fair 360 survey as leaders in workplace fairness;
- Earned a Top 10 ranking on the 100 Best Corporate Citizens in 2023 by 3BL Media for the sixth year in a row; and
- Recognized as one of the "2023 World's Most Ethical Companies" by Ethisphere Institute for the sixth year in a row.

More information about Owens Corning's approach to human capital and other social issues can be found in our Sustainability Report on our website[1].

AVAILABILITY OF INFORMATION

Owens Corning makes available, free of charge, through its website, the Company's Annual Report on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K and all amendments to those reports as soon as reasonably practicable after such material is electronically filed with or furnished to the Securities and Exchange Commission. These documents are available through the Investor Relations page of the Company's website at www.owenscorning.com. Copies of any materials we file with the SEC can also be obtained free of charge through the SEC's website at http://www.sec.gov.

[1] The information on our website, including our Sustainability Report, is not, and will not be deemed to be, a part of this Annual Report on Form 10-K or incorporated into any of our other filings with the SEC.

ITEM 1A. RISK FACTORS

In an enterprise as diverse as ours, a wide range of factors could affect future performance. We discuss in this section some of the risk factors that could materially and adversely affect our business, financial condition, value and results of operations. You should not interpret the disclosure of any risk factor to imply that the risk has not already materialized. You should consider these risk factors in connection with evaluating the forward-looking statements contained in this Annual Report on Form 10-K because these risk factors could cause our actual results and financial condition to differ materially from those projected in forward-looking statements.

The Company maintains processes that aim to manage enterprise risks through identification and mitigation of those risks. Despite our efforts, we may fail to identify or mitigate certain risks, which could have a material and adverse impact on our business, financial condition, value and results of operations in future periods.

MACROECONOMIC, MARKET AND OPERATIONAL RISKS

Low levels of residential, commercial or industrial construction activity can have a material adverse impact on our business and results of operations.

A large portion of our products are used in the markets for residential and commercial construction and repair and remodeling. Demand for certain of our products is affected in part by the level of new residential construction in the United States and elsewhere, although typically not until a number of months after the change in the level of construction. Lower demand in the regions and markets where our products are sold could result in lower revenues and lower profitability. Historically, construction activity has been cyclical and is influenced by prevailing economic conditions, including the level of interest rates and availability of financing, inflation, employment levels, consumer spending habits, consumer confidence and other macroeconomic factors outside our control. Interest rates increased substantially in fiscal years 2022 and 2023, and may continue to increase. The combination of high interest rates and high levels of inflation have reduced the affordability of mortgages and other financing options, and increased the cost of home improvement projects. These trends have likely resulted in reduced levels of repair and remodel as well as new construction activity and demand for our products, and we expect these trends may continue for the foreseeable future. We cannot predict if or when interest rates or inflation levels will stabilize or decline or the impact that any such decline may have on repair and remodel activity, new construction activity, demand for our products, our business generally, or our financial condition.

Residential and commercial construction is also affected by the cost and availability of skilled labor, which could impact both the cost and pace of construction activity, as well as the construction methods used, all of which could adversely affect demand for our products.

Some of our products, particularly in our Insulation business, are used in industrial applications, such as piping and storage tanks. Lower levels of industrial production and other macroeconomic factors affecting industrial construction activity could lessen demand for those products and lead to lower revenues or profitability.

We may be exposed to cost increases or reduced availability of raw materials or transportation, which could reduce our margins and have a material adverse impact on our business, financial condition and results of operations.

Our business relies heavily on certain commodities and raw materials used in our manufacturing processes. Additionally, we spend a significant amount on inputs that are influenced by energy prices, such as asphalt, chemicals, resins, and transportation. Price increases for these inputs could raise costs and reduce our margins if we are not able to offset them by increasing the prices of our products, improving productivity or hedging, where appropriate.

Availability of certain of the raw materials we use has occasionally been limited, and our sourcing of some of these raw materials from a limited number of suppliers, and in some cases a sole supplier, increases the risk of unavailability. For example, if one of the raw materials important to our business is sourced from a sole supplier, our production could be interrupted regardless of whether we have a long-term supply contract for the material. Global economic conditions may also result in global or regional supply chain issues that adversely impact our access to raw materials and supplies. Despite our contractual supply agreements with many of our suppliers, and despite any programs we may undertake to mitigate supply risks, it is possible that we could experience a lack of certain raw materials that limits our ability to manufacture our products, thereby materially and adversely impacting our business, financial condition and results of operations.

In addition, we are dependent on third-party freight carriers to transport some of our raw materials and products. We may be unable to transport our raw materials or products in a timely manner or at economically favorable rates in certain circumstances, particularly in cases of adverse market conditions or disruptions to transportation infrastructure.

Supply constraints and increases in the cost of energy could have a material adverse impact on our business or results of operations.

The cost of producing our products is sensitive to the price of energy, including its impact on transport costs which is subject to factors outside of our control. Energy prices, in particular oil and natural gas, have fluctuated in recent years. For example, natural gas forms the primary energy source for our European operations and our European operations can be directly affected by volatility in the cost and availability of natural gas. Natural gas supply shortages could lead to additional price increases, energy supply rationing, or temporary reduction in our European operations, which could have a material adverse impact on our business or results of operations.

We are subject to risks and uncertainties associated with our international operations.

We sell products and operate plants throughout the world. Our international sales and operations are subject to risks and uncertainties, including:

- difficulties and costs associated with complying with a wide variety of complex and changing laws, including securities laws, climate-related laws, tax laws, employment and pension-related laws, competition laws, U.S. and foreign export and trading laws, and laws governing improper business practices, treaties and regulations;
- limitations on our ability to enforce legal rights and remedies;
- adverse domestic or international economic and political conditions, business interruption, war and civil disturbance;
- changes to tax, currency, or other laws or policies that may adversely impact our ability to repatriate cash from non-United States subsidiaries, make cross-border investments, or engage in other intercompany transactions;
- future tax legislation, regulations, or related guidance or interpretations;
- changes to tariffs or other import or export restrictions, penalties or sanctions, including modification or elimination of international agreements covering trade or investment;
- costs and availability of shipping and transportation;
- nationalization or forced relocation of properties by foreign governments;
- currency exchange rate fluctuations between the United States Dollar and foreign currencies; and
- uncertainty with respect to any potential changes to laws, regulations and policies that could exacerbate the risks described above.

As we continue to expand our business globally, we may have difficulty anticipating and effectively managing these and other risks that our international operations may face, which may adversely impact our business, financial condition and results of operations.

In addition, we operate in many parts of the world that have experienced governmental corruption and we could be adversely affected by violations of the Foreign Corrupt Practices Act (“FCPA”) and similar worldwide anti-corruption laws. The FCPA and similar anti-corruption laws in other jurisdictions generally prohibit companies and their intermediaries from making improper payments to officials for the purpose of obtaining or retaining business. Although we mandate compliance with these anti-corruption laws and maintain an anti-corruption compliance program, these measures may not prevent our employees or agents from violating these laws. If we were found liable for violations of anti-corruption laws, we could be liable for criminal or civil penalties or other sanctions, which could have a material adverse impact on our business, financial condition and results of operations.

Our sales may fall rapidly in response to declines in demand because we do not operate under long-term volume agreements to supply our customers and because of customer concentration in certain segments.

Many of our customer volume commitments are short-term; therefore, we do not have a significant manufacturing backlog. As a result, we do not benefit from the visibility provided by long-term volume contracts against downturns in customer demand and sales. Further, we are not able to immediately adjust our costs in response to declines in sales. Our ability to sell some of the products in our Insulation and Roofing segments is dependent on a limited number of customers, who account for a significant portion of such sales. In 2023, we had one customer that represented 11% of our annual sales. The loss of key customers for these products, a consolidation of key customers or a significant reduction in sales to those customers, could significantly reduce our revenues from these products. In addition, if key customers experience financial pressure or

consolidate, they could attempt to demand more favorable contractual terms, which would place additional pressure on our margins and cash flows. Lower demand for our products, loss of key customers and material changes to contractual terms could materially and adversely impact our business, financial condition and results of operations. Furthermore, some of our sales are concentrated in certain geographic areas, and market growth that is skewed to other geographic areas may negatively impact our rate of growth or market share.

Worldwide economic conditions and credit tightening could have a material adverse impact on the Company.

The Company's business may be materially and adversely impacted by changes in United States or global economic conditions, including global industrial production rates, inflation, deflation, interest rates, availability of capital, consumer spending rates, energy availability and commodity prices, trade laws, and the effects of governmental initiatives to manage economic conditions. Changes in and/or new laws, regulations and policies that may be enacted in the United States or elsewhere could also materially impact economic conditions and the Company's business and results of operations. These changes and conditions could materially and adversely impact the Company's operations, financial results and/or liquidity, including:

- the financial stability of our customers or suppliers may be compromised, which could result in reduced demand for our products, additional bad debts for the Company or non-performance by suppliers;
- one or more of the financial institutions associated with our credit facilities could cease to fulfill their funding obligations, or the amount of eligible receivables under our receivables securitization facility could decrease, which could materially and adversely impact our liquidity;
- it may become more expensive or difficult to obtain financing or refinance the Company's debt in the future;
- the value of the Company's assets held in pension plans may decline; and
- the Company's assets may be impaired or subject to write-down or write-off.

With the volatility in the current global economic climate, inflation and geopolitical events around the world, including the Russian invasion of Ukraine and the Israel-Hamas conflict, it is difficult for us to predict the complete impact of the foregoing matters on our business and results of operations.

Uncertainty about global economic conditions may also cause consumers of our products to reduce or postpone spending in response to tighter credit, negative financial news and/or declines in income or asset values. This could have a material adverse impact on the demand for our products and on our financial condition and operating results. A deterioration of economic conditions may exacerbate these adverse effects and could result in a wide-ranging and prolonged impact on general business conditions, thereby negatively impacting our operations, financial results and/or liquidity.

We are subject to risks relating to our information technology systems (including cybersecurity) risks, and any failure to adequately protect our critical information technology systems could materially affect our operations.

We rely on information technology systems across our operations, including for management, supply chain and financial information and various other processes and transactions. Our ability to effectively manage our business depends on the security, reliability and capacity of these systems. Our information technology systems, some of which are dependent on services provided by third parties, may be vulnerable to damage, interruption, or shutdown due to any number of causes outside of our control such as catastrophic events, natural disasters, fires, power outages, systems failures, telecommunications failures, employee error or malfeasance, security breaches, computer viruses or other malicious codes, ransomware, unauthorized access attempts, denial of service attacks, phishing, hacking, and other cyberattacks. In addition, our operations in certain geographic locations may be particularly vulnerable to cybersecurity attacks or other problems. Any such damage, interruption, or shutdown could cause delays or cancellation of customer orders or impede the manufacture or shipment of products, processing of transactions or reporting of financial results. An attack or other problem with our systems could also result in the disclosure of proprietary information about our business or confidential information concerning our customers or employees, which could result in significant damage to our business and our reputation.

We have established a range of security measures that are designed to protect against the unauthorized access to and misappropriation of our information, corruption of data, intentional or unintentional disclosure of confidential information, or disruption of operations. However, advanced cybersecurity threats, such as malware, ransomware, and phishing attacks, attempts to access information, and other security breaches, are persistent and continue to evolve, making them increasingly difficult to identify and prevent. Protecting against these threats may require significant resources, and we may not be able to implement measures that will protect against all of the significant risks to our information technology systems. In addition, we rely on a number of third-party service providers to execute certain business processes and maintain certain information technology systems and infrastructure, and any breach of security on their part could impair our ability to effectively operate.

Any breach of our security measures, or those of our third-party service providers, could result in unauthorized access to and misappropriation of our information, corruption of data or disruption of operations or transactions, any of which could have a material adverse effect on our business strategy, results of operations or financial condition.

Additionally, we regularly move data across national borders to conduct our operations and, consequently, are subject to a variety of laws and regulations in the United States and other jurisdictions regarding privacy, data protection, and data security, including those related to the collection, storage, handling, use, disclosure, transfer, and security of personal data, including the European Union General Data Protection Regulation. Our efforts to comply with privacy and data protection laws may impose significant costs and challenges that are likely to increase over time.

Our efforts in acquiring and integrating other businesses, establishing joint ventures, expanding our production capacity or divesting assets are subject to a number of risks.

Some of the ways we have historically grown or restructured our business have been through acquisitions, joint ventures, the expansion of our production capacity and divestitures. Our ability to grow or restructure our business depends upon our ability to identify, negotiate and finance suitable arrangements. If we cannot successfully execute on such arrangements or receive any required regulatory approvals on a timely basis, we may be unable to generate desired returns, and our expectations of future results of operations, including cost savings and synergies, may not be achieved. Acquisitions, joint ventures, production capacity expansions and divestitures involve substantial risks, including:

- unforeseen difficulties in operations, technologies, products, services, accounting and personnel;
- increased cybersecurity threats or incidents;
- diversion of financial and management resources from existing operations;
- unforeseen difficulties related to entering geographic regions, markets or product lines where we do not have prior experience;
- risks relating to obtaining sufficient financing;
- difficulty in integrating the acquired business' standards, processes, procedures and controls with our existing operations;
- potential loss of key employees;
- unanticipated competitive responses;
- potential loss of customers or suppliers; and
- undisclosed or undiscovered liabilities or claims, or retention of unpredictable future liabilities.

Our failure to address these risks or other problems encountered in connection with our past or future acquisitions, including the planned acquisition of Masonite, investments and divestitures could cause us to fail to realize the anticipated benefits of such transactions, incur unanticipated liabilities, and harm our business generally. Future acquisitions and investments could also result in dilutive issuances of our equity securities, the incurrence of debt, contingent liabilities, or amortization expenses, or write-offs of goodwill, any of which could have a material adverse impact on our business, financial condition and results of operations. Also, the anticipated benefits of our investments may not materialize.

We recently announced that we have decided to review strategic alternatives for our global glass reinforcements business, consistent with our strategy to expand our building materials offering and focus on products and applications where we can build market-leading positions. While a range of options are under consideration, including a potential sale, spin-off or other strategic option, there can be no assurance that the strategic review will result in any transaction or other outcome, or that we will realize our strategic and other objectives in connection with any such transaction or outcome.

We face significant competition in the markets we serve and we may not be able to compete successfully.

All of the markets we serve are highly competitive. We compete with manufacturers and distributors, both within and outside the United States. Some of our competitors may have superior financial, technical, marketing and other resources. In some cases, we face competition from manufacturers in countries able to produce similar products at lower costs. Price competition or overcapacity may limit our ability to raise prices for our products, may force us to reduce prices and may also result in reduced levels of demand for our products and cause us to lose market share. We also face competition from the introduction by competitors of new products or technologies that may address our customers' needs in a better manner, whether based on considerations of pricing, usability, effectiveness, sustainability, quality or other features or benefits. In addition, to effectively

compete, we must continue to develop new products that meet changing consumer preferences and successfully develop, manufacture and market these new products. If we are not able to successfully commercialize our innovation efforts, we may lose market share. Our inability to effectively compete could result in the loss of customers and reduce the sales of our products, which could have a material adverse impact on our business, financial condition and results of operations.

Emerging issues related to our development, integration and use of artificial intelligence ("AI") could give rise to legal or regulatory action, damage our reputation or otherwise materially harm our business.

Our development, integration and use of AI technology in our operations remains in the early phases. We have started to assess the use of AI technology to drive productivity and data analytics. While we aim to develop, integrate and use AI responsibly, we may ultimately be unsuccessful in identifying or resolving issues, such as accuracy issues, cybersecurity risks, unintended biases, and discriminatory outputs, before they arise. AI is a new and emerging technology in early stages of commercial use and presents a number of risks inherent in its use, including, but not limited to, ethical considerations, public perception, intellectual property protection, regulatory compliance, privacy concerns and data security, all of which could have a material adverse effect on our business, results of operations and financial position. As a result, we cannot predict future developments in AI and related impacts to our business and our industry. If we are unable to successfully and accurately develop, integrate and use AI technology, as well as address the risks and challenges associated with AI, our business, results of operations and financial position could be negatively impacted. Additionally, if the content, analyses, or recommendations that AI applications assist in producing are or are alleged to be deficient, inaccurate, or biased, our reputation, business, financial condition, and results of operations may be adversely affected.

Climate change, weather conditions and storm activity could have a material adverse impact on our business, financial condition and results of operations.

Climate change could have an impact on several aspects of our business, financial condition and results of operations.Weather phenomena associated with climate change, such as flooding or altered storm activity, may impact our ability to operate our manufacturing facilities in some locations. In addition, customer preferences for lower-carbon and more environmentally friendly solutions could impact demand for our products. Although we believe that some of our product categories, such as insulation and composites, could experience increased demand due to environmental benefits, such as energy efficiency and renewable energy, the timing and impact of such increased demand is uncertain.

Weather conditions and the level of severe storms can have a significant impact on the markets for residential and commercial construction, repair and improvement projects. These factors could impact our business as follows:

- generally, any weather conditions that slow or limit residential or commercial construction activity can adversely impact demand for our products; and
- a portion of our annual product demand is attributable to the repair of damage caused by severe storms. In periods with below average levels of severe storms, demand for such products could be reduced.

Lower demand for our products as a result of either of these weather-related scenarios could have a material adverse impact on our business, financial condition and results of operations. Additionally, severely low or high temperatures may lead to significant and immediate spikes in costs of natural gas, electricity and other commodities that could negatively affect our results of operations.

We will not be insured against all potential losses and could be seriously harmed by natural disasters, catastrophes, pandemics, theft or sabotage.

Many of our business activities globally involve substantial investments in manufacturing facilities and many products are produced at a limited number of locations. These facilities could be materially damaged by natural disasters such as floods, tornados, hurricanes, fires, earthquakes, pandemics or by theft or sabotage. We could incur uninsured losses and liabilities arising from such events, including damage to our reputation, and/or suffer material losses in operational capacity, which could have a material adverse impact on our business, financial condition and results of operations.

LEGAL, REGULATORY AND COMPLIANCE RISKS

We could face potential product liability and warranty claims, we may not accurately estimate costs related to such claims, and we may not have sufficient insurance coverage available to cover such claims.

Our products are used and have been used in a wide variety of residential, commercial and industrial applications. We face an inherent business risk of exposure to product liability or other claims in the event our products are alleged to be defective or that

the use of our products is alleged to have resulted in harm to others or to property. We may, in the future, incur liability if product liability lawsuits against us are successful. Moreover, any such lawsuits, whether or not successful, could result in adverse publicity to us, which could cause our sales to decline. We maintain insurance coverage to protect us against product liability claims, but that coverage may not be adequate to cover all claims that may arise or we may not be able to maintain adequate insurance coverage in the future at an acceptable cost. Any liability not covered by insurance or that exceeds our established reserves could materially and adversely impact our business, financial condition and results of operations.

For example, during the second quarter of 2023, the Company's subsidiary, Paroc Group OY ("Paroc"), notified the appropriate European maritime regulatory authorities that specific insulation products in its marine insulation product line may not meet certain fire safety requirements in accordance with their certifications. Paroc voluntarily withdrew these specific products from the market, issued recalls, and suspended distribution and sales of these products. Paroc is cooperating with the applicable regulatory and government authorities, and continues to work with its customers and end-users to assist with remediation. Although we established an estimated liability for expected future costs related to this matter, it is reasonably possible that additional product recall costs could be incurred that exceed the estimated liability by amounts that could be material to our consolidated financial statements. These actions may also result in harm to our reputation and results of operations.

In addition, consistent with industry practice, we provide warranties on many of our products. We may experience costs of warranty claims when the product is not performing to the satisfaction of the claimant even though it has not caused harm to others or property. We estimate our future warranty costs based on historical trends and product sales, but we may fail to accurately estimate those costs and thereby fail to establish adequate warranty reserves for them. Warranty claims are not insurable.

We may be subject to liability under and may make substantial future expenditures to comply with environmental and emerging product-based laws and regulations.

Our manufacturing facilities are subject to numerous foreign, federal, state and local laws and regulations relating to the presence of hazardous materials, pollution and the protection of the environment, including those governing emissions to air, discharges to water, use, storage and transport of hazardous materials, storage, treatment and disposal of waste, remediation of contaminated sites and protection of worker health and safety. We are also subject to laws, rules and regulations relating to certain raw materials used in our business or in our products.

Liability under these laws involves inherent uncertainties. Environmental liability estimates may be affected by changing determinations of what constitutes an environmental exposure or an acceptable level of cleanup. For example, remediation activities generally involve a potential range of activities and costs related to soil and groundwater contamination. This can include pre-cleanup activities, such as fact finding and investigation, risk assessment, feasibility studies, remedial action design and implementation (where actions may range from monitoring to removal of contaminants, to installation of longer-term remediation systems). Please see "Item 1 - Business - Environmental Control" for information on costs and accruals related to environmental remediation. To the extent that the required remediation procedures or timing of those procedures change, additional contamination is identified, or the financial condition of other potentially responsible parties is adversely affected, the estimate of our environmental liabilities may change. Change in required remediation procedures or timing of those procedures at existing legacy sites, or discovery of contamination at additional sites, could result in increases to our environmental obligations. Violations of environmental, health and safety laws are subject to civil, and, in some cases, criminal sanctions.

As a result of these uncertainties, we may incur unexpected interruptions to operations, fines, penalties or other reductions in income which could adversely impact our business, financial condition and results of operations. It is possible that new laws and regulations will specifically address climate change, toxic air emissions, ozone forming emissions and fine particulate matter. New environmental and chemical regulations could impact our ability to expand production or construct new facilities in every geographic region in which we operate. Continued and increased government and public emphasis on environmental issues is expected to result in increased future investments for environmental controls at ongoing operations, which will be charged against income from future operations. Present and future environmental laws and regulations applicable to our operations, and changes in their interpretation, may require substantial capital expenditures or may require or cause us to modify or curtail our operations, which may have a material adverse impact on our business, financial condition and results of operations. Although emerging in nature, an increasing number of laws and regulations focused on product and chemical hazards, including regulations concerning the impact of product manufacturing and use on climate change, and resulting preferential product selection could also impact our ability to manufacture and sell certain products or require significant research and development investment and capital expenditures to meet regulatory requirements.

Proposed or future laws or regulations aimed at addressing climate change, including, but not limited to, local building codes, Environmental Protection Agency regulations on greenhouse gas emissions ("GHG"), laws or regulations impacting energy supply, and other laws or regulations, may materially impact demand for our products or our cost of doing business.

We believe it is likely that the scientific and political attention to issues concerning the extent and causes of climate change will continue, with new and more restrictive laws and regulations focusing on environmental, social and governance ("ESG") initiatives that could affect our financial condition, results of operations and cash flows. Foreign, federal, state and local regulatory and legislative bodies have enacted or proposed various legislative and regulatory measures relating to increased transparency and standardization of reporting matters that may include climate change, regulating GHG emissions, water usage, recycling of plastic materials, and energy policies, including waste tax, and other governmental charges and mandates. As a result, we expect to be subject to overlapping, yet distinct, climate-related disclosure requirements in multiple jurisdictions. Compliance with foreign, federal, state and local legislation and regulations concerning climate-related disclosures, including compliance with the European Commission's Corporate Sustainability Reporting Directive and the SEC's proposed climate disclosure requirements, may result in additional costs and capital expenditures, and the failure to comply with such legislation and regulations could result in fines to us and could affect our business, financial condition, results of operations and cash flows. We could also face increased costs related to defending and resolving legal claims and other litigation related to climate change and the alleged impact of our operations on climate change. In addition, energy prices could increase as a result of climate change legislation or other environmental mandates, which could have an adverse effect on our results of operations.

In addition, from time to time, we establish targets, strategies and expectations related to climate change and other environmental matters. Our ability to achieve any such targets, strategies or expectations is subject to risks and uncertainties, many of which are outside of our control. These risks and uncertainties include, but are not limited to, our ability to execute our strategies and achieve our goals within the currently projected costs and expected timeframes, availability, use and success of on and off-site renewable energy, evolving regulatory and other standards, processes, and assumptions, the pace of scientific and technological developments, increased costs and availability of requisite financing, market trends that may alter business opportunities, the conduct of third-party manufacturers and suppliers, constraints or disruptions to our supply chain, and changes in carbon markets. There are no assurances that we will be able to successfully execute our strategies and achieve our targets. Failures or delays (whether actual or perceived) to achieve our targets or strategies related to climate change and other environmental matters could damage our reputation, customer and investor relationships, adversely affect our business, operations and increase risk of litigation.

Our intellectual property rights may not provide meaningful commercial protection for our products or brands and third parties may assert that we violate their intellectual property rights, which could have a material adverse impact on our business, financial condition and results of operations.

We rely on our intellectual property, including numerous patents, trademarks, trade secrets, confidential information, as well as our licensed intellectual property, to differentiate our products and brands in the marketplace. We monitor and protect against activities that might infringe, dilute, or otherwise harm our intellectual property and rely on the laws of the United States and other countries to protect our rights. However, in some instances, we may be unaware of unauthorized use of our intellectual property. To the extent we cannot protect our innovations or are unable to enforce our intellectual property rights, unauthorized use and misuse of our intellectual property or innovations could harm our competitive position and have a material adverse impact on our business, financial condition and results of operations. In addition, the laws of some foreign jurisdictions provide less protection for our proprietary rights than the laws of the United States and we therefore may not be able to effectively enforce our intellectual property rights in these jurisdictions. If we are unable to maintain certain exclusive licenses, our brand recognition and sales could be adversely impacted. Current employees, contractors and suppliers have, and former employees, contractors and suppliers may have, access to trade secrets and confidential information regarding our operations that could be disclosed improperly and in breach of contract to our competitors or otherwise used to harm us.

Third parties may also claim that we are infringing upon their intellectual property rights. If we are unable to successfully defend or license such alleged infringing intellectual property or if we are required to substitute similar technology from another source, our operations could be adversely affected. Even if we believe that such intellectual property claims are without merit, defending such claims can be costly, time consuming and require significant resources. Claims of intellectual property infringement also may require us to redesign affected products, pay costly damage awards, or face injunctions prohibiting us from manufacturing, importing, marketing or selling certain of our products. Even if we have agreements to indemnify us, indemnifying parties may be unable or unwilling to do so.

We are subject to various legal and regulatory proceedings, including litigation in the ordinary course of business, and uninsured judgments or a rise in insurance premiums may have a material adverse impact on our business, financial condition and results of operations.

In the ordinary course of business, we are subject to various legal and regulatory proceedings, which may include but are not limited to those involving antitrust, tax, trade, environmental, intellectual property, data privacy and other matters, including general commercial litigation. Any claims raised in legal and regulatory proceedings, whether with or without merit, could be time consuming and expensive to defend and could divert management's attention and resources. Additionally, the outcome of legal and regulatory proceedings may differ from our expectations because the outcomes of these proceedings are often difficult to predict reliably. Various factors and developments can lead to changes in our estimates of liabilities and related insurance receivables, where applicable, or may require us to make additional estimates, including new or modified estimates, that may be appropriate due to a judicial ruling or judgment, a settlement, regulatory developments or changes in applicable law. A future adverse ruling, settlement or unfavorable development could result in charges that could have a material adverse effect on our results of operations in any particular period.

In accordance with customary practice, we maintain insurance against some, but not all, of these potential claims. In the future, we may not be able to maintain insurance at commercially acceptable premium levels. In addition, the levels of insurance we maintain may not be adequate to fully cover any and all losses or liabilities. If any significant judgment or claim is not fully insured or indemnified against, it could have a material adverse impact on our business, financial condition and results of operations.

FINANCIAL RISKS

Our level of indebtedness could adversely impact our business, financial condition or results of operations.

At December 31, 2023, we had total debt of approximately $3.0 billion. On February 8, 2024, we entered into a definitive agreement to acquire Masonite International Corporation ("Masonite"), subject to the satisfaction or waiver of specified conditions. We expect to incur approximately $3.0 billion of debt to pay a substantial portion of the purchase price for the acquisition of Masonite, as well as assume up to $875 million of Masonite's senior unsecured notes.

Our debt level and degree of leverage, particularly if we complete the Masonite acquisition, could have important consequences, including the following:

- our ability to obtain additional debt or equity financing for working capital, capital expenditures, debt service requirements, acquisitions and general corporate or other purposes may be limited;
- a substantial portion of our cash flow could be required for the payment of principal and interest on our indebtedness, and may not be available for other business purposes;
- certain of our available borrowings are at variable rates of interest, exposing us to the risk of increased interest rates to borrow in the future;
- if due to liquidity needs we must replace any indebtedness upon maturity, we would be exposed to the risk that we may not be able to refinance such indebtedness;
- our ability to adjust to changing market conditions may be limited and place us at a competitive disadvantage compared to our competitors if they have less debt; and
- we may be vulnerable in a downturn in general economic conditions or in our business, or we may be unable to carry out important capital spending.

The credit agreement governing our senior revolving credit facility, the indentures governing our senior notes, and the receivables purchase agreement governing our receivables securitization facility contain various covenants that impose operating and financial restrictions on us and our subsidiaries. Additionally, instruments and agreements governing our future indebtedness may impose other restrictive conditions or covenants that could restrict our ability to conduct our business operations or pursue growth strategies.

Downgrades of our credit ratings could adversely impact us.

Our credit ratings are important to our cost of capital. The major debt rating agencies routinely evaluate our debt based on a number of factors, which include financial strength and business risk as well as transparency with rating agencies and timeliness of financial reporting. A downgrade in our debt rating could result in increased interest on our existing variable

interest rate debt, increased interest and other expenses for future borrowings, and reduced ability for our suppliers to utilize supply chain financing programs. Downgrades in our debt rating could also restrict our access to capital markets and affect the value and marketability of our outstanding senior notes.

Our operations require substantial capital, leading to high levels of fixed costs that will be incurred regardless of our level of business activity.

Our businesses are capital intensive, and regularly require capital expenditures to expand operations, maintain equipment, increase operating efficiency and comply with applicable laws and regulations, leading to high fixed costs, including depreciation expense. Increased regulatory requirements for our operations could lead to additional or higher fixed costs in the future. We are limited in our ability to reduce fixed costs quickly in response to reduced demand for our products and these fixed costs may not be fully absorbed, resulting in higher average unit costs and lower gross margins if we are not able to offset this higher unit cost with price increases. Alternatively, we may be limited in our ability to quickly respond to unanticipated increased demand for our products, which could result in an inability to satisfy demand for our products and loss of market share.

Our ongoing efforts to increase productivity and reduce costs may not result in anticipated savings in operating costs.

Our cost reduction and productivity efforts, including those related to our existing operations, production capacity expansions, new manufacturing platforms, or other capital expenditures, may not produce anticipated results. Our ability to achieve cost savings and other benefits within expected time frames is subject to many estimates and assumptions. These estimates and assumptions are subject to significant economic, competitive, legal and other uncertainties, some of which are beyond our control. If these estimates and assumptions are incorrect, if we experience delays, or if other unforeseen events occur, our business, financial condition and results of operations could be adversely impacted.

Our results of operations in a given period may be impacted by price volatility in certain renewable-generated energy markets.

In connection with our sustainability goals to reduce GHG and toxic air emissions, we entered into contracts pursuant to which we have agreed to purchase renewable-generated electricity from third parties. Under these contracts, we do not take physical delivery of renewable-generated electricity. The generated electricity is instead sold by our counterparties to local grid operators at the prevailing market price and we obtain the associated non-tax renewable energy credits. The prevailing market pricing for renewable-generated electricity can be affected by factors beyond our control and is subject to significant period over period volatility. For example, renewable-generated energy output fluctuates due to climactic and other factors beyond our control and can be constrained by available transmission capacity, thereby significantly impacting pricing. Due to this potential volatility, it is possible that these contracts, or similar contracts we execute in the future, could have an impact on our results of operations in a given reporting period.

Our hedging activities to address energy price fluctuations may not be successful in offsetting increases in those costs or may reduce or eliminate the benefits of any decreases in those costs.

To mitigate short-term variation in our operating results due to commodity price fluctuations in certain geographic markets, we may hedge a portion of our near-term exposure to the cost of energy. The results of our hedging practices could be positive, neutral or negative in any period depending on price changes of the hedged exposures.

Our hedging activities are not designed to mitigate long-term commodity price fluctuations and, therefore, would not protect us from long-term commodity price increases. In addition, in the future, our hedging positions may not correlate to our actual energy costs, which would cause acceleration in the recognition of unrealized gains and losses on our hedging positions in our operating results.

If we were required to write down all or part of our goodwill or other indefinite-lived intangible assets, our results of operations or financial condition could be materially adversely affected in a particular period.

Declines in our business may result in an impairment of our tangible and intangible assets, which could result in a material non-cash charge. A significant or prolonged decrease in our market capitalization, including a decline in stock price, a negative long-term performance outlook, or an increase in discount rates could result in an impairment of our tangible and intangible assets which results when the carrying value of the Company’s assets exceed their fair value.

At least annually, we assess our goodwill and intangible assets for impairment. When we utilize a discounted cash flow methodology to calculate the fair value of our reporting units, weak demand for a specific product line or business could result

in an impairment. Accordingly, any determination requiring the write-off of a significant portion of goodwill or intangible assets could negatively impact our results of operations.

HUMAN CAPITAL RISKS

We depend on our senior management team and other skilled and experienced personnel to operate our business effectively, and the loss of any of these individuals or the failure to attract additional qualified personnel could adversely impact our business, financial condition and results of operations.

We are highly dependent on the skills and experience of our senior management team and other skilled and experienced personnel. These individuals possess sales, marketing, manufacturing, logistical, financial, business strategy and administrative skills that are important to the operation of our business. We cannot assure that we will be able to retain all of our existing senior management personnel and skilled and experienced personnel. The loss of any of these individuals or an inability to attract additional qualified personnel could prevent us from implementing our business strategy and could adversely impact our business and our future financial condition or results of operations. The current and future labor markets may impact our ability to retain these individuals.

Labor shortages and increased turnover rates, increased employee-related costs, and labor disputes could have a material adverse impact on our operations, results of operations, liquidity and cash flows.

Our operations depend on the availability and relative costs of labor and maintaining good relations with our personnel and the labor unions. Several factors have had and may continue to have adverse effects on the labor force available to us, including general economic uncertainty, government regulations, laws and regulations related to workers' health and safety, inflation, wage and hour practices and immigration. Labor shortages and increased turnover rates within our personnel have led to and could in the future lead to increased costs, such as increased costs associated with training new employees and increased wage rates to attract and retain employees. An overall or prolonged labor shortage, lack of skilled labor, increased turnover or labor inflation could have a material adverse impact on our operations, results of operations, liquidity and cash flows.

We are also subject to the risk that labor strikes or other types of conflicts with personnel may arise or that we may become the subject of union organizing activity at additional facilities. Renewal of collective bargaining agreements typically involves negotiation, with the potential for work stoppages or increased costs at affected facilities.

Significant changes in the factors and assumptions used to measure our defined benefit plan obligations, actual investment returns on pension assets and other factors could have a negative impact on our financial condition or liquidity.

We have certain defined benefit pension plans and other post-employment benefit ("OPEB") plans. Our future funding requirements for defined benefit pension and OPEB plans depend upon a number of factors and assumptions, including our actual experience against assumptions with regard to interest rates used to determine funding levels, return on plan assets, benefit levels, participant experience (e.g., mortality and retirement rates), health care cost trends, and applicable regulatory changes. To the extent actual results are less favorable than our assumptions, there could be a material adverse impact on our financial condition and results of operations.

Additional risks exist due to the nature and magnitude of our investments, including the implementation of or changes to the investment policy, insufficient market capacity to absorb a particular investment strategy or high-volume transactions, and the inability to quickly rebalance illiquid and long-term investments.

If our cash flows and capital resources are insufficient to fund our pension or OPEB obligations, we could be forced to reduce or delay investments and capital expenditures, seek additional capital, or restructure or refinance our indebtedness.

RISKS RELATED TO OUR PLANNED ACQUISITION OF MASONITE

Our planned acquisition of Masonite may not occur at all or may not occur in the expected time frame, which may negatively affect the trading prices of our stock and our future business and financial results.

Completion of the planned acquisition of Masonite is subject to the satisfaction or waiver of customary and other closing conditions. The acquisition is not assured and is subject to risks and uncertainties, including the risk that the necessary regulatory approvals or shareholder approval will not be obtained or that other closing conditions will not be satisfied. We cannot predict whether and when such approvals will be received, or such conditions will be satisfied.

Our obligation to complete the planned acquisition of Masonite is not subject to a financing condition.

Our obligation to complete the planned acquisition of Masonite is not subject to a financing condition. We have obtained committed financing for $3.0 billion to pay a substantial portion of the purchase price for the acquisition of Masonite. If any of the banks in the committed financing facility are unable to perform their commitments, we may be required to finance a portion of the purchase price of the planned acquisition at interest rates higher than currently expected.

We may not realize the growth opportunities and cost synergies that are anticipated from the planned acquisition of Masonite.

The benefits that are expected to result from the planned acquisition of Masonite will depend, in part, on our ability to realize the anticipated growth opportunities and cost synergies as a result of the planned acquisition. Our success in realizing these growth opportunities and cost synergies, and the timing of this realization, depends on the successful integration of Masonite. There can be no assurance that we will successfully or cost-effectively integrate Masonite. The failure to do so could have a material adverse effect on our business, financial condition, and results of operations.

Even if we are able to integrate Masonite successfully, this integration may not result in the realization of the full benefits of the growth opportunities and cost synergies that we currently expect from this integration, and we cannot guarantee that these benefits will be achieved within anticipated time frames or at all. For example, we may not be able to eliminate duplicative costs. Additionally, we may incur substantial expenses in connection with the integration of Masonite. While it is anticipated that certain expenses will be incurred to achieve cost synergies, such expenses are difficult to estimate accurately, and may exceed current estimates. Accordingly, the benefits from the planned acquisition may be offset by costs incurred to, or delays in, integrating the businesses.

RISKS RELATED TO OWNERSHIP OF OUR COMMON STOCK

The market price of our common stock is subject to volatility.

The market price of our common stock could be subject to wide fluctuations in response to numerous factors, many of which are beyond our control. These factors include actual or anticipated variations in our operational results and cash flow, our earnings relative to our competition, changes in financial estimates by securities analysts, trading volume, sales by holders of large amounts of our common stock, short selling, market conditions within the industries in which we operate, seasonality of our business operations, the general state of the securities markets and the market for stocks of companies in our industry, governmental legislation or regulation and currency and exchange rate fluctuations, as well as general economic and market conditions, such as recessions.

We are a holding company with no operations of our own and depend on our subsidiaries for cash.

As a holding company, most of our assets are held by our direct and indirect subsidiaries and we will primarily rely on dividends and other payments or distributions from our subsidiaries to meet our debt service and other obligations and to enable us to pay dividends. The ability of our subsidiaries to pay dividends or make other payments or distributions to us will depend on their respective operating results and may be restricted by, among other things, the laws of their jurisdiction of organization (which may limit the amount of funds available for the payment of dividends or other payments), agreements of those subsidiaries, agreements with any co-investors in non-wholly-owned subsidiaries, the terms of our credit and receivables facilities and senior notes and the covenants of any future indebtedness we or our subsidiaries may incur.

Provisions in our amended and restated certificate of incorporation and bylaws or Delaware law may discourage, delay or prevent a change in control of the Company or changes in our management and therefore depress the trading price of our common stock.

Our amended and restated certificate of incorporation and bylaws contain provisions that could depress the trading price of our common stock through provisions that may discourage, delay or prevent a change in control of the Company or changes in our management that our stockholders may deem advantageous.

Additionally, we are subject to Section 203 of the Delaware General Corporation Law, which generally prohibits a Delaware corporation from engaging in any of a broad range of business combinations with any “interested” stockholder for a period of three years following the date on which the stockholder became an “interested” stockholder and which may discourage, delay or prevent a change in control of our company.

ITEM 1A. RISK FACTORS (continued)

Dividend payments on our common stock are not guaranteed and are declared at the discretion of our Board of Directors.

Since February 2014, our Board of Directors has declared a quarterly dividend on our common stock. The payment of any future cash dividends to our stockholders is not guaranteed and will depend on decisions that will be made by our Board of Directors and will depend on then-existing conditions, including our operating results, financial conditions, contractual restrictions, corporate law restrictions, capital agreements, applicable laws of the State of Delaware and business prospects.

ITEM 1B. UNRESOLVED STAFF COMMENTS

None.

ITEM 1C. CYBERSECURITY

Risk Management and Strategy

We have a range of security measures that are designed to protect against the unauthorized access to and misappropriation of our information, corruption of data, intentional or unintentional disclosure of confidential information, or disruption of operations. These security measures include controls, security processes and monitoring of our manufacturing systems. We have cloud security tools and governance processes designed to assess, identify and manage material risks from cybersecurity threats. In addition, we maintain an information security training program designed to address phishing and email security, password security, data handling security, cloud security, operational technology security processes, and cyber-incident response and reporting processes.

Our cybersecurity strategy includes defense in depth, zero trust, and standards-based controls intended to protect our information technology systems. We perform incident response tabletop exercises that include members of the Company's senior management team to validate, test, and assess the effectiveness and adequacy of certain roles and decision-making processes in the event of a cybersecurity incident. We also assess, identify, and manage cyber risk associated with divestiture and merger and acquisition activities.

The oversight of our cybersecurity risk management process is integrated into our overall risk management process. The risk committee is responsible for overseeing and monitoring our risk assessment and mitigation-related actions, including with respect to cybersecurity risks. The risk committee is not a committee of our Board of Directors. It is a cross-functional committee that includes members across many areas of expertise and is structurally independent of our business lines. The risk committee's membership is designed to provide diversity of thought and perspective related to risk, including cybersecurity risks. The risk committee identifies risks and mitigation strategies, and it provides key updates to executive officers and the Audit Committee of our Board of Directors.

We use third-party service providers to execute certain business processes, maintain certain information systems and infrastructure, evaluate defenses, and implement recommendations. We periodically have external information security assessments performed by third parties to analyze our information technology systems and to stay informed of information security risks. Additionally, we have a supplier validation process, which provides for review and approval by our cybersecurity group for cloud services.

Although we experience cybersecurity incidents from time to time as part of our operations, we have not experienced any information security breach that had, or is reasonably likely to have, a material impact on our business strategy, results of operations or financial condition. Any breach of our security measures, or those of our third-party service providers, could result in unauthorized access to and misappropriation of our information, corruption of data or disruption of systems, operations or transactions, any of which could have a material adverse effect on our business strategy, results of operations or financial condition. See "Risk Factors" on page 9 of this Form 10-K for further discussion of the risks related to cybersecurity threats.

Governance

The Board of Directors is responsible for overseeing risk for the Company and has delegated to the Audit Committee responsibility for overseeing the cybersecurity risk management strategy for the Company. The Audit Committee receives regular updates on our cybersecurity risk management process from members of management, including our Chief Information Officer ("CIO"). The Audit Committee review our comprehensive cybersecurity framework, including reviewing our cybersecurity reporting protocol that provides for the notification, escalation and communication of significant cybersecurity events to a crisis management team and appropriate levels of management, including our CIO, as well as to the Audit Committee. Management also provides the Audit Committee with a cybersecurity dashboard, which the full Board of Directors can access as well. Additionally, the Audit Committee regularly provides updates to the Board on the status of the Company's cybersecurity risk management process.

The Company's cybersecurity program is overseen by our CIO, who is responsible for global information technology, including cybersecurity. Our Vice President, Global Information Security, is primarily responsible for assessing and managing material risks from cybersecurity threats, including monitoring the measures used for prevention, detection, mitigation and remediation of cybersecurity incidents. The information security organization is comprised of internal Owens Corning employees and external security suppliers who provide security monitoring and response. Our Global Information Services team is regularly engaged in cybersecurity training and awareness and incorporates relevant reviews in technology design and development.

ITEM 1C. CYBERSECURITY (continued)

Our CIO has 19 years of experience in the information technology industry, including engagement with cybersecurity strategy and oversight. Our CIO reports directly to our Chief Executive Officer.

Our Vice President, Global Information Security has 27 years of experience in the cybersecurity industry, including previous experience in the U.S. Air Force, consulting, and 21 years with Owens Corning, and reports directly to our CIO.

ITEM 2. PROPERTIES

Roofing

Our Roofing segment operates out of 33 manufacturing facilities. This number separately counts multiple roofing and asphalt manufacturing facilities that are located at the same site. In connection with our exit of the Protective Packaging business, the Company has ceased operations at the Qingdao, China facility. Principal manufacturing facilities for our Roofing segment, all of which are owned by the Company, include the following:

Brookville, Indiana	Minneapolis, Minnesota
Denver, Colorado	Portland, Oregon
Irving, Texas	Savannah, Georgia
Kearny, New Jersey	Silvassa, India
Medina, Ohio	Summit, Illinois
Memphis, Tennessee	

Insulation

Our Insulation segment operates out of 41 manufacturing facilities. The Company ceased operations at the Wabash, Indiana facility in the fourth quarter of 2023. Principal manufacturing facilities for our Insulation segment, all of which are owned by the Company, include the following:

Delmar, New York	Rockford, Illinois
Edmonton, Alberta, Canada	Sedalia, Missouri
Fairburn, Georgia	Tallmadge, Ohio
Guangzhou, Guandong, China	Tessenderlo, Belgium
Hällekis, Sweden	Toronto, Ontario, Canada
Joplin, Missouri	Trzemeszno, Poland
Kansas City, Kansas	Vilnius, Lithuania
Mexico City, Mexico	Waxahachie, Texas
Newark, Ohio	

Composites

Our Composites segment operates out of 29 manufacturing facilities. Principal manufacturing facilities for our Composites segment, all of which are owned by the Company, include the following:

Aiken, South Carolina	Hangzhou, China
Amarillo, Texas	Jackson, Tennessee
Anderson, South Carolina	Kimchon, Korea
Apeldoorn, The Netherlands	L'Ardoise, France
Danville, Illinois	Rio Claro, Brazil
Fort Smith, Arkansas	Taloja, India
Gastonia, North Carolina	Tlaxcala, Mexico

We believe that these properties are in good condition and well maintained, and are suitable and adequate to carry on our business. The capacity of each plant varies depending upon product mix.

Our principal executive offices are located at the Owens Corning World Headquarters in Toledo, Ohio, an owned facility of approximately 400,000 square feet. Our research and development activities are primarily conducted at our Science and Technology Center, located on approximately 500 acres of land owned by the Company outside of Granville, Ohio. It consists of approximately 20 structures totaling more than 650,000 square feet. In addition, we have application development and other product and market focused research and development centers in various locations.

ITEM 3. LEGAL PROCEEDINGS

Environmental Legal Proceedings

None.

Litigation, Other Regulatory Proceedings and Environmental Matters

Additional information required by this item is incorporated by reference to Note 16, Contingent Liabilities and Other Matters.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

INFORMATION ABOUT OUR EXECUTIVE OFFICERS

The name, age and business experience during the past five years of Owens Corning's executive officers as of January 1, 2024 are set forth below. Each executive officer holds office until his or her successor is elected and qualified or until his or her earlier resignation, retirement or removal. All of the listed executive officers have been employees of Owens Corning during the past five years except as indicated below.

Name and Age	Position*
Gina A. Beredo (49)	Executive Vice President, General Counsel and Corporate Secretary since June 2021; formerly Executive Vice President, General Counsel and Corporate Secretary of Nordson Corporation (a precision technology manufacturing company) (NASDAQ: NDSN) (2018)
Brian D. Chambers (57)	Board Chair, President and Chief Executive Officer since April 2020; formerly President and Chief Executive Officer (2019); formerly President and Chief Operating Officer (2018)
Nicolas Del Monaco (46)	President, Insulation since September 2023; formerly Senior Vice President and Managing Director, Europe (2021); formerly Vice President for Non-Wovens and Glass Reinforcements Europe (2018)
Mari K. Doerfler (41)	Vice President and Controller since April 2023; formerly Assistant Controller (2021); formerly Americas Accounting Director (2019); formerly Global Internal Controls Leader (2016)
Todd W. Fister (49)	Executive Vice President and Chief Financial Officer since September 2023; formerly President, Insulation (2019); formerly Vice President of Global Insulation and Strategy (2019); formerly Vice President and Managing Director for Europe Insulation and Global Foamglas® (2018)
José L. Méndez-Andino (50)	Executive Vice President, Chief Research and Development Officer since April 2021; formerly Vice President of Science and Technology for Insulation and Roofing (2019); formerly Vice President of Science and Technology for Insulation (2015)
Paula J. Russell (46)	Executive Vice President, Chief Human Resources Officer since January 2021; formerly Senior Vice President, Chief Human Resources Officer (December 2019); formerly Vice President, Chief Human Resources Officer (April 2019); formerly Vice President of Total Rewards and Center of Excellence (2018)
Marcio A. Sandri (60)	President, Composites since May 2018
Gunner S. Smith (50)	President, Roofing since August 2018

* Information in parentheses indicates year during the past five years in which service in position began. The last item listed for each individual represents the position held by such individual at the beginning of the five-year period.

Part II

ITEM 5. MARKET FOR OWENS CORNING'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES

Market Information

Owens Corning's common stock trades on the New York Stock Exchange under the symbol "OC."

Holders of Common Stock

The number of stockholders of record of Owens Corning's common stock on February 9, 2024 was 55.

Cash Dividends

The payment of any future cash dividends to our stockholders will depend on decisions that will be made by our Board of Directors and will depend on then existing conditions, including our operating results, financial conditions, contractual restrictions, corporate law restrictions, capital agreements, applicable laws of the State of Delaware and business prospects.

Under the credit agreement applicable to our senior revolving credit facility, the Company may not declare a cash dividend if a default or event of default exists or would come to exist at the time of declaration or if a dividend declaration violates the provisions of our formation documents or other material agreements.

The Company's subsidiaries are subject to certain restrictions on their ability to pay dividends under the agreements governing our senior revolving credit facility and our receivables securitization facility.

Recent Sales of Unregistered Securities; Use of Proceeds from Registered Securities

None.

Issuer Purchases of Equity Securities

The following table provides information about Owens Corning's purchases of its common stock during the three months ended December 31, 2023:

Period	Total Number of Shares (or Units) Purchased*	Average Price Paid per Share (or Unit)	Total Number of Shares (or Units) Purchased as Part of Publicly Announced Plans or Programs**	Maximum Number of Shares (or Units) that May Yet Be Purchased Under the Plans or Programs**
October 1-31, 2023	1,001	$ 132.31	—	10,767,634
November 1-30, 2023	1,492,377	129.00	1,485,065	9,282,569
December 1-31, 2023	344,062	143.08	337,367	8,945,202
Total	1,837,440	$ 131.64	1,822,432	8,945,202

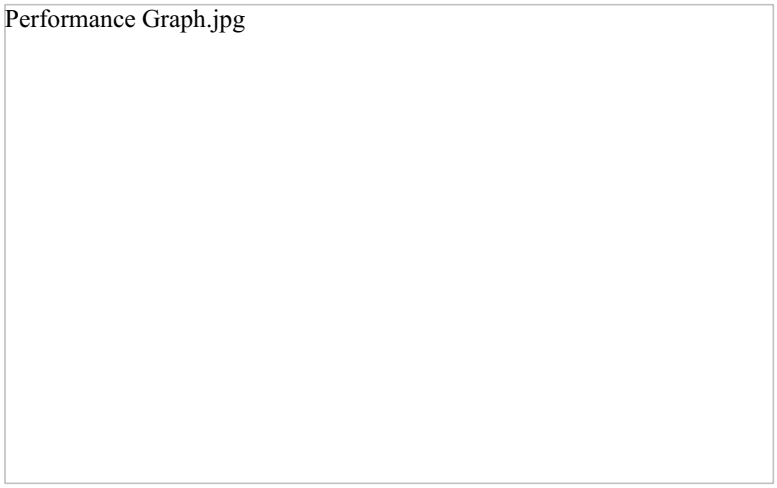
* The Company retained 15,008 shares surrendered to satisfy tax withholding obligations in connection with the vesting of restricted stock units granted to our employees.

** The Board of Directors approved two share repurchase programs in 2022 under which the Company is authorized to repurchase up to an aggregate of 20 million shares of the Company's outstanding common stock (the "Repurchase Authorization"). The Repurchase Authorization enables the Company to repurchase shares through the open market, privately negotiated or other transactions. The actual number of shares repurchased will depend on timing, market conditions and other factors and will be at the Company's discretion. The Company repurchased 1.8 million shares of its common stock for $238 million, inclusive of applicable taxes, during the three months ended December 31, 2023 under the Repurchase Authorization. As of December 31, 2023, 8.9 million shares remain available for repurchase under the Repurchase Authorization.

ITEM 5. MARKET FOR OWENS CORNING'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES (continued)

Performance Graph

The annual changes for the five-year period shown in the graph on this page are based on the assumption that $100 had been invested in Owens Corning (OC) stock, the Standard & Poor's 500 Stock Index ("S&P 500"), and a peer group index on December 31, 2018, and that all quarterly dividends were reinvested. The total cumulative dollar returns shown on the graph represent the value that such investments would have had on December 31, 2023. We chose to use a self-selected peer group consisting of the companies noted below to include in the performance graph as we believe this peer group aligns with our specific industry, markets, and global exposure. The criteria used in determining this peer group included the size of the companies (measured in terms of annual revenue and market capitalization), industries and geographies in which the companies operate, stock price correlation and volatility relative to Owens Corning, and increased representation of comparator companies used by shareholder advisory firms.



Performance Graph

	2018	2019	2020	2021	2022	2023
OC	$ 100	$ 151	$ 178	$ 215	$ 206	$ 365
S&P 500	$ 100	$ 131	$ 156	$ 200	$ 164	$ 207
Peer Group	$ 100	$ 140	$ 171	$ 235	$ 166	$ 222

The peer group index is comprised of the following companies: A.O. Smith Corporation; Advance Drainage Systems, Inc.; Allegion plc; Armstrong World Industries, Inc.; Ball Corporation; Builders FirstSource, Inc.; Carlisle Companies Incorporated; Carrier Global Corporation; Celanese Corporation; Eastman Chemical Company; Fortune Brands Innovations, Inc.; Greif, Inc.; JELD-WEN Holding, Inc.; Johnson Controls International plc; Lennox International Inc.; Louisiana-Pacific Corporation; Masco Corporation; Masonite International Corporation; Mohawk Industries, Inc.; O-I Glass, Inc.; PPG Industries, Inc.; Resideo Technologies, Inc.; RPM International Inc.; Stanley Black & Decker, Inc.; The Sherwin-Williams Company; Trane Technologies; Trex Company, Inc.; and UFP Industries, Inc.

ITEM 6. RESERVED

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

This Management's Discussion and Analysis ("MD&A") is intended to help investors understand Owens Corning, our operations and our present business environment. MD&A is provided as a supplement to, and should be read in conjunction with, our Consolidated Financial Statements and the accompanying Notes thereto contained in this report. Unless the context requires otherwise, the terms "Owens Corning," "Company," "we," "its," and "our" in this Annual Report on Form 10-K refer to Owens Corning and its subsidiaries.

This section of this Annual Report on Form 10-K generally discusses 2023 and 2022 items and year-to-year comparisons between 2023 and 2022. Discussions of 2021 items and year-to-year comparisons between 2022 and 2021 that are not included in this Form 10-K can be found in "Management's Discussion and Analysis of Financial Condition and Results of Operations" in Part II, Item 7 of the Company's Annual Report on Form 10-K for the fiscal year ended December 31, 2022.

GENERAL

Owens Corning is a global building and construction materials leader committed to building a sustainable future through material innovation. The Company has three reporting segments: Roofing, Insulation and Composites. Through these lines of business, the Company manufactures and sells products worldwide. We are a market leader in many of our major product categories.

EXECUTIVE OVERVIEW

Net earnings attributable to Owens Corning were $1,196 million in 2023, compared to $1,241 million in 2022. The Company generated $1,805 million in adjusted earnings before interest and taxes ("Adjusted EBIT") in 2023 compared to $1,762 million in 2022. See the Adjusted Earnings Before Interest and Taxes paragraph of the MD&A for further information regarding Adjusted EBIT, including the reconciliation to net earnings attributable to Owens Corning. Segment earnings before interest and taxes ("EBIT") performance compared to 2022 increased $343 million in our Roofing segment, increased $7 million in our Insulation segment and decreased $256 million in our Composites segment. Within our Corporate, Other and Eliminations category, General corporate expenses and other increased by $51 million.

Cash and cash equivalents were $1.6 billion as of December 31, 2023, compared to $1.1 billion as of December 31, 2022. In 2023, the Company's operating activities provided $1,719 million of cash flow, compared to $1,760 million in 2022.

On February 8, 2024, the Company entered into a definitive agreement to purchase all of the outstanding shares of Masonite. The purchase price for the acquisition of Masonite is approximately $3.9 billion in cash, which we expect to fund with cash on hand and new committed financing. Masonite is a leading global designer, manufacturer, marketer and distributor of interior and exterior doors and door systems for the new construction and repair, renovation and remodeling sectors of the residential and non-residential building construction markets. The transaction was unanimously approved by the board of directors of both companies and is expected to close mid-2024, subject to regulatory and other customary closing conditions, including the approval of Masonite shareholders.

On February 9, 2024, the Company announced the decision to review strategic alternatives for its global glass reinforcements ("GR") business, consistent with our strategy to focus on building and construction materials. The GR business, which operates within our Composites segment, supplies a wide variety of glass fiber products for applications in wind energy, infrastructure, industrial, transportation, and consumer markets. The GR business generates annual revenues of approximately $1.3 billion and has operations in 11 countries, with 18 manufacturing facilities. While a range of options are under consideration, including a potential sale, spin-off or other strategic option, there can be no assurance that the strategic review will result in any transaction or other outcome.

In the fourth quarter of 2023, the Company entered into two agreements to purchase non-participating annuity contracts from insurance companies to transfer $291 million of the Company's outstanding pension projected benefit obligations related to certain U.S. and non-U.S. pension plans. These transactions were funded with pension plan assets of $268 million. As a result of these transactions, the Company recognized a pre-tax settlement charge of $145 million in the fourth quarter of 2023 from the accelerated recognition of a pro rata portion of plan actuarial losses. This charge was recorded in Non-operating expense (income), net on the Consolidated Statements of Earnings. These transactions did not have a material effect on the plans' funded statuses.

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS (continued)

During the second quarter of 2023, the Company's subsidiary, Paroc Group OY ("Paroc"), which the Company acquired in 2018, notified the appropriate European maritime regulatory authorities that specific products in its marine insulation product line may not meet certain fire safety requirements in accordance with their certifications. Paroc voluntarily withdrew these specific products from the market, issued recalls, and suspended distribution and sales of these products. Paroc continues to cooperate with the applicable regulatory and government authorities and work with its customers and end-users to assist with remediation. During 2023, the Company established an estimated liability for expected future costs related to the marine recall on our Consolidated Balance Sheet as of December 31, 2023.

As part of its review of the Paroc insulation product portfolio, the Company discovered potential nonconformances relating to certain ventilation duct insulation products. In January 2024, Paroc suspended sales of the affected insulation products as a precautionary measure while it reviews the potential nonconformances. The Company is continuing its review.

In May 2023, the Company made the decision to exit the Protective Packaging business within the Roofing segment, including the production and sale of wood packaging, metal packaging and custom products. Exiting Protective Packaging will allow the Company to focus resources on the growth of its building materials products, which supports the future growth aspirations of the enterprise. With the exit of the Protective Packaging business, the Company closed its plants in Dorval, Quebec and Mission, British Columbia, Canada. The Company also ceased operations at its Qingdao, China facility. In connection with the exit of the Protective Packaging business, the Company estimates that it will incur cash charges of approximately $15 million, primarily related to severance and other exit costs. Additionally, the Company expects to incur total non-cash charges in the range of $70 to $75 million, primarily related to accelerated depreciation of property, plant and equipment and accelerated amortization of definite-lived intangibles. The Company has exited the majority of the business and expects to generate savings of approximately $7 million annually beginning 2024. During the twelve months ended 2023, the Company recorded $78 million of charges, primarily related to accelerated depreciation, accelerated amortization and severance.

In March 2023, the Company finalized the sale of its Insulation site in Santa Clara, California for total proceeds of $234 million, net of transaction fees. Total proceeds included a non-refundable deposit of $50 million received in the third quarter 2021. As a result, the Company recognized a pre-tax gain of $189 million in the first quarter of 2023, which is recorded in Gain on sale of site on the Consolidated Statements of Earnings.

In 2023, the Company repurchased 5.4 million shares of the Company's common stock for $629 million, inclusive of applicable taxes, under previously announced repurchase authorizations. As of December 31, 2023, 8.9 million shares remained available for repurchase under the repurchase authorizations.

RESULTS OF OPERATIONS

Consolidated Results (in millions)

	Twelve Months Ended December 31,		
	2023	2022	2021
Net sales	$ 9,677	$ 9,761	$ 8,498
Gross margin	$ 2,683	$ 2,616	$ 2,217
% of net sales	*28 %*	*27 %*	*26 %*
Marketing and administrative expenses	$ 831	$ 803	$ 757
Gain on equity method investment	$ —	$ (130)	$ —
Gain on sale of site	$ (189)	$ —	$ —
Other expense (income), net	$ 106	$ 123	$ (69)
Non-operating expense (income), net	$ 145	$ (9)	$ (10)
Earnings before interest and taxes	$ 1,667	$ 1,723	$ 1,448
Interest expense, net	$ 76	$ 109	$ 126
Loss on extinguishment of debt	$ —	$ —	$ 9
Income tax expense	$ 401	$ 373	$ 319
Net earnings attributable to Owens Corning	$ 1,196	$ 1,241	$ 995

The Consolidated Results discussion below provides a summary of our results and the trends affecting our business, and should be read in conjunction with the more detailed Segment Results discussion that follows.

NET SALES

Net sales decreased $84 million in 2023 compared to 2022. The decrease in net sales was driven by lower sales volumes in both Insulation and Composites segments, partially offset by higher selling prices across all three segments. The remaining variance was driven by favorable customer mix, which was partially offset by the unfavorable net impact of acquisitions and divestitures.

GROSS MARGIN

Gross margin increased $67 million in 2023 compared to 2022. The increase in gross margin was driven by higher selling prices across all three segments, which was partially offset by lower sales volumes in both Insulation and Composites segments and higher production downtime. Favorable delivery and favorable customer and product mix more than offset higher input costs and the unfavorable net impact of acquisitions and divestitures.

MARKETING AND ADMINISTRATIVE EXPENSES

Marketing and administrative expenses increased $28 million in 2023 compared to 2022. The increase was driven primarily by ongoing inflationary pressures, as well as higher general corporate expenses.

GAIN ON EQUITY METHOD INVESTMENT

In 2022, the Company recognized a non-cash gain of $130 million from the remeasurement of the previously held equity method investment in Fiberteq, LLC upon the Company's acquisition of the remaining 50% of the joint venture with IKO.

GAIN ON SALE OF SITE

In the first quarter of 2023, the Company finalized the sale of the Company's Insulation site in Santa Clara, California resulting in the recognition of a pre-tax gain of $189 million.

OTHER EXPENSE (INCOME), NET

Other expense (income), net decreased $17 million in 2023 compared to 2022. Higher restructuring costs, lower gains on the sale of precious metals and the establishment of the estimated liability for the Paroc marine recall matter in 2023 were more than offset by the favorable comparison year-over-year to indefinite-lived intangible asset impairment charges of $96 million and the net loss from divestiture related activities.

NON-OPERATING EXPENSE (INCOME), NET

Non-operating expense (income), net increased $154 million in 2023 compared to 2022. The increase was driven by the pension settlement loss in the fourth quarter of 2023.

INTEREST EXPENSE, NET

Interest expense, net decreased $33 million in 2023 compared to 2022. The decrease was driven by higher interest income related to the increase in cash and interest rates, as well as higher capitalized interest resulting from higher construction in progress balances.

INCOME TAX EXPENSE

Income tax expense for 2023 was $401 million compared to $373 million in 2022. The Company's effective tax rate for 2023 was 25% on pre-tax income of $1,591 million. The difference between the 25% effective tax rate and the U.S. federal statutory tax rate of 21% is primarily due to U.S. state and local income tax expense.

The Company's effective tax rate for 2022 was 23% on pre-tax income of $1,614 million. The difference between the 23% effective tax rate and the U.S. federal statutory tax rate of 21% is primarily attributable to U.S. state and local income tax expense, adjustments to R&D tax credits, and other adjustments.

See Note 20 for additional information.

Restructuring, Acquisition and Divestiture-Related Costs

The Company has incurred restructuring, transaction and integration costs related to acquisitions and divestitures, along with restructuring and other exit costs in connection with its global cost reduction, product line and productivity initiatives and growth strategy. These costs are recorded within Corporate, Other and Eliminations. Please refer to Note 12 of the Consolidated Financial Statements for further information on the nature of these costs.

The following table presents the impact and respective location of total restructuring, acquisition and divestiture-related costs on the Consolidated Statements of Earnings (in millions):

		Twelve Months Ended December 31,		
	Location	**2023**	**2022**	**2021**
Restructuring costs	Cost of sales	$ (102)	$ (42)	$ (14)
Restructuring costs	Marketing and administrative expenses	(2)	—	(2)
Severance	Other expense (income), net	(34)	(1)	(11)
Other exit costs	Other expense (income), net	(31)	(5)	(5)
Gain on sale of land in India	Other expense (income), net	—	—	15
Restructuring costs	Non-operating (income) expense	—	—	(2)
Recognition of acquisition inventory fair value step-up	Cost of sales	—	—	(1)
Acquisition and divestiture-related costs	Marketing and administrative expenses	—	(7)	—
Gain on sale of Santa Clara, California site	Gain on sale of site	189	—	—
Gain on sale of Shanghai, China facility	Other expense (income), net	—	27	—
Loss on sale of Chambery, France DUCS business	Other expense (income), net	—	(30)	—
Gain on remeasurement of Fiberteq equity investment	Gain on equity method investment	—	130	—
Loss on sale of Russian operations	Other expense (income), net	—	(33)	—
Total restructuring, acquisition and divestiture-related gains (costs)		$ 20	$ 39	$ (20)

Adjusted Earnings Before Interest and Taxes ("Adjusted EBIT")

Adjusted EBIT is a non-GAAP measure that excludes certain items that management does not allocate to our segment results because it believes they are not representative of the Company's ongoing operations. Adjusted EBIT is used internally by the Company for various purposes, including reporting results of operations to the Board of Directors of the Company, analysis of performance and related employee compensation measures. Although management believes that these adjustments result in a measure that provides a useful representation of our operational performance, the adjusted measure should not be considered in isolation or as a substitute for Net earnings (loss) attributable to Owens Corning as prepared in accordance with accounting principles generally accepted in the United States.

Adjusting (expense) income items to EBIT are shown in the table below (in millions):

	Twelve Months Ended December 31,		
	2023	2022	2021
Restructuring costs	$ (169)	$ (48)	$ (34)
Gain on sale of land in India	—	—	15
Gains on sale of certain precious metals	2	18	53
Intangible assets impairment charge	—	(96)	—
Recognition of acquisition inventory fair value step-up	—	—	(1)
Pension settlement losses	(145)	—	—
Acquisition and divestiture-related costs	—	(7)	—
Gain on sale of Santa Clara, California site	189	—	—
Gain on sale of Shanghai, China facility	—	27	—
Gain on remeasurement of Fiberteq equity investment	—	130	—
Paroc marine recall	(15)	—	—
Loss on sale of Chambery, France DUCS business	—	(30)	—
Loss on sale of Russian operations	—	(33)	—
Total adjusting items	$ (138)	$ (39)	$ 33

The reconciliation from Net earnings (loss) attributable to Owens Corning to EBIT and Adjusted EBIT is shown in the table below (in millions):

	Twelve Months Ended December 31,		
	2023	2022	2021
NET EARNINGS ATTRIBUTABLE TO OWENS CORNING	$ 1,196	$ 1,241	$ 995
Net loss attributable to non-redeemable and redeemable noncontrolling interests	(3)	—	—
NET EARNINGS	1,193	1,241	995
Equity in net earnings of affiliates	3	—	1
Income tax expense	401	373	319
EARNINGS BEFORE TAXES	1,591	1,614	1,313
Interest expense, net	76	109	126
Loss on extinguishment of debt	—	—	9
EARNINGS BEFORE INTEREST AND TAXES	1,667	1,723	1,448
Less: Adjusting items from above	(138)	(39)	33
ADJUSTED EBIT	$ 1,805	$ 1,762	$ 1,415

Segment Results

EBIT by segment consists of net sales less related costs and expenses and is presented on a basis that is used internally for evaluating segment performance. Certain items, such as general corporate expenses or income and certain other expense or income items, are excluded from the internal evaluation of segment performance. Accordingly, these items are not reflected in EBIT for our reportable segments and are included in the Corporate, Other and Eliminations category, which is presented following the discussion of our reportable segments.

Earnings before interest, taxes, depreciation and amortization ("EBITDA") by segment is a non-GAAP measure that consists of EBIT plus depreciation and amortization. Segment EBITDA is used internally by the Company for analysis of our performance.

Roofing

The table below provides a summary of net sales, EBIT, depreciation and amortization expense, and EBITDA for the Roofing segment (in millions):

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
Net sales	$ 4,030	$ 3,658	$ 3,209
% change from prior year	*10 %*	*14 %*	*19 %*
EBIT	$ 1,174	$ 831	$ 753
EBIT as a % of net sales	*29 %*	*23 %*	*23 %*
Depreciation and amortization expense	$ 64	$ 62	$ 59
EBITDA	$ 1,238	$ 893	$ 812
EBITDA as a % of net sales	*31 %*	*24 %*	*25 %*

NET SALES

In our Roofing segment, net sales increased $372 million in 2023 compared to 2022 due to higher sales volumes of approximately 5% and higher selling prices of $166 million. Favorable product and customer mix were partially offset by lower third-party asphalt sales of $44 million.

EBIT

In our Roofing segment, EBIT increased $343 million in 2023 compared to 2022 driven primarily by higher selling prices of $166 million. The remaining improvement was driven by favorable input costs and delivery of $80 million, higher sales volumes, and favorable customer and product mix of $48 million, which were partially offset by higher selling, general and administrative expenses and $8 million of higher production costs.

OUTLOOK

In our Roofing segment, the Company expects North American new residential construction market to temporarily remain soft. Other uncertainties that may impact Roofing demand include demand from storms and other weather-related events, demand from repair and remodeling activity, competitive pricing pressure and the cost and availability of raw materials, particularly asphalt. The Company will continue to focus on managing costs, capital expenditures and working capital.

Insulation

The table below provides a summary of net sales, EBIT, depreciation and amortization expense and EBITDA for the Insulation segment (in millions):

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
Net sales	$ 3,668	$ 3,714	$ 3,184
% change from prior year	*-1 %*	*17 %*	*22 %*
EBIT	$ 619	$ 612	$ 446
EBIT as a % of net sales	*17 %*	*16 %*	*14 %*
Depreciation and amortization expense	$ 210	$ 206	$ 208
EBITDA	$ 829	$ 818	$ 654
EBITDA as a % of net sales	*23 %*	*22 %*	*21 %*

NET SALES

In our Insulation segment, 2023 net sales decreased $46 million compared to 2022. The decrease was driven by lower sales volumes of approximately 10%, which more than offset higher selling prices of $245 million and favorable customer and product mix. The favorable net impact of acquisitions and divestitures and $5 million of favorable impact of translating sales denominated in foreign currencies into United States dollars also contributed to the offset of decreased volumes.

EBIT

In our Insulation segment, EBIT increased $7 million in 2023 compared to 2022. Higher selling prices of $245 million more than offset lower sales volumes and $57 million of input cost inflation. Higher manufacturing costs of $29 million and higher production downtime were partially offset by favorable delivery of $21 million and favorable customer and product mix. The remaining variance was driven by the $7 million negative impact of translating profits denominated in foreign currencies into United States dollars and higher start-up costs.

OUTLOOK

The outlook for Insulation demand is driven by North American new residential construction, remodeling and repair activity, as well as commercial and industrial construction activity in the United States, Canada, Europe, Asia-Pacific and Latin America. Demand in commercial and industrial insulation markets is most closely correlated to industrial production growth and overall economic activity in the global markets we serve. Demand for residential insulation is most closely correlated to U.S. housing starts.

During the fourth quarter of 2023, the average Seasonally Adjusted Annual Rate ("SAAR") of U.S. housing starts was approximately 1.454 million starts, which is up from 1.403 million starts in the fourth quarter of 2022.

The Company expects both the North American new residential construction market and global commercial and industrial construction markets to temporarily remain soft with the weaker macro-economic outlook, higher interest rates and continued input cost inflation. The Company remains focused on managing costs, capital expenditures, and working capital.

Composites

The table below provides a summary of net sales, EBIT, depreciation and amortization expense and EBITDA for the Composites segment (in millions):

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
Net sales	$ 2,286	$ 2,660	$ 2,341
% change from prior year	*-14 %*	*14 %*	*19 %*
EBIT	$ 242	$ 498	$ 376
EBIT as a % of net sales	*11 %*	*19 %*	*16 %*
Depreciation and amortization expense	$ 172	$ 175	$ 162
EBITDA	$ 414	$ 673	$ 538
EBITDA as a % of net sales	*18 %*	*25 %*	*23 %*

NET SALES

Net sales in our Composites segment decreased $374 million in 2023 compared to 2022. The decrease was primarily driven by lower sales volumes of approximately 12% and the net unfavorable impact of divestitures and acquisitions. Unfavorable customer mix of $16 million was partially offset by higher selling prices of $9 million and the favorable impact of translating sales denominated in foreign currencies into United States dollars.

EBIT

EBIT in our Composites segment decreased $256 million in 2023 compared to 2022. The decrease was driven by lower sales volumes, $83 million of higher production downtime and the net unfavorable impact of divestitures and acquisitions of $37 million. Higher input cost inflation of $41 million was offset by favorable delivery and higher selling prices. The remaining variance was driven by unfavorable customer mix, higher rebuild costs and the $5 million negative impact of translating profits denominated in foreign currencies into United States dollars, which was partially offset by favorable manufacturing costs.

OUTLOOK

Global glass reinforcements market demand has several economic indicators, including residential, non-residential construction and manufacturing production indices, as well as global wind installations. The Company anticipates continued impacts of economic uncertainty in a dynamic global environment, as well as competitive pricing pressure. The Company remains focused on managing costs, capital expenditures, and working capital.

Corporate, Other and Eliminations

The table below provides a summary of EBIT and depreciation and amortization expense for the Corporate, Other and Eliminations category (in millions):

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
Restructuring costs	$ (169)	$ (48)	$ (34)
Gain on sale of land in India	—	—	15
Gains on sale of certain precious metals	2	18	53
Intangible assets impairment charge	—	(96)	—
Recognition of acquisition inventory fair value step-up	—	—	(1)
Pension settlement losses	(145)	—	—
Acquisition and divestiture-related costs	—	(7)	—
Gain on sale of Santa Clara, California site	189	—	—
Gain on sale of Shanghai, China facility	—	27	—
Gain on remeasurement of Fiberteq equity investment	—	130	—
Paroc marine recall	(15)	—	—
Loss on sale of Chambery, France DUCS business	—	(30)	—
Loss on sale of Russian operations	—	(33)	—
General corporate expense and other	(230)	(179)	(160)
EBIT	$ (368)	$ (218)	$ (127)
Depreciation and amortization	$ 163	$ 88	$ 73

EBIT

The impact on EBIT from Corporate, Other and Eliminations in 2023 was $150 million higher compared to 2022. The increase was primarily driven by pension settlement losses and higher restructuring costs, partially offset by the gain on sale of the Santa Clara, California site.

General corporate expense and other in 2023 was $51 million higher than in 2022.

OUTLOOK

In 2024, we expect general corporate expenses to range between $240 and $250 million, without considering the effect of the planned acquisition of Masonite.

LIQUIDITY, CAPITAL RESOURCES AND OTHER RELATED MATTERS

Liquidity

The Company's primary sources of liquidity are its balance of Cash and cash equivalents of $1.6 billion as of December 31, 2023, its Senior Revolving Credit Facility and its Receivables Securitization Facility (each as defined below).

The Company has an $800 million senior revolving credit facility (the "Senior Revolving Credit Facility") that has been amended from time to time, which matures in July 2026.

The Company has a $280 million securitization facility (the "Receivables Securitization Facility") that has been amended from time to time, which matures in April 2024.

The following table shows how the Company utilized its primary sources of liquidity (in millions):

	As of December 31, 2023	
	Senior Revolving Credit Facility	Receivables Securitization Facility
Facility size	$ 800	$ 280
Collateral capacity limitation on availability	N/A	—
Outstanding borrowings	—	—
Outstanding letters of credit	4	1
Availability on facility	$ 796	$ 279

The Receivables Securitization Facility and Senior Revolving Credit Facility mature in 2024 and 2026, respectively. The Company's 4.2% senior notes mature in the fourth quarter of 2024. As of December 31, 2023, the Company had $3.0 billion of total debt and cash and cash equivalents of $1.6 billion. The agreements governing our Senior Revolving Credit Facility and Receivables Securitization Facility contain various covenants that we believe are usual and customary. These covenants include a maximum allowed leverage ratio. We were in compliance with these covenants as of December 31, 2023.

On February 8, 2024, the Company entered into a commitment letter with Morgan Stanley Senior Funding, Inc. ("MSSF"), pursuant to which MSSF has committed to provide, subject to the satisfaction of customary closing conditions, a 364-day senior unsecured term loan facility in an aggregate principal amount of up to $3.0 billion for purposes of funding a substantial portion of the Masonite acquisition. We expect to assume up to $875 million of Masonite's outstanding senior unsecured notes. On February 9, 2024, the three major credit rating agencies reaffirmed our investment-grade debt ratings.

Cash and cash equivalents held by foreign subsidiaries may be subject to foreign withholding taxes upon repatriation to the U.S. As of December 31, 2023 and December 31, 2022, the Company had $114 million and $188 million, respectively, in cash and cash equivalents in certain of its foreign subsidiaries. The Company continues to assert indefinite reinvestment in accordance with Accounting Standards Codification ("ASC") 740 based on the laws as of enactment of the tax legislation commonly known as the U.S. Tax Cuts and Jobs Act of 2017.

As a holding company, we have no operations of our own and most of our assets are held by our direct and indirect subsidiaries. Dividends and other payments or distributions from our subsidiaries will be used to meet our debt service and other obligations and to enable us to pay dividends to our stockholders. Please refer to the Risk Factors disclosed in Item 1A of this Annual Report on Form 10-K for details on the factors that could inhibit our subsidiaries' abilities to pay dividends or make other distributions to the parent company.

We have no material off-balance sheet arrangements that have or are reasonably likely to have a material current or future effect on our financial condition, results of operations, liquidity, capital expenditures or other resources.

Material Cash Requirements

Our anticipated uses of cash include capital expenditures, working capital needs, share repurchases, meeting financial obligations, payments of any dividends authorized by our Board of Directors, acquisitions, including the planned acquisition of Masonite, restructuring actions and pension contributions. We expect that our cash on hand, coupled with future cash flows from operations and other available sources of liquidity, including our Senior Revolving Credit Facility and our Receivables Securitization Facility, will provide ample liquidity to enable us to meet our cash requirements for at least the next 12 months and foreseeable future thereafter. We expect to use cash on hand and new committed financing to fund the purchase price of the Masonite acquisition and for any required repurchases of Masonite's outstanding senior unsecured notes.

The following discussion of material cash requirements evaluates known contractual and other obligations, but does not include amounts that are contingent on events or other factors that are uncertain or unknown at this time including legal contingencies, and uncertain tax positions among others. The amounts presented are based on various estimates, including estimates regarding the timing of payments, prevailing interest rates, the occurrence of certain events and other factors. Actual results may vary materially from the amounts discussed below.

Capital Expenditures: Our capital expenditures are primarily related to the maintenance and rebuild of our long-term assets, as well as investing in projects that support growth and innovation to further our enterprise strategy. Our capital expenditures on a

cash basis were $526 million in 2023. Without considering the effect of the planned acquisition of Masonite, we expect to have capital expenditures on a cash basis of approximately $550 million in 2024. The anticipated increase in capital expenditures in 2024 is primarily driven by growth, manufacturing productivity and sustainability projects across all three segments. We expect that capital expenditures will be funded through cash flows from operations. See Note 2 and Note 6 of the Consolidated Financial Statements for additional information on property, plant and equipment.

Long-term debt obligations, including current portion of long-term debt: As of December 31, 2023, total long-term debt of $3.0 billion primarily consists of various outstanding senior notes. The current portion of long-term debt includes $399 million of 4.2% senior notes maturing in the fourth quarter of 2024. Further discussion of the amount and timing of the future scheduled maturities of our senior notes can be found in Note 13 of the Consolidated Financial Statements. There were no borrowings on our Senior Revolving Credit Facility or our Receivables Securitization Facility as of December 31, 2023.

Interest on debt: We are obligated to make periodic interest payments at fixed rates, depending on the terms of the applicable debt agreements. Based on interest rates and scheduled maturities as of December 31, 2023, these interest obligations range from $99 million to $130 million annually over the next five years.

Finance lease obligations: Our finance lease obligations primarily consist of real estate, oxygen plants, computers and software, and fleet vehicles. As of December 31, 2023 we had a total of $196 million of minimum finance lease payments. Further discussion of the future maturities of these lease liabilities can be found in Note 9 of the Consolidated Financial Statements.

Operating lease obligations: Our operating lease obligations primarily consist of real estate and material handling equipment. As of December 31, 2023, we had a total of $248 million of minimum operating lease payments. Further discussion of the future maturities of these lease liabilities can be found in Note 9 of the Consolidated Financial Statements.

Purchase obligations: Purchase obligations are commitments to suppliers to purchase goods or services, and include take-or-pay arrangements, capital expenditures, and contractual commitments to purchase equipment. As of December 31, 2023, the total of these obligations was $328 million, inclusive of $241 million payable in the next 12 months. The Company did not include ordinary course of business purchase orders in this amount as the majority of such purchase orders may be canceled and are reflected in historical operating cash flow trends. The Company does not believe such purchase orders will adversely affect our liquidity position.

Pension Contributions: The Company has several defined benefit pension plans. The Company made cash contributions of $18 million and $8 million to the plans during the twelve months ended December 31, 2023 and 2022, respectively. The Company expects to contribute $20 million in cash to its pension plans during 2024. Actual contributions to the plans may change as a result of several factors, including changes in laws that impact funding requirements. The ultimate cash flow impact to the Company, if any, of the pension plan liability and the timing of any such impact will depend on numerous variables, including future changes in actuarial assumptions, legislative changes to pension funding laws, and market conditions. Further discussion of the Company's defined benefit pension plans can be found in Note 14 of the Consolidated Financial Statements.

Other Strategic Uses of Cash: We have outstanding share repurchase authorizations and will evaluate and consider repurchasing shares of our common stock, as well as payments of any dividends authorized by our Board of Directors, strategic acquisitions, joint ventures, debt repurchases or repayments and other transactions to create stockholder value and enhance financial performance. Such transactions may require cash expenditures beyond current sources of liquidity or generated proceeds.

Supplier Finance Programs

We review supplier terms and conditions on an ongoing basis, and have negotiated payment terms extensions in recent years in connection with our efforts to reduce working capital and improve cash flow. Separate from those terms extension actions, certain of our subsidiaries have entered into paying agency agreements with third-party administrators. These voluntary supply chain finance programs (collectively, the "Programs") generally give participating suppliers the ability to sell, or otherwise pledge as collateral, their receivables from the Company to the participating financial institutions, at the sole discretion of both the suppliers and financial institutions. The Company is not a party to the arrangements between the suppliers and the financial institutions. The Company's obligations to its suppliers, including amounts due and scheduled payment dates, are not impacted by the suppliers' decisions to sell, or otherwise pledge as collateral, amounts under these arrangements. The Company's payment terms to the financial institutions, including the timing and amount of payments, are based on the original supplier invoices. One of our Programs includes a parent guarantee to the participating financial institution for a certain U.S. subsidiary that, at the time of the respective Program's inception in 2015, was a guarantor subsidiary of the Company's Credit Agreement. The obligations are presented as Accounts payable within Total current liabilities on the Consolidated Balance Sheets and all activity related to the obligations is presented within operating activities on the Consolidated Statements of Cash Flow.

The desire of suppliers and financial institutions to participate in the Programs could be negatively impacted by, among other factors, the availability of capital committed by the participating financial institutions, the cost and availability of our suppliers' capital, a credit rating downgrade or deteriorating financial performance of the Company or its participating subsidiaries, or other changes in financial markets beyond our control. We do not expect these risks, or potential long-term growth of our Programs, to materially affect our overall financial condition, as we expect a significant portion of our payments to continue to be made outside of the Programs. Accordingly, we do not believe the Programs have materially impacted our current period liquidity, and do not believe that the Programs are reasonably likely to materially affect liquidity in the future.

Please refer to the *Supplier Finance Programs* section in Note 1 of the Consolidated Financial Statements for a rollforward of outstanding obligations under the supplier finance programs.

Cash Flows

The following table presents a summary of our cash balance, cash flows, and availability on credit facilities (in millions):

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
Cash and cash equivalents	$ 1,615	$ 1,099	$ 959
Net cash flow provided by operating activities	$ 1,719	$ 1,760	$ 1,503
Net cash flow used for investing activities	$ (356)	$ (623)	$ (377)
Net cash flow used for financing activities	$ (877)	$ (974)	$ (881)
Availability on the Senior Revolving Credit Facility	$ 796	$ 796	$ 796
Availability on the Receivables Securitization Facility	$ 279	$ 279	$ 279

Operating activities: In 2023, the Company generated $1,719 million of cash from operating activities compared to $1,760 million in 2022. The decrease in cash provided by operating activities was primarily due to reductions in payables and lower earnings in 2023, which were partially offset by inventory reductions.

Investing activities: The cash used for investing activities in 2023 was $356 million compared to $623 million in 2022. This decrease was due to lower spending on acquisitions in 2023 compared to 2022 (see Note 7 for additional information). This was partially offset by higher capital spending and lower cash from derivative settlements compared to the prior year.

Financing activities: Net cash used for financing activities in 2023 was $877 million compared to $974 million in 2022. The year-over-year decrease was primarily due to lower purchases of treasury stock which were partially offset by higher 2023 dividend payments.

Derivatives

Please refer to Note 4 of the Consolidated Financial Statements.

Fair Value Measurement

Please refer to Notes 1, 4, 13, 14 and 15 of the Consolidated Financial Statements.

CRITICAL ACCOUNTING ESTIMATES

Our discussion and analysis of our financial condition and results of operations is based upon our Consolidated Financial Statements, which have been prepared in accordance with accounting principles generally accepted in the United States. The preparation of these financial statements requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and the disclosure of contingent assets and liabilities at the date of the financial statements and the reported amounts of revenues and expenses during the reporting period. On an ongoing basis, management evaluates its estimates and judgments related to these assets, liabilities, revenues and expenses. We believe these estimates to be reasonable under the circumstances. Management bases its estimates and judgments on historical experience, expected future outcomes, and on various other factors that are believed to be reasonable under the circumstances, the results of which form the basis for making judgments about the carrying values of assets and liabilities that are not readily apparent from other sources. Actual results may differ from these estimates.

The Company believes that the following accounting estimates are critical to our financial results:

Tax Estimates. The determination of our tax provision is complex due to operations in several tax jurisdictions outside the United States. We apply a more-likely-than-not recognition threshold for all tax uncertainties. Such uncertainties include any claims by the Internal Revenue Service for income taxes, interest, and penalties attributable to audits of open tax years.

In addition, we record a valuation allowance to reduce our deferred tax assets to the amount that we believe is more likely than not to be realized. We estimate future taxable income and the effect of tax planning strategies in our consideration of whether deferred tax assets will more likely than not be realized. In the event we were to determine that we would not be able to realize all or part of our net deferred tax assets in the future, an adjustment to reduce the net deferred tax assets would be charged to earnings in the period such determination was made. Conversely, if we were to determine that we would be able to realize our net deferred tax assets in the future in excess of their currently recorded amount, an adjustment to increase the net deferred tax assets would be credited to earnings in the period such determination was made.

Impairment of Assets. The Company exercises judgment in evaluating assets for impairment. Goodwill and other indefinite-lived intangible assets are tested for impairment annually, or when circumstances arise which indicate there may be an impairment. Long-lived assets are tested for impairment when economic conditions or management decisions indicate an impairment may exist. These tests require comparing recorded values to estimated fair values for the assets under review.

The Company has recorded its goodwill and conducted testing for potential goodwill impairment at a reporting unit level. Our reporting units represent a business for which discrete financial information is available and segment management regularly reviews the operating results. The Company has three reporting units: Roofing, Insulation and Composites.

2023 Annual Goodwill Impairment Assessment

Goodwill is an intangible asset that is not subject to amortization; however, annual tests are required to be performed to determine whether impairment exists. Prior to performing the impairment testing process described in ASC 350-20, the guidance permits companies to assess qualitative factors to determine if it is more likely than not that a reporting unit's fair value is less than its carrying value. If, based on the review of the qualitative factors, we determine it is not more likely than not that the fair value of a reporting unit is less than its carrying value, we would bypass the quantitative impairment test. Events and circumstances we consider in performing the qualitative assessment include macro-economic conditions, market and industry conditions, internal cost factors, and the operational stability and the overall financial performance of the reporting units. If it is more likely than not that a reporting unit's fair value is less than or close to its carrying value, then the quantitative impairment test must be performed to determine if impairment is required.

When it is determined necessary for the Company to perform the quantitative impairment process for goodwill, we estimate fair values using a discounted cash flow approach from the perspective of a market participant. Significant assumptions used in the discounted cash flow approach are the revenue growth rates and EBIT margins used in estimating discrete period cash flow forecasts of the reporting unit, the discount rate, the reporting unit tax rate and the long-term revenue growth rate and EBIT margin used in estimating the terminal business value. The cash flow forecasts of the reporting unit are based upon management's long-term view of our markets and are the forecasts that are used by senior management and the Board of Directors to evaluate operating performance. The discount rate utilized is management's estimate of what the market's weighted average cost of capital is for a company with a similar debt rating and stock volatility, as measured by beta. The reporting unit specific tax rate is based on blended global historical rates. The terminal business value is determined by applying the long-term growth rate to the latest year for which a forecast exists. As part of our goodwill quantitative testing process, the Company evaluates whether there are reasonably likely changes to management's estimates that would have a material impact on the results of the goodwill impairment testing.

Our annual test of goodwill for impairment was conducted as of October 1, 2023. The Company elected to perform the qualitative approach on all of its reporting units: Roofing, Insulation and Composites. After evaluating and weighing all relevant events and circumstances, we concluded it is more likely than not that the fair value of the Roofing and Insulation reporting units exceeds their respective carrying value amounts. Consequently, we did not perform a quantitative analysis for the Roofing and Insulation reporting units and determined that their goodwill was not impaired for 2023.
For the Composites reporting unit, based on the qualitative assessment we concluded that it is more likely than not that the fair value of the reporting unit was less than its carrying amount. Therefore, we performed a quantitative analysis as described above. As a result of this test, we determined that no impairment existed for the reporting unit. Testing indicated that the business enterprise value for the Composites reporting unit exceeded its carrying value by approximately 5%. There is uncertainty surrounding the macroeconomic factors that impact this reporting unit and a sustained downturn in these factors or a change in the long-term revenue growth or profitability for this reporting unit could increase the likelihood of a future impairment. The most significant assumptions used in our analysis to determine the fair value of the Composites reporting unit are the discount rate and long-term growth rate. If all other assumptions remain constant, a 50 basis point increase in the selected discount rate of 11% would decrease the fair value of the Composites reporting unit by approximately 5%, and a 50 basis point decrease in the selected long-term growth rate of 2.5% would decrease the fair value of the Composites reporting unit by approximately 4%.

The following table summarizes the segment allocation of recorded goodwill on our Consolidated Balance Sheet as of December 31, 2023 (in millions):

Segment		December 31, 2023	Percent of Total
Roofing	$	395	28 %
Insulation		572	41 %
Composites		425	31 %
Total goodwill	$	1,392	100 %

Annual 2023 Indefinite-lived Intangible Asset Impairment Assessment

Fair values used in testing for potential impairment of our trademarks and trade names are calculated by applying an estimated market value royalty rate to the forecasted revenues of the businesses that utilize those assets. The assumed cash flows from this calculation are discounted at a rate based on a market-participant discount rate. Our annual test of indefinite-lived intangibles was conducted as of October 1, 2023. The fair value of each of our indefinite-lived intangible assets exceeded the carrying value as of the date of our assessment.

Testing indicated that the fair values of a trade name used by our European building and technical insulation business and a trademark used on global cellular glass insulation products exceeded their carrying values by 2% and 1%, respectively. A change in the estimated long-term revenue growth rate or increase in the discount rate assumption could increase the likelihood of a future impairment for these assets. For the trade name used by our European building and technical insulation business, if all other assumptions remain constant, a 50 basis point increase in the selected discount rate of 12.5% would decrease the fair value by approximately 5%, and a 50 basis point decrease in the selected long-term growth rate of 2.0% would decrease the fair value by approximately 4%. For the trademark used on global cellular glass insulation products, if all other assumptions remain constant, a 50 basis point increase in the selected discount rate of 12.0% would decrease the fair value by approximately 5%, and a 50 basis point decrease in the selected long-term growth rate of 2.0% would decrease the fair value by approximately 4%.

The carrying values of the European building and technical insulation trade name and the global cellular glass insulation trademark are $90 million and $80 million, respectively. Both of these assets are included within the Insulation segment.

The fair value of the remaining assets substantially exceeded their carrying value as of the date of our assessment.

Long-lived Asset Recoverability Assessment

Fair values for long-lived asset testing are calculated by estimating the undiscounted cash flows from the use and ultimate disposition of the asset or by estimating the amount that a willing third party would pay. For impairment testing, long-lived assets are grouped at the lowest level for which identifiable cash flows are largely independent of the cash flows of other groups of assets and liabilities. The Company groups long-lived assets based on manufacturing facilities that produce similar products either globally or within a geographic region. Management tests asset groups for potential impairment whenever events or changes in circumstances indicate that the carrying value may not be recoverable. We evaluated and concluded that there are not any reasonably likely changes to management's estimates that would indicate that the carrying value of our long-lived assets is unrecoverable.

However, changes in management intentions, market conditions, operating performance and other similar circumstances could affect the assumptions used in these impairment tests. Changes in the assumptions could result in impairment charges that could be material to our Consolidated Financial Statements in any given period.

Product Warranty: The Company records a liability for warranty obligations at the date the related products are sold. Most significant are the standard warranties on our roofing products. The standard warranties generally provide full coverage of labor and materials for a period of 5-10 years from the original installation date and prorated materials for the remaining life of the roof.

Our estimated cost of our standard warranty obligations is calculated using a 5-year historical average of claims paid for each major product category, the estimated future cost to manufacture the replacement shingles, and the estimated future cost for contractor labor, subject to the applicable warranty coverage, for a 20-year period from the date of installation.

Additionally, the Company sells contractors extended warranties that extend coverage beyond our standard product warranty. The extended warranties revenue is deferred and recognized over the related coverage period, ranging from 16 to 20 years.

Pensions and Other Postretirement Benefits. Accounting for pensions and other postretirement benefits involves estimating the cost of benefits to be provided well into the future and attributing that cost over the time period each employee works. To accomplish this, extensive use is made of assumptions about investment returns, discount rates, inflation, mortality, turnover, and medical costs. Changes in assumptions used could result in a material impact to our Consolidated Financial Statements in any given period.

Two key assumptions that could have a significant impact on the measurement of pension liabilities and pension expense are the discount rate and the expected return on plan assets. For our largest plan, the United States plan, the discount rate used for the December 31, 2023 measurement date is based on a yield curve approach where the expected future benefit payments are matched with a yield curve derived from certain AA-rated corporate bonds.

The result supported a discount rate of 5.00% at December 31, 2023 compared to 5.15% at December 31, 2022. A 25 basis point increase (decrease) in the discount rate would (decrease) increase the December 31, 2023 projected benefit obligation for the United States pension plan by approximately $9 million. A 25 basis point increase (decrease) in the discount rate would (decrease) increase 2024 net periodic pension cost by less than $1 million.

The expected return on plan assets in the United States was derived by taking into consideration the target plan asset allocation, historical rates of return on those assets, projected future asset class returns and net outperformance of the market by active investment managers and plan related and investment related expenses paid from the plan trust. The Company uses the target plan asset allocation because we rebalance our portfolio to target on at least a quarterly basis. An asset return model was used to develop an expected range of returns on plan investments over a 20-year period, with the expected rate of return selected from a best estimate range within the total range of projected results. This process resulted in the selection of an expected return of 5.75% at the December 31, 2023 measurement date, which is used to determine net periodic pension cost for the year 2024. The expected return selected at the December 31, 2022 measurement date was 5.75%, which was used to determine the net periodic pension cost for the year 2023. A 25 basis point increase (decrease) in return on plan assets assumption would result in a respective decrease (increase) of 2024 net periodic pension cost by approximately $1 million.

The discount rate for our United States postretirement plan was selected using the same method as described for the pension plan. The result supported a discount rate of 4.90% at December 31, 2023 compared to 5.10% at December 31, 2022. A 25 basis point increase (decrease) in the discount rate would (decrease) increase the United States postretirement benefit obligation by approximately $2 million and (decrease) increase 2024 net periodic postretirement benefit cost by less than $1 million.

The methods corresponding to those described above are used to determine the discount rate and expected return on assets for non-U.S. pension and postretirement plans, to the extent applicable.

RECENT ACCOUNTING PRONOUNCEMENTS

Please refer to Note 1 of the Consolidated Financial Statements.

ENVIRONMENTAL MATTERS

Please refer to Note 16 of the Consolidated Financial Statements.

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS (continued)

CAUTIONARY STATEMENT CONCERNING FORWARD-LOOKING STATEMENTS

Our disclosures and analysis in this report, including Management's Discussion and Analysis of Financial Condition and Results of Operations, contain forward-looking statements within the meaning of Section 27A of the Securities Act of 1933 and Section 21E of the Securities Exchange Act of 1934 (the "Exchange Act"). Forward-looking statements present our current forecasts and estimates of future events. These statements do not strictly relate to historical or current results and can be identified by words such as "anticipate," "appear," "assume," "believe," "estimate," "expect," "forecast," "intend," "likely," "may," "plan," "project," "seek," "should," "strategy," "will" and other terms of similar meaning or import in connection with any discussion of future operating, financial or other performance. These forward-looking statements are subject to risks, uncertainties and other factors and actual results may differ materially from those results projected in the statements. These risks, uncertainties and other factors include, without limitation:

- levels of residential and commercial or industrial construction activity;
- demand for our products;
- industry and economic conditions including, but not limited to, supply chain disruptions, recessionary conditions, inflationary pressures, interest rate and financial markets volatility, and the viability of banks and other financial institutions;
- availability and cost of energy and raw materials;
- levels of global industrial production;
- competitive and pricing factors;
- relationships with key customers and customer concentration in certain areas;
- issues related to acquisitions, divestitures and joint ventures or expansions, including the planned acquisition of Masonite;
- climate change, weather conditions and storm activity;
- legislation and related regulations or interpretations, in the United States or elsewhere;
- domestic and international economic and political conditions, policies or other governmental actions, as well as war and civil disturbance;
- changes to tariff, trade or investment policies or laws;
- uninsured losses, including those from natural disasters, catastrophes, pandemics, theft or sabotage;
- environmental, product-related or other legal and regulatory liabilities, proceedings or actions;
- research and development activities and intellectual property protection;
- issues involving implementation and protection of information technology systems;
- foreign exchange and commodity price fluctuations;
- our level of indebtedness; including the planned acquisition of Masonite;
- our liquidity and the availability and cost of credit;
- our ability to achieve expected synergies, cost reductions and/or productivity improvements;
- the level of fixed costs required to run our business;
- levels of goodwill or other indefinite-lived intangible assets;
- price volatility in certain wind energy markets in the U.S.;
- loss of key employees and labor disputes or shortages;
- our ability to complete and successfully integrate the Masonite acquisition;
- any material adverse changes in the business of Masonite
- the ability to obtain required regulatory, shareholder or other third-party approvals and consents and otherwise complete the Masonite acquisition;
- our ability to achieve the strategic and other objectives relating to the Masonite acquisition, including any expected synergies, and the strategic review of our GR business; and
- defined benefit plan funding obligations.

All forward-looking statements in this Annual Report on Form 10-K should be considered in the context of the risks and other factors described herein, and in Item 1A above, and as detailed from time to time in the Company's filings with the U.S. Securities and Exchange Commission. Users of this Annual Report on Form 10-K should not interpret the disclosure of any risk factor to imply that the risk has not already materialized. Any forward-looking statements speak only as of the date the statement is made and we undertake no obligation to update or revise any forward-looking statements, whether as a result of new information, future events or otherwise, except as required by federal securities laws. It is not possible to identify all of the risks, uncertainties and other factors that may affect future results. In light of these risks and uncertainties, the forward-looking events and circumstances discussed in this Annual Report on Form 10-K may not occur and actual results may differ materially from those anticipated or implied in the forward-looking statements. Accordingly, users of this Annual Report on Form 10-K are cautioned not to place undue reliance on the forward-looking statements.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

The Company is exposed to the impact of changes in foreign currency exchange rates, interest rates and the prices of various commodities used in the normal course of business. To mitigate some of the near-term volatility in our earnings and cash flows, the Company manages certain of our exposures through the use of financial contracts, contracts for physical delivery of a particular commodity, and derivative financial instruments. The Company's objective with these instruments is to reduce exposure to near-term fluctuations in earnings and cash flows. The Company's policy enables the use of foreign currency, interest rate and commodity derivative financial instruments only to the extent necessary to manage exposures as described above. The Company does not enter into such transactions for trading purposes.

A discussion of the Company's accounting policies for derivative financial instruments, as well as the Company's exposure to market risk, is included in Notes 1 and 4 to the Consolidated Financial Statements. Please refer to Note 4 for details of the fair values of derivative financial instruments and their classification on the Consolidated Balance Sheets.

For purposes of disclosing the market risk inherent in its derivative financial instruments the Company uses sensitivity analysis disclosures that express the potential loss in fair values of market rate sensitive instruments resulting from changes in interest rates, foreign currency exchange rates, and commodity prices that assume instantaneous, parallel shifts in exchange rates, interest rate yield curves, and commodity prices. The following analysis provides such quantitative information regarding market risk. There are certain shortcomings inherent in the sensitivity analysis presented, primarily due to the assumption that exchange rates change instantaneously and that interest rates change in a parallel fashion. In addition, the analyses are unable to reflect the complex market reactions that normally would arise from the market shifts modeled.

Foreign Exchange Rate Risk

The Company has transactional foreign currency exposures related to buying, selling, and financing in currencies other than the local currencies in which it operates. The Company enters into various forward contracts, which change in value as foreign currency exchange rates change, to preserve the carrying amount of foreign currency-denominated assets, liabilities, commitments, and certain anticipated foreign currency transactions. Exposures are related to the United States Dollar primarily relative to the Brazilian Real, Indian Rupee, Chinese Yuan, Hong Kong Dollar, South Korean Won, and the European Euro exchange rates. Also, there are additional exposures related to the European Euro primarily versus the Polish Złoty, British Pound Sterling, and the U.S. Dollar. These transactional risks are mitigated through the use of derivative financial instruments and balancing of cash deposits and loans. The net fair value of derivative financial instruments used to limit exposure to foreign currency risk was a liability of less than $1 million and a liability of $1 million as of December 31, 2023 and 2022, respectively. As of December 31, 2023, the potential change in fair value for such financial instruments from an increase (decrease) of 10% in the quoted foreign currency exchange rates would be a (decrease) increase of approximately $4 million and $3 million, respectively. As of December 31, 2022, the potential change in fair value for such financial instruments from an increase (decrease) of 10% in the quoted foreign currency exchange rates would be a (decrease) increase of approximately $7 million and $5 million, respectively.

We have translation exposure resulting from translating the financial statements of foreign subsidiaries into United States Dollars. Our most significant translation exposures are the Canadian Dollar, Chinese Yuan, European Euro, Indian Rupee, and Polish Złoty in relation to the United States Dollar.

Interest Rate Risk

The Company is subject to market risk from exposure to changes in interest rates due to its financing, investing, and cash management activities. The Company has a Senior Revolving Credit Facility, Receivables Securitization Facility, other floating rate debt and cash and cash equivalents which are exposed to floating interest rates and may impact cash flow. As of December 31, 2023, the Company had no borrowings on its Senior Revolving Credit Facility or Receivables Securitization Facility, with the balance of other floating-rate debt of $1 million. As of December 31, 2022, the Company had no borrowings on its Senior Revolving Credit Facility or Receivables Securitization Facility, with the balance of other floating rate debt of $1 million. Cash and cash equivalents were $1.6 billion and $1.1 billion at December 31, 2023 and 2022, respectively. Based on the year-end outstanding balances on floating rate debt, a one percentage point increase (decrease) in interest rates at December 31, 2023 and 2022 would increase (decrease) our annual net interest expense by less than $1 million for each year.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK (continued)

The fair market value of the Company's senior notes are subject to interest rate risk. The following table shows how a one percentage point increase / decrease in interest rates would impact the fair market value of the senior notes:

	Senior Notes Maturity Year						
As of December 31, 2023:	**2024**	**2026**	**2029**	**2030**	**2036**	**2047**	**2048**
Increase in interest rates							
Decrease in fair value	1%	2%	5%	5%	8%	13%	13%
Decrease in interest rates							
Increase in fair value	1%	3%	5%	6%	9%	16%	16%

	Senior Notes Maturity Year						
As of December 31, 2022:	**2024**	**2026**	**2029**	**2030**	**2036**	**2047**	**2048**
Increase in interest rates							
Decrease in fair value	2%	3%	6%	6%	8%	12%	12%
Decrease in interest rates							
Increase in fair value	2%	3%	6%	7%	10%	15%	15%

Commodity Price Risk

The Company is exposed to changes in prices of commodities used in its operations, primarily associated with energy, such as natural gas, and raw materials, such as asphalt and polystyrene. The Company enters into cash-settled natural gas swap contracts in certain markets to protect against changes in natural gas prices that mature within 15 months; however, no financial instruments are currently used to protect against changes in raw material costs. At December 31, 2023 and 2022, the net fair value of such swap contracts was a liability of $15 million and a liability of $30 million, respectively. The potential change in fair value at December 31, 2023 and 2022 resulting from an increase (decrease) of 10% in the underlying commodity prices would be an increase (decrease) of $4 million for 2023 and an increase (decrease) of $8 million for 2022. This amount excludes the offsetting impact of the price risk inherent in the physical purchase of the underlying commodities.

ITEM 8. FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

Pages 61 through 115 of this filing are incorporated herein by reference.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A. CONTROLS AND PROCEDURES

The Company maintains (a) disclosure controls and procedures (as such term is defined in Rules 13a-15(e) and 15d-15(e) under the Exchange Act), and (b) internal control over financial reporting (as such term is defined in Rules 13a-15(f) and 15d-15(f) under the Exchange Act).

The Company's management, with the participation of the Company's Chief Executive Officer and Chief Financial Officer, has evaluated the effectiveness of the Company's disclosure controls and procedures as of the end of the period covered by this Annual Report on Form 10-K. Based on such evaluation, the Company's Chief Executive Officer and Chief Financial Officer have concluded that, as of the end of such period, the Company's disclosure controls and procedures are effective.

There has been no change in the Company's internal control over financial reporting during the quarter ended December 31, 2023 that materially affected, or is reasonably likely to materially affect, the Company's internal control over financial reporting.

A report of the Company's management on the Company's internal control over financial reporting is contained on page 58 hereof and is incorporated here by reference. PricewaterhouseCoopers LLP's report on the effectiveness of internal control over financial reporting is included in the Report of Independent Registered Public Accounting Firm beginning on page 59 hereof.

ITEM 9B. OTHER INFORMATION

10b5-1 Plans

On October 31, 2023, Gunner Smith, the Company's President, Roofing, entered into a written plan for the sale of shares of Company common stock, intended to satisfy the affirmative defense conditions of Rule 10b5-1(c) under the Securities Exchange Act of 1934 (the "Exchange Act"). Mr. Smith's plan provides for the sale of 12,515 shares of Company common stock in the aggregate underlying future vesting restricted stock units and performance stock units ("PSUs") that are expected to vest during the term of the plan. The amount of shares disclosed above has not been reduced by the number of shares that may be withheld for income taxes, and assumes that the PSUs will vest at 100% attainment. The actual number of PSUs that may vest can vary between 0% - 200% of the target award amount, subject to the achievement of certain performance conditions as set forth in the PSU award agreement, and the number of shares sold pursuant to Mr. Smith's plan may increase or decrease accordingly. This plan is scheduled to terminate no later than October 25, 2024.

On December 13, 2023, José Méndez-Andino, the Company's Executive Vice President, Chief Research and Development Officer, entered into a written plan for the sale of up to 2,833 shares of Company common stock, intended to satisfy the affirmative defense conditions of Rule 10b5-1(c) under the Exchange Act. This plan is scheduled to terminate no later than December 13, 2024.

On December 14, 2023, Paula Russell, the Company's Executive Vice President, Chief Human Resources Officer, entered into a written plan for the sale of up to3,765 shares of Company common stock, intended to satisfy the affirmative defense conditions of Rule 10b5-1(c) under the Exchange Act. This plan is scheduled to terminate no later than November 29, 2024.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not applicable.

Part III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE

Information with respect to directors, corporate governance, and compliance with Section 16(a) of the Exchange Act will be presented in the 2024 Proxy Statement in the sections titled “Information Concerning Directors,” “Governance Information,” and “Delinquent Section 16(a) Reports,” and such information is incorporated herein by reference.

Information with respect to our executive officers is included herein under Part I, “Information about our Executive Officers”.

Code of Ethics

Owens Corning has adopted an Ethics Policy for Chief Executive and Senior Financial Officers (“Ethics Policy”) that applies to our Chief Executive Officer, Chief Financial Officer and Controller. This Ethics Policy is available on our website (www.owenscorning.com) under the “Corporate Governance” tab located in the “Investing in Owens Corning” section and print copies will be made available free of charge upon request to the Corporate Secretary of the Company. To the extent required by applicable SEC rules or New York Stock Exchange listing standards, the Company intends to post any amendments or waivers to the above referenced codes of ethics to our website, under the tab entitled “Corporate Governance.”

ITEM 11. EXECUTIVE COMPENSATION

Information regarding executive officer and director compensation will be presented in the 2024 Proxy Statement under the section titled “Executive Compensation,” exclusive of the subsection titled “Compensation Committee Report,” and the section titled “2023 Non-Management Director Compensation,” and such information is incorporated herein by reference.

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

Information regarding security ownership of certain beneficial owners and management and related stockholder matters, as well as equity compensation plan information, will be presented in the 2024 Proxy Statement under the sections titled “Beneficial Ownership of Shares,” “Security Ownership of Executive Officers and Directors” and “Equity Compensation Plan Information,” and such information is incorporated herein by reference.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE

Information regarding certain relationships and related transactions and director independence will be presented in the 2024 Proxy Statement under the sections titled “Review of Transactions with Related Persons,” “Director Qualifications Standards” and “Director Independence,” and such information is incorporated herein by reference.

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES

Information regarding principal accounting fees and services will be presented in the 2024 Proxy Statement under the sections titled “Principal Accountant Fees and Services,” and such information is incorporated herein by reference.

Part IV

ITEM 15. EXHIBITS AND FINANCIAL STATEMENT SCHEDULES

(a) DOCUMENTS FILED AS PART OF THIS REPORT

1. See Index to Consolidated Financial Statements on page 57 hereof.
2. See Index to Financial Statement Schedules on page 116 hereof.

EXHIBIT INDEX

Pursuant to the rules and regulations of the SEC, the Company has filed or incorporated by reference certain agreements as exhibits to this Annual Report on Form 10-K. These agreements may contain representations and warranties by the parties. These representations and warranties have been made solely for the benefit of the other party or parties to such agreements and (i) may have been qualified by disclosures made to such other party or parties, (ii) were made only as of the date of such agreements or such other date(s) as may be specified in such agreements and are subject to more recent developments, which may not be fully reflected in the Company's public disclosure, (iii) may reflect the allocation of risk among the parties to such agreements and (iv) may apply materiality standards different from what may be viewed as material to investors. Accordingly, these representations and warranties may not describe the Company's actual state of affairs at the date hereof and should not be relied upon.

Exhibit Number	Description
2.1	Arrangement Agreement, dated as of February 8, 2024, among OwensCorning, Masonite International Corporation and MT Acquisition Co ULC (incorporated by reference to Exhibit 2.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed February 9, 2024).
3.1	Amended and Restated Certificate of Incorporation (incorporated by reference to Exhibit 3.1 of Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended March 31, 2016).
3.2	Fourth Amended and Restated Bylaws of Owens Corning (as adopted on June 15, 2023) (incorporated by reference to Exhibit 3.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed June 22, 2023).
4.1	Indenture, dated as of October 31, 2006, by and among Owens Corning, each of the guarantors named therein and LaSalle Bank, National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed November 2, 2006).
4.2	Form of 7.000% Senior Notes due 2036 (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed November 2, 2006).
4.3	First Supplemental Indenture, dated as of April 13, 2007, by and among Owens Corning, each of the guarantors named therein and LaSalle Bank National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed April 13, 2007).
4.4	Second Supplemental Indenture, dated as of December 12, 2007, by and among Owens Corning, each of the guarantors named therein and LaSalle Bank National Association, as trustee (incorporated by reference to Exhibit 4.3 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100), for the year ended December 31, 2007).
4.5	Third Supplemental Indenture, dated as of April 24, 2008, by and among Owens Corning, each of the guarantors named therein and LaSalle Bank National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended June 30, 2008).
4.6	Fourth Supplemental Indenture, dated as of May 26, 2010, by and among Owens Corning, each of the guarantors named therein and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed May 28, 2010).

4.7	Fifth Supplemental Indenture, dated as of October 3, 2016, by and among Owens Corning, each of the guarantors named therein and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.7 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100) for the year ended December 31, 2017).
4.8	Sixth Supplemental Indenture, dated as of February 27, 2017, by and among Owens Corning, each of the guarantors named therein and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.8 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100), for the year ended December 31, 2017).
4.9	Seventh Supplemental Indenture, dated as of August 23, 2017, by and among Owens Corning, the guarantor named therein and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.5 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended September 30, 2017).
4.10	Indenture, dated as of June 2, 2009, by and among Owens Corning, certain of Owens Corning's subsidiaries and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Registration Statement on Form S-3 (File No. 333-159689), filed June 3, 2009).
4.11	Third Supplemental Indenture, dated as of October 22, 2012, by and among Owens Corning, certain subsidiaries, and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Form 8-K (File No. 1-33100), filed October 22, 2012).
4.12	Fourth Supplemental Indenture, dated as of November 12, 2014, by and among Owens Corning, the guarantors named therein and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed November 12, 2014).
4.13	Form of 4.200% Senior Notes due 2024 (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed November 12, 2014).
4.14	Fifth Supplemental Indenture, dated as of August 8, 2016, by and among the Owens Corning, the guarantors party thereto and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed August 8, 2016).
4.15	Form of 3.400% Senior Notes due 2026 (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed August 8, 2016).
4.16	Sixth Supplemental Indenture, dated as of October 3, 2016, by and among Owens Corning, the guarantors party thereto and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.9 to Post-Effective Amendment No. 1 to Owens Corning's Registration Statement on Form S-3 (Registration No. 333-202011), filed June 21, 2017).
4.17	Seventh Supplemental Indenture, dated as of February 27, 2017, by and among Owens Corning, the guarantors party thereto and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.10 to Post-Effective Amendment No. 1 to Owens Corning's Registration Statement on Form S-3 (Registration No. 333-202011), filed June 21, 2017).
4.18	Eighth Supplemental Indenture, dated as of June 26, 2017, by and among Owens Corning, the guarantors party thereto and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed June 26, 2017).
4.19	Form of 4.300% Senior Notes due 2047 (incorporated by reference to Exhibit 4.2 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed June 26, 2017).
4.20	Ninth Supplemental Indenture, dated as of August 23, 2017, by and among Owens Corning, the guarantor named therein and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.6 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended September 30, 2017).

4.21	Tenth Supplemental Indenture, dated as of January 25, 2018, by and among Owens Corning, the guarantors party thereto and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed January 25, 2018).
4.22	Form of 4.400% Senior Notes due 2048 (incorporated by reference to Exhibit 4.2 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed January 25, 2018).
4.23	Eleventh Supplemental Indenture, dated as of August 12, 2019, by and between Owens Corning and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed August 12, 2019).
4.24	Form of 3.950% Senior Notes due 2029 (incorporated by reference to Exhibit 4.2 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed August 12, 2019).
4.25	Twelfth Supplemental Indenture, dated as of May 12, 2020, by and between Owens Corning and Wells Fargo Bank, National Association, as trustee (incorporated by reference to Exhibit 4.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed May 12, 2020).
4.26	Form of 3.875% Senior Notes due 2030 (incorporated by reference to Exhibit 4.2 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed May 12, 2020).
4.27	Description of Securities Registered under Section 12 of the Securities Exchange Act of 1934 (incorporated by reference to Exhibit 4.26 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100), for the year ended December 31, 2019).
10.1	Amended and Restated Credit Agreement, dated as of July 23, 2021, by and among Owens Corning, as borrower, the lenders signatory thereto and Wells Fargo Bank, National Association, as administrative agent (incorporated by reference to Exhibit 10.1 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended September 30, 2021).
10.2	First Amendment to Amended and Restated Credit Agreement dated as of June 13, 2022, by and among Owens Corning and Wells Fargo Bank, National Association (incorporated by reference to Exhibit 10.1 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended June 30, 2022).
10.3	Second Amendment to Amended and Restated Credit Agreement, dated as of May 24, 2023, by and among Owens Corning and Wells Fargo Bank, National Association (incorporated by references to Exhibit 10.1 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended June 30, 2023).
10.4	Second Amended and Restated Receivables Purchase Agreement, dated as of May 5, 2017 (incorporated by reference to Exhibit 10.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed May 9, 2017).
10.5	First Amendment, dated April 12, 2018, related to the Second Amended and Restated Receivables Purchase Agreement, dated as of May 5, 2017 (incorporated by reference to Exhibit 10.1 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended March 31, 2018).
10.6	Second Amendment to Second Amended and Restated Receivables Purchase Agreement, dated April 8, 2019 (incorporated by reference to Exhibit 10.1 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-133100), for the quarter ended June 30, 2019).
10.7	Third Amendment to Second Amended and Restated Receivables Purchase Agreement, dated as of April 26, 2021, by and among Owens Corning Sales, LLC, Owens Corning Receivables LLC, PNC Bank, National Association and the other parties thereto (incorporated by reference to Exhibit 10.1 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended June 30, 2021).
10.8	Purchase and Sale Agreement, dated as of March 31, 2011, between Owens Corning Sales, LLC and Owens Corning Receivables, LLC (incorporated by reference to Exhibit 10.2 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed April 5, 2011).

10.9	First Amendment to Purchase and Sale Agreement, dated as of May 5, 2017, by and between Owens Corning Sales, LLC and Owens Corning Receivables LLC (incorporated by reference to Exhibit 10.2 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended June 30, 2017).
10.10	Second Amendment to Purchase and Sale Agreement, dated as of April 26, 2021, by and between Owens Corning Sales, LLC and Owens Corning Receivables, LLC (incorporated by reference to Exhibit 10.2 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended June 30, 2021).
10.11	Amended and Restated Performance Guaranty, dated as of May 5, 2017 (incorporated by reference to Exhibit 10.3 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended June 30, 2017).
10.12	Form of Key Management Severance Agreement for Executive Officers (incorporated by reference to Exhibit 10.10 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100), for the year ended December 31, 2013).*
10.13	Form of Directors' Indemnification Agreement (incorporated by reference to Exhibit 10.2 of Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed November 2, 2006).
10.14	Owens Corning Executive Supplemental Benefit Plan, 2009 Restatement (incorporated by reference to Exhibit 10.28 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100), for the year ended December 31, 2008).*
10.15	Owens Corning Supplemental Executive Retirement Plan, as amended and restated, effective as of January 1, 2009 (incorporated by reference to Exhibit 10.30 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100), for the year ended December 31, 2008).*
10.16	Owens Corning 2021 Corporate Incentive Plan (incorporated by reference to Exhibit 10.16 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100), for the year ended December 31, 2020).*
10.17	Owens Corning Amended and Restated Deferred Compensation Plan, effective as of January 1, 2021 (incorporated by reference to Exhibit 10.17 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100), for the year ended December 31, 2020).*
10.18	Owens Corning 2010 Stock Plan (incorporated by reference to Exhibit 10.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed April 23, 2010).*
10.19	Owens Corning 2013 Stock Plan (incorporated by reference to Annex C to Owens Corning's Definitive Proxy Statement (File No 1-33100), filed March 14, 2013).*
10.20	Owens Corning 2016 Stock Plan (incorporated by reference to Exhibit 10.39 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended March 31, 2016).*
10.21	Owens Corning 2019 Stock Plan (incorporated by reference to Exhibit 10.1 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended March 31, 2019).*
10.22	Owens Corning 2023 Stock Plan (incorporated by reference to Exhibit 10.1 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended March 31, 2023.*
10.23	Amended and Restated Owens Corning Employee Stock Purchase Plan, effective April 16, 2020 (incorporated by reference to Exhibit 10.1 to Owens Corning's Current Report on Form 8-K (File No. 1-33100), filed April 21, 2020).*
10.24	Form of Owens Corning 2013 Long Term Incentive Program Award Agreement for Option Award (incorporated by reference to Exhibit 10.27 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100), for the year ended December 31, 2013).*

10.25	Form of Owens Corning 2018 Long Term Incentive Program Award Agreement for Performance Share Units (incorporated by reference to Exhibit 10.2 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended March 31, 2018). *
10.26	Form of Owens Corning 2018 Long Term Incentive Program Award Agreement for Restricted Stock (incorporated by reference to Exhibit 10.3 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended March 31, 2018).*
10.27	Form of Owens Corning 2019 Long Term Incentive Program Award Agreement pursuant to the Owens Corning 2016 Stock Plan for Restricted Stock Unit Award (incorporated by reference to Exhibit 10.2 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended March 31, 2019).*
10.28	Form of Deferred Stock Unit Award Agreement for Directors (incorporated by reference to Exhibit 10.32 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended June 30, 2015).*
10.29	Form of Long Term Incentive Program Award Agreement for Restricted Stock Unit (incorporated by reference to Exhibit 10.33 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended June 30, 2015).*
10.30	Form of Long Term Incentive Program Award Agreement for Performance Share Unit (incorporated by reference to Exhibit 10.34 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended June 30, 2015).*
10.31	Form of Owens Corning 2020 Long Term Incentive Program Award Agreement pursuant to the Owens Corning 2019 Stock Plan for Performance Share Unit Award (incorporated by reference to Exhibit 10.30 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100), for the year ended December 31, 2020).*
10.32	Form of Long Term Incentive Program Award Agreement for Restricted Stock (incorporated by reference to Exhibit 10.35 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended June 30, 2015).*
10.33	Form of Owens Corning 2020 Long Term Incentive Program Award Agreement pursuant to the Owens Corning 2019 Stock Plan for Restricted Stock Unit Award (incorporated by reference to Exhibit 10.32 to Owens Corning's Annual Report on Form 10-K (File No. 1-33100), for the year ended December 31, 2020).*
10.34	Form of Owens Corning 2022 Long Term Incentive Program Award Agreement pursuant to the Owens Corning 2019 Stock Plan for Restricted Stock Unit Award (incorporated by reference to Exhibit 10.1 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended March 31, 2022).*
10.35	Form of Owens Corning 2022 Long Term Incentive Program Award Agreement pursuant to the Owens Corning 2019 Stock Plan for Performance Share Unit Award (incorporated by reference to Exhibit 10.2 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended March 31, 2022).*
10.36	Form of Owens Corning Restricted Stock Unit Award Agreement (incorporated by reference to Exhibit 10.2 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended September 30, 2023.*
10.37	Retirement Transition Agreement, dated as of August 4, 2023, by and between Owens Corning and Dan Smith (incorporated by reference to Exhibit 10.1 to Owens Corning's Quarterly Report on Form 10-Q (File No. 1-33100), for the quarter ended September 30, 2023.*
21.1	Subsidiaries of Owens Corning (filed herewith).
23.1	Consent of PricewaterhouseCoopers LLP (filed herewith).
31.1	Certification of Chief Executive Officer pursuant to Exchange Act Rules 13a-14(a) and 15d-14(a) (filed herewith).

31.2	Certification of Chief Financial Officer pursuant to Exchange Act Rules 13a-14(a) and 15d-14(a) (filed herewith).
32.1	Certification of Chief Executive Officer pursuant to 18 U.S.C. Section 1350 (furnished herewith).
32.2	Certification of Chief Financial Officer pursuant to 18 U.S.C. Section 1350 (furnished herewith).
97.1	Owens Corning Clawback Policy (filed herewith).
101	The following materials from the Annual Report on Form 10-K for Owens Corning for the period ended December 31, 2023, formatted in iXBRL (Inline Extensible Business Reporting Language): (i) Consolidated Statements of Earnings; (ii) Consolidated Statements of Comprehensive Earnings; (iii) Consolidated Balance Sheets; (iv) Consolidated Statements of Stockholders' Equity, (v) Consolidated Statements of Cash Flows; (vi) related notes to these financial statements; and (vii) document and entity information.
104	The cover page from this Annual Report on Form 10-K, formatted as Inline XBRL

\+ Schedules and similar attachments have been omitted from this filing pursuant to Item 601(a)(5) of Regulation S-K. A copy of any omitted schedule or similar attachment will be furnished to the Securities and Exchange Commission upon request.

* Denotes management contract or compensatory plan or arrangement required to be filed as an exhibit pursuant to Form 10-K.

Owens Corning agrees to furnish to the U.S. Securities and Exchange Commission, upon request, copies of all instruments defining the rights of holders of long-term debt of Owens Corning where the total amount of securities authorized under each issue does not exceed 10% of the total assets of Owens Corning and its subsidiaries on a consolidated basis.

ITEM 16. FORM 10-K SUMMARY

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

OWENS CORNING

By	/s/ Brian D. Chambers	February 14, 2024
	Brian D. Chambers Chief Executive Officer (Principal Executive Officer)	

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the Registrant and in the capacities and on the dates indicated.

/s/ Brian D. Chambers	February 14, 2024
Brian D. Chambers, Chief Executive Officer and Director (Principal Executive Officer)	
/s/ Todd W. Fister	February 14, 2024
Todd W. Fister, Chief Financial Officer (Principal Financial Officer)	
/s/ Mari K. Doerfler	February 14, 2024
Mari K. Doerfler, Vice President and Controller	
/s/ Eduardo E. Cordeiro	February 14, 2024
Eduardo E. Cordeiro, Director	
/s/ Adrienne D. Elsner	February 14, 2024
Adrienne D. Elsner, Director	
/s/ Alfred E. Festa	February 14, 2024
Alfred E. Festa, Director	
/s/ Edward F. Lonergan	February 14, 2024
Edward F. Lonergan, Director	
/s/ Maryann T. Mannen	February 14, 2024
Maryann T. Mannen, Director	
/s/ Paul E. Martin	February 14, 2024
Paul E. Martin, Director	

/s/ W. Howard Morris	February 14, 2024
W. Howard Morris, Director	
/s/ Suzanne P. Nimocks	February 14, 2024
Suzanne P. Nimocks, Director	
/s/ John D. Williams	February 14, 2024
John D. Williams, Director	

INDEX TO CONSOLIDATED FINANCIAL STATEMENTS

ITEM	PAGE
Management's Report on Internal Control Over Financial Reporting	58
Report of Independent Registered Public Accounting Firm (PCAOB ID 238)	59
Consolidated Statements of Earnings	61
Consolidated Statements of Comprehensive Earnings	62
Consolidated Balance Sheets	63
Consolidated Statements of Stockholders' Equity	64
Consolidated Statements of Cash Flows	65
Notes to Consolidated Financial Statements	66
1. Business and summary of significant accounting policies	66
2. Segment information	73
3. Inventories	77
4. Derivative financial instruments	77
5. Goodwill and other intangible assets	80
6. Property, plant and equipment	82
7. Acquisitions	83
8. Divestitures	84
9. Leases	85
10. Other current liabilities	87
11. Warranties	87
12. Restructuring, acquisition and divestiture-related costs	88
13. Debt	92
14. Pension plans	95
15. Postemployment and postretirement benefits other than pensions	102
16. Contingent liabilities and other matters	105
17. Stock compensation	106
18. Changes in accumulated other comprehensive deficit	110
19. Earnings per share	111
20. Income taxes	112
21. Subsequent events	115



Management's Report on Internal Control Over Financial Reporting

Management of the Company is responsible for establishing and maintaining adequate internal control over financial reporting as such term is defined in Rules 13a-15(f) and 15d-15(f) under the Securities Exchange Act of 1934.

Management has assessed the effectiveness of the Company's internal control over financial reporting as of December 31, 2023 based on criteria established in the Internal Control-Integrated Framework in 2013 issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

PricewaterhouseCoopers LLP has audited the effectiveness of the internal controls over financial reporting as of December 31, 2023 as stated in their Report of Independent Registered Public Accounting Firm on page 59 hereof.

Based on our assessment, management determined that, as of December 31, 2023, the Company's internal control over financial reporting was effective.

/s/ Brian D. Chambers	February 14, 2024
Brian D. Chambers, Chief Executive Officer (Principal Executive Officer)	
/s/ Todd W. Fister	February 14, 2024
Todd W. Fister, Chief Financial Officer (Principal Financial Officer)	

Report of Independent Registered Public Accounting Firm

To the Board of Directors and Stockholders of Owens Corning

Opinions on the Financial Statements and Internal Control over Financial Reporting

We have audited the accompanying consolidated balance sheets of Owens Corning and its subsidiaries (the "Company") as of December 31, 2023 and 2022, and the related consolidated statements of earnings, of comprehensive earnings, of stockholders' equity and of cash flows for each of the three years in the period ended December 31, 2023, including the related notes and schedule of valuation and qualifying accounts and reserves for each of the three years in the period ended December 31, 2023 appearing on page 117 (collectively referred to as the "consolidated financial statements"). We also have audited the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control - Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Company as of December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2023 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control - Integrated Framework (2013) issued by the COSO.

Basis for Opinions

The Company's management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Report on Internal Control Over Financial Reporting. Our responsibility is to express opinions on the Company's consolidated financial statements and on the Company's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Definition and Limitations of Internal Control over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matters

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that (i) relates to accounts or disclosures that are material to the consolidated financial statements and (ii) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Goodwill Impairment Assessment – Composites Reporting Unit

As described in Notes 1 and 5 to the consolidated financial statements, the Company's consolidated goodwill balance was $1,392 million as of December 31, 2023, and the goodwill associated with the Composites reporting unit was $425 million. Management tests goodwill for impairment as of October 1 each year, or more frequently should circumstances change or events occur that would more likely than not reduce the fair value of a reporting unit below its carrying amount. Management estimates fair value using a discounted cash flow approach from the perspective of a market participant. Significant assumptions used in the discounted cash flow approach are revenue growth rates and earnings before interest and taxes ("EBIT") margins used in estimating the discrete period cash flow forecasts of the reporting unit, the discount rate, and the long-term revenue growth rate and EBIT margin used in estimating the terminal business value.

The principal considerations for our determination that performing procedures relating to the goodwill impairment assessment of the Composites reporting unit is a critical audit matter are (i) the significant judgment by management when developing the fair value estimate of the Composites reporting unit; (ii) a high degree of auditor judgment, subjectivity and effort in performing procedures and evaluating management's significant assumptions related to the revenue growth rates and EBIT margins used in estimating the discrete period cash flow forecasts of the reporting unit, the discount rate, and the long-term revenue growth rate and EBIT margin used in estimating the terminal business value; and (iii) the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to management's goodwill impairment assessment, including controls over the valuation of the Composites reporting unit. These procedures also included, among others (i) testing management's process for developing the fair value estimate; (ii) evaluating the appropriateness of the discounted cash flow approach used by management; (iii) testing the completeness and accuracy of underlying data used in the discounted cash flow approach; and (iv) evaluating the reasonableness of the significant assumptions used by management related to the revenue growth rates and EBIT margins used in estimating the discrete period cash flow forecasts, the discount rate, and the long-term revenue growth rate and EBIT margin used in estimating the terminal business value. Evaluating management's assumptions related to the revenue growth rates and EBIT margins used in estimating the discrete period cash flow forecasts and EBIT margin used in estimating the terminal business value involved evaluating whether the assumptions used by management were reasonable considering (i) the current and past performance of the reporting unit; (ii) the consistency with external market and industry data; and (iii) whether the assumptions were consistent with evidence obtained in other areas of the audit. Professionals with specialized skill and knowledge were used to assist in evaluating (i) the appropriateness of management's discounted cash flow approach and (ii) the reasonableness of the discount rate assumption and the long-term revenue growth rate assumption.

/s/ PricewaterhouseCoopers LLP
Toledo, Ohio
February 14, 2024

We have served as the Company's auditor since 2002.

OWENS CORNING AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF EARNINGS
(in millions, except per share amounts)

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
NET SALES	$ 9,677	$ 9,761	$ 8,498
COST OF SALES	6,994	7,145	6,281
Gross margin	2,683	2,616	2,217
OPERATING EXPENSES			
Marketing and administrative expenses	831	803	757
Science and technology expenses	123	106	91
Gain on sale of site	(189)	—	—
Gain on equity method investment	—	(130)	—
Other expense (income), net	106	123	(69)
Total operating expenses	871	902	779
OPERATING INCOME	1,812	1,714	1,438
Non-operating expense (income), net	145	(9)	(10)
EARNINGS BEFORE INTEREST AND TAXES	1,667	1,723	1,448
Interest expense, net	76	109	126
Loss on extinguishment of debt	—	—	9
EARNINGS BEFORE TAXES	1,591	1,614	1,313
Income tax expense	401	373	319
Equity in net earnings of affiliates	3	—	1
NET EARNINGS	1,193	1,241	995
Net loss attributable to non-redeemable and redeemable noncontrolling interests	(3)	—	—
NET EARNINGS ATTRIBUTABLE TO OWENS CORNING	$ 1,196	$ 1,241	$ 995
EARNINGS PER COMMON SHARE ATTRIBUTABLE TO OWENS CORNING COMMON STOCKHOLDERS			
Basic	$ 13.27	$ 12.85	$ 9.61
Diluted	$ 13.14	$ 12.70	$ 9.54
WEIGHTED AVERAGE COMMON SHARES			
Basic	90.1	96.6	103.5
Diluted	91.0	97.7	104.3

The accompanying Notes to the Consolidated Financial Statements are an integral part of these Statements.

OWENS CORNING AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF COMPREHENSIVE EARNINGS
(in millions)

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
NET EARNINGS	$ 1,193	$ 1,241	$ 995
Other comprehensive income (loss), net of tax			
Currency translation adjustment (net of tax of $(2), $(1) and $(3), for the periods ended December 31, 2023, 2022 and 2021, respectively)	61	(104)	(59)
Pension and other postretirement adjustment (net of tax of $(36), $(3) and $(18), for the periods ended December 31, 2023, 2022 and 2021, respectively)	105	17	54
Hedging adjustment (net of tax of $(4), $6 and $(4), for the periods ended December 31, 2023, 2022 and 2021, respectively)	11	(16)	12
Total other comprehensive income (loss), net of tax	177	(103)	7
TOTAL COMPREHENSIVE EARNINGS	1,370	1,138	1,002
Comprehensive loss attributable to non-redeemable and redeemable noncontrolling interests	(4)	(3)	—
COMPREHENSIVE EARNINGS ATTRIBUTABLE TO OWENS CORNING	$ 1,374	$ 1,141	$ 1,002

The accompanying Notes to the Consolidated Financial Statements are an integral part of these Statements.

OWENS CORNING AND SUBSIDIARIES
CONSOLIDATED BALANCE SHEETS
(in millions)

ASSETS	**December 31, 2023**	**December 31, 2022**
CURRENT ASSETS		
Cash and cash equivalents	$ 1,615	$ 1,099
Receivables, less allowances of $11 at December 31, 2023 and 2022	987	961
Inventories	1,198	1,334
Assets held for sale	—	45
Other current assets	117	117
Total current assets	3,917	3,556
Property, plant and equipment, net	3,841	3,729
Operating lease right-of-use assets	222	204
Goodwill	1,392	1,383
Intangible assets, net	1,528	1,602
Deferred income taxes	24	16
Other non-current assets	313	262
TOTAL ASSETS	$ 11,237	$ 10,752
LIABILITIES AND EQUITY		
CURRENT LIABILITIES		
Accounts payable	$ 1,216	$ 1,345
Current operating lease liabilities	62	52
Long-term debt – current portion	431	28
Other current liabilities	615	679
Total current liabilities	2,324	2,104
Long-term debt, net of current portion	2,615	2,992
Pension plan liability	69	78
Other employee benefits liability	112	118
Non-current operating lease liabilities	165	152
Deferred income taxes	427	388
Other liabilities	315	299
Total liabilities	6,027	6,131
Redeemable noncontrolling interest	25	25
OWENS CORNING STOCKHOLDERS' EQUITY		
Preferred stock, par value $0.01 per share (a)	—	—
Common stock, par value $0.01 per share (b)	1	1
Additional paid in capital	4,166	4,139
Accumulated earnings	4,794	3,794
Accumulated other comprehensive deficit	(503)	(681)
Cost of common stock in treasury (c)	(3,292)	(2,678)
Total Owens Corning stockholders' equity	5,166	4,575
Noncontrolling interests	19	21
Total equity	5,185	4,596
TOTAL LIABILITIES AND EQUITY	$ 11,237	$ 10,752

(a) 10 shares authorized; none issued or outstanding at December 31, 2023 and December 31, 2022
(b) 400 shares authorized; 135.5 issued and 87.2 outstanding at December 31, 2023; 135.5 issued and 91.9 outstanding at December 31, 2022
(c) 48.3 shares at December 31, 2023 and 43.6 shares at December 31, 2022

The accompanying Notes to the Consolidated Financial Statements are an integral part of these Statements.

OWENS CORNING AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY
(in millions)

	Common Stock Outstanding		Treasury Stock						
	Shares	Par Value	Shares	Cost	APIC (a)	Accumulated Earnings	AOCI (b)	NCI (c)	Total
Balance at December 31, 2020	**105.6**	**$ 1**	**29.9**	**$ (1,400)**	**$ 4,059**	**$ 1,829**	**$ (588)**	**$ 40**	**$ 3,941**
Net earnings attributable to Owens Corning	—	—	—	—	—	995	—	—	995
Currency translation adjustment	—	—	—	—	—	—	(59)	(1)	(60)
Pension and other postretirement adjustment (net of tax)	—	—	—	—	—	—	54	—	54
Deferred gain on hedging transactions (net of tax)	—	—	—	—	—	—	12	—	12
Issuance of common stock under share-based payment plans	1.0	—	(1.0)	48	(17)	—	—	—	31
Purchases of treasury stock	(6.2)	—	6.2	(570)	—	—	—	—	(570)
Stock-based compensation expense	—	—	—	—	50	—	—	—	50
Dividends declared (d)	—	—	—	—	—	(118)	—	—	(118)
Balance at December 31, 2021	**100.4**	**$ 1**	**35.1**	**$ (1,922)**	**$ 4,092**	**$ 2,706**	**$ (581)**	**$ 39**	**$ 4,335**
Net earnings attributable to Owens Corning	—	—	—	—	—	1,241	—	—	1,241
Net earnings attributable to non-redeemable noncontrolling interests	—	—	—	—	—	—	—	2	2
Redeemable noncontrolling interest adjustment to redemption value	—	—	—	—	(2)	—	—	—	(2)
Currency translation adjustment	—	—	—	—	—	—	(101)	(3)	(104)
Pension and other postretirement adjustment (net of tax)	—	—	—	—	—	—	17	—	17
Deferred loss on hedging transactions (net of tax)	—	—	—	—	—	—	(16)	—	(16)
Purchases of noncontrolling interest	—	—	—	—	8	—	—	(17)	(9)
Issuance of common stock under share-based payment plans	0.7	—	(0.7)	39	(10)	—	—	—	29
Purchases of treasury stock	(9.2)	—	9.2	(795)	—	—	—	—	(795)
Stock-based compensation expense	—	—	—	—	51	—	—	—	51
Dividends declared (d)	—	—	—	—	—	(153)	—	—	(153)
Balance at December 31, 2022	**91.9**	**$ 1**	**43.6**	**$ (2,678)**	**$ 4,139**	**$ 3,794**	**$ (681)**	**$ 21**	**$ 4,596**
Net earnings attributable to Owens Corning	—	—	—	—	—	1,196	—	—	1,196
Net earnings attributable to non-redeemable noncontrolling interests	—	—	—	—	—	—	—	1	1
Redeemable noncontrolling interest adjustment to redemption value	—	—	—	—	(2)	—	—	—	(2)
Dividends distributed to non-redeemable noncontrolling interests	—	—	—	—	—	—	—	(2)	(2)
Currency translation adjustment	—	—	—	—	—	—	62	(1)	61
Pension and other postretirement adjustment (net of tax)	—	—	—	—	—	—	105	—	105
Deferred gain on hedging transactions (net of tax)	—	—	—	—	—	—	11	—	11
Issuance of common stock under share-based payment plans	1.1	—	(1.1)	48	(22)	—	—	—	26
Purchases of treasury stock	(5.8)	—	5.8	(662)	—	—	—	—	(662)
Stock-based compensation expense	—	—	—	—	51	—	—	—	51
Dividends declared (d)	—	—	—	—	—	(196)	—	—	(196)
Balance at December 31, 2023	**87.2**	**$ 1**	**48.3**	**$ (3,292)**	**$ 4,166**	**$ 4,794**	**$ (503)**	**$ 19**	**$ 5,185**

(a) Additional Paid in Capital (APIC)
(b) Accumulated Other Comprehensive Earnings (Deficit) ("AOCI")
(c) Noncontrolling Interest ("NCI")
(d) Dividend declarations of $2.16 per share as of December 31, 2023, $1.57 per share as of December 31, 2022, and $1.13 per share as of December 31, 2021.

The accompanying Notes to the Consolidated Financial Statements are an integral part of these Statements.

OWENS CORNING AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF CASH FLOWS
(in millions)

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
NET CASH FLOW PROVIDED BY OPERATING ACTIVITIES			
Net earnings	$ 1,193	$ 1,241	$ 995
Adjustments to reconcile net earnings to cash provided by operating activities			
Depreciation and amortization	609	531	502
Deferred income taxes	26	37	44
Pension annuity settlement charge	145	—	—
Stock-based compensation expense	51	51	50
Intangible assets impairment charge	—	96	—
Loss on extinguishment of debt	—	—	9
Gains on sale of certain precious metals	(2)	(18)	(53)
Gain on equity method investment	—	(130)	—
Gain on sale of site	(189)	—	—
Net loss on sale of assets or affiliates	—	36	—
Other adjustments to reconcile net earnings to cash provided by operating activities	(44)	4	11
Change in operating assets and liabilities:			
Changes in receivables, net	(26)	(14)	(28)
Changes in inventories	148	(287)	(227)
Changes in accounts payable and accrued liabilities	(158)	363	302
Changes in other operating assets and liabilities	3	(81)	(65)
Pension fund contributions	(18)	(8)	(21)
Payments for other employee benefits liabilities	(11)	(11)	(13)
Other	(8)	(50)	(3)
Net cash flow provided by operating activities	1,719	1,760	1,503
NET CASH FLOW USED BY INVESTING ACTIVITIES			
Cash paid for property, plant and equipment	(526)	(446)	(416)
Derivative settlements	—	44	(4)
Proceeds from the sale of assets or affiliates	194	212	89
Investment in subsidiaries and affiliates, net of cash acquired	(6)	(417)	(42)
Other	(18)	(16)	(4)
Net cash flow used by investing activities	(356)	(623)	(377)
NET CASH FLOW USED BY FINANCING ACTIVITIES			
Payments on long-term debt	—	—	(193)
Purchase of noncontrolling interest	—	(9)	—
Dividends paid	(188)	(136)	(108)
Purchases of treasury stock	(657)	(795)	(570)
Finance lease payments	(33)	(30)	(23)
Other	1	(4)	13
Net cash flow used by financing activities	(877)	(974)	(881)
Effect of exchange rate changes on cash	30	(22)	(3)
Net increase in cash, cash equivalents and restricted cash	516	141	242
Cash, cash equivalents and restricted cash at beginning of period	1,107	966	724
CASH, CASH EQUIVALENTS AND RESTRICTED CASH AT END OF PERIOD	$ 1,623	$ 1,107	$ 966
DISCLOSURE OF CASH FLOW INFORMATION			
Cash paid during the year for income taxes	$ 428	$ 319	$ 244
Cash paid during the year for interest	$ 135	$ 123	$ 133

The accompanying Notes to the Consolidated Financial Statements are an integral part of these Statements.

OWENS CORNING AND SUBSIDIARIES

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

1. BUSINESS AND SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES

Description of Business

Owens Corning, a Delaware corporation, is a global building and construction materials leader committed to building a sustainable future through material innovation. The Company operates within three segments: Roofing, Insulation and Composites. Through these lines of business, Owens Corning manufactures and sells products worldwide. The Company maintains leading market positions in many of its major product categories.

General

On February 1, 2024, the Board of Directors declared a quarterly dividend of $0.60 per common share payable on April 4, 2024 to shareholders of record as of March 4, 2024.

Basis of Presentation

Unless the context requires otherwise, the terms “Owens Corning,” “Company,” “we” and “our” in these notes refer to Owens Corning and its subsidiaries.

The accompanying Consolidated Financial Statements have been prepared in accordance with accounting principles generally accepted in the United States.

Principles of Consolidation

The Consolidated Financial Statements of the Company include the accounts of majority-owned subsidiaries. Intercompany accounts and transactions are eliminated.

Reclassifications

Certain reclassifications have been made to the 2022 and 2021 Consolidated Financial Statements and Notes to the Consolidated Financial Statements to conform to the classifications used in 2023.

Use of Estimates and Assumptions

The preparation of financial statements in conformity with accounting principles generally accepted in the United States requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent assets and liabilities at the date of the financial statements and the reported amounts of revenues and expenses during the reporting period. Actual results could differ materially from those estimates.

Revenue Recognition

We recognize revenue as the amount of consideration that we expect to receive in exchange for transferring promised goods or services to customers. We do not adjust the transaction price for the effects of a significant financing component, as the time period between control transfer of goods and services and expected payment is one year or less. At the time of sale, we estimate provisions for different forms of variable consideration (discounts, rebates, returns and other refund liabilities) based on historical experience, current conditions and contractual obligations, as applicable. The estimated transaction price is typically not subject to significant reversals. We adjust these estimates when the most likely amount of consideration we expect to receive changes, although these changes are typically minor. Sales, value-added and other similar taxes that we collect are excluded from revenue.

Many of our customer volume commitments are short-term and our performance obligations are generally limited to single purchase orders. Substantially all of our revenue is recognized at a point-in-time when control of goods transfers to the customer. Control transfer typically occurs when goods are shipped from our facilities or at other predetermined control transfer points (for instance, destination terms or consignment arrangements).

1. BUSINESS AND SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES (continued)

Revenue Recognition (continued)

We typically do not satisfy performance obligations without obtaining an unconditional right to payment from customers and, therefore, do not carry contract asset balances on the Consolidated Balance Sheets. Contract liability balances are recorded separately from receivables on the Consolidated Balance Sheets in either Total current liabilities or Other liabilities, depending on the timing of performance obligation satisfaction.

We sell separately-priced warranties that extend certain product and workmanship coverages beyond our standard product warranty, which is described in Note 11. The up-front consideration on extended warranty contracts is deferred and recognized as revenue over time, based on the respective coverage period, ranging from 16 to 20 years. On an annual basis, we expect to recognize approximately $7 million of revenue associated with these extended warranty contracts. Additionally, in certain limited cases, we receive consideration before goods or services are transferred to the customer. These customer down payments and deposits are deferred, and typically recognized as revenue in the following quarters when we satisfy the related performance obligations.

As of December 31, 2023, our contract liability balances (for extended warranties, down payments and deposits, collectively) totaled $101 million. As of December 31, 2022, our contract liability balances totaled $89 million, of which $18 million was recognized as revenue throughout 2023. As of December 31, 2021, our contract liability balances (for extended warranties, down payments and deposits, collectively) totaled $76 million, of which $17 million was recognized as revenue throughout 2022. As of December 31, 2020, our contract liability balances (for extended warranties, down payments and deposits, collectively) totaled $66 million, of which $17 million was recognized as revenue throughout 2021.

As a practical expedient, we recognize incremental costs of obtaining a contract, if any, as an expense when incurred if the amortization period of the asset would have been one year or less. We do not have any costs to obtain or fulfill a contract that are capitalized under Accounting Standards Codification ("ASC") 606.

Cost of Sales

Cost of sales includes material, labor, energy and manufacturing overhead costs, including depreciation and amortization expense associated with the manufacture and distribution of the Company's products. Provisions for warranties are provided in the same period that the related sales are recorded and are based on historical experience, current conditions and contractual obligations, as applicable. Distribution costs include inbound freight costs; purchasing and receiving costs; inspection costs; warehousing costs; shipping and handling costs, which include costs incurred relating to preparing, packaging, and shipping products to customers; and other costs of the Company's distribution network. We account for shipping and handling activities that occur after control of the related good transfers as fulfillment activities instead of performance obligations. All shipping and handling costs billed to the customer are included as net sales in the Consolidated Statements of Earnings.

Marketing and Advertising Expenses

Marketing and advertising expenses are included in Marketing and administrative expenses. These costs include advertising and marketing communications, which are expensed the first time the advertisement takes place. Marketing and advertising expenses for the years ended December 31, 2023, 2022 and 2021 were $134 million, $125 million and $110 million, respectively.

Science and Technology Expenses

The Company incurs certain expenses related to science and technology. These expenses include salaries, building and equipment costs, utilities, administrative expenses, materials and supplies associated with the improvement and development of the Company's products and manufacturing processes. These costs are expensed as incurred.

Earnings per Share

Basic earnings per share are computed using the weighted-average number of common shares outstanding during the period. Diluted earnings per share reflect the dilutive effect of common equivalent shares and increased shares that would result from the conversion of equity securities. The effects of anti-dilution are not presented.

1. BUSINESS AND SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES (continued)

Cash, Cash Equivalents and Restricted Cash

The Company defines cash and cash equivalents as cash and time deposits with maturities of three months or less when purchased.On the Consolidated Statements of Cash Flows, the total of Cash, cash equivalents and restricted cash includes restricted cash of $8 million, $8 million and $7 million as of December 31, 2023, 2022 and 2021, respectively. Restricted cash primarily represents amounts received from a counterparty related to its performance assurance on an executory contract, and is included in Other current assets on the Consolidated Balance Sheets. These amounts are contractually required to be set aside, and the counterparty can exchange the cash for another form of performance assurance at its discretion.

Accounts Receivable

Trade accounts receivable are recorded at the invoiced amount and do not bear interest. Consistent with the requirements of ASU 2016-13, “Financial Instruments - Credit Losses (Topic 236),” the allowance for credit losses is based on the Company’s assessment of the expected losses of customer accounts. The Company regularly reviews the allowance by considering factors such as historical experience, credit quality, the age of the accounts receivable balances, and current economic conditions that may affect a customer’s ability to pay. Account balances are charged off against the allowance when the Company believes it is probable the receivable will not be recovered.

Inventory Valuation

Inventory costs include material, labor, and manufacturing overhead costs, including depreciation and amortization expense associated with the manufacture and distribution of the Company’s products. Inventories are stated at lower of cost or net realizable value and expense estimates are made for excess and obsolete inventories. Cost is determined by the first-in, first-out (“FIFO”) method.

Investments in Affiliates

The Company accounts for investments in affiliates of 20% to 50% ownership when the Company does not have a controlling financial interest using the equity method under which the Company’s share of earnings and losses of the affiliate is reflected in earnings, and dividends are credited against the investment in affiliate when declared. Investments in affiliates are recorded in Other non-current assets on the Consolidated Balance Sheets, and as of December 31, 2023 and 2022, the total value of investments was $29 million and $27 million, respectively.

Goodwill and Other Intangible Assets

Goodwill assets are not amortized but are tested for impairment on at least an annual basis. The Company has the option to use a qualitative approach to determine whether it is more likely than not that the fair value of a reporting unit is less than its carrying value as a basis for determining whether it is necessary to perform a quantitative test. In the current year, as part of the annual assessment, the Company used both a qualitative and quantitative approach to determine whether the fair value of a reporting unit was less than its carrying amount.

Events and circumstances we consider in performing the qualitative assessment include macro-economic conditions, market and industry conditions, internal cost factors, and the operational stability and the overall financial performance of the reporting units. When it is determined necessary for the Company to perform the quantitative testing process for goodwill, the Company estimates fair values using a discounted cash flow approach from the perspective of a market participant. Significant assumptions used in the discounted cash flow approach are revenue growth rates and earnings before interest and taxes (“EBIT”) margins used in estimating discrete period cash flow forecasts of the reporting unit, the discount rate, and the long-term revenue growth rate and EBIT margin used in estimating the terminal business value. The cash flow forecasts of the reporting units are based upon management’s long-term view of our markets and are the forecasts that are used by senior management and the Board of Directors to evaluate operating performance. The discount rate utilized is management’s estimate of what the market’s weighted average cost of capital is for a company with a similar debt rating and stock volatility, as measured by beta. The terminal business value is determined by applying the long-term growth rate to the latest year for which a forecast exists. As part of our goodwill quantitative testing process, we would evaluate whether there are reasonably likely changes to management’s estimates that would have a material impact on the results of the goodwill impairment testing.

1. BUSINESS AND SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES (continued)

Goodwill and Other Intangible Assets (continued)

Other indefinite-lived intangible assets are not amortized but are tested for impairment on at least an annual basis or when determined to have a finite useful life. Substantially all of the indefinite-lived intangible assets are in trademarks and trade names. The Company uses the royalty relief approach to determine whether it is more likely than not that the fair value of these assets is less than its carrying amount. This review is performed annually, or when circumstances arise which indicate there may be impairment. When applying the royalty relief approach, the Company performs a discounted cash flow analysis based on the value derived from owning these trademarks and trade names and being relieved from paying royalty to third parties. Significant assumptions used include the discrete period revenue growth rates, long-term revenue growth rate, royalty rates, discount rates and terminal value.

The inputs for the goodwill and indefinite-lived intangible tests are considered Level 3 inputs under the fair value hierarchy as they are the Company's own data, and are unobservable in the marketplace. Indefinite-lived intangible assets purchased through acquisitions are generally tested qualitatively for impairment in the first year following the acquisition before transitioning to the standard methodology described herein in subsequent years.

Please refer to Note 5 for additional disclosures related to Goodwill and Other Intangible Assets.

Emissions Rights

The Company is allotted carbon emission credit allowances ("emissions rights") from several of the governments under which it operates. These emissions rights are recorded at market value as of the date of issuance and are classified as Intangible assets on the Consolidated Balance Sheets. When the Company emits more than the allotted amounts, additional emissions rights must be purchased.

Properties and Depreciation

Property, plant and equipment are stated at cost and depreciated over their estimated useful lives using the straight-line method. When assets are disposed or otherwise retired, Property, plant and equipment accounts are relieved of the cost and related accumulated depreciation and any gain or loss is included in the Consolidated Statements of Earnings.

Precious metals used in our production tooling are included in property, plant and equipment and are depleted as they are consumed during the production process. Depletion typically represents an annual expense of 2% of the outstanding value and is recorded in Cost of sales on the Consolidated Statements of Earnings.

The range of useful lives for the major components of the Company's plant and equipment is as follows:	
Buildings and leasehold improvements	15 – 40 years
Machinery and equipment	
Furnaces	4 – 15 years
Information systems	5 – 10 years
Equipment	5 – 20 years

Expenditures for normal maintenance and repairs are expensed as incurred.

Asset Impairments

The Company evaluates tangible and intangible long-lived assets for impairment when triggering events have occurred. This requires significant assumptions including projected cash flows, projected income tax rate and terminal business value. These inputs are considered Level 3 inputs under the fair value hierarchy as they are the Company's own data, and are unobservable in the marketplace. Changes in management intentions, market conditions or operating performance could indicate that impairment charges might be necessary that could be material to the Company's Consolidated Financial Statements in any given period. Please refer to Note 5 for additional detail on impairment charges recorded in 2022.

1. BUSINESS AND SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES (continued)

Supplier Finance Programs

We review supplier terms and conditions on an ongoing basis, and have negotiated payment terms extensions in recent years in connection with our efforts to reduce working capital and improve cash flow. Separate from those terms extension actions, certain of our subsidiaries have entered into paying agency agreements with third-party administrators. These voluntary supply chain finance programs (collectively, the "Programs") generally give participating suppliers the ability to sell, or otherwise pledge as collateral, their receivables from the Company to the participating financial institutions, at the sole discretion of both the suppliers and financial institutions. The Company is not a party to the arrangements between the suppliers and the financial institutions. The Company's obligations to its suppliers, including amounts due and scheduled payment dates, are not impacted by the suppliers' decisions to sell, or otherwise pledge as collateral, amounts under these arrangements. The Company's payment terms to the financial institutions, including the timing and amount of payments, are based on the original supplier invoices. One of our programs includes a parent guarantee to the participating financial institution for a certain U.S. subsidiary that, at the time of the respective program's inception in 2015, was a guarantor subsidiary of the Company's Credit Agreement. The obligations are presented as Accounts payable within Total current liabilities on the Consolidated Balance Sheets and all activity related to the obligations is presented within operating activities on the Consolidated Statements of Cash Flow.

The Company's outstanding obligations under the programs for the twelve months ended December 31, 2023 and 2022 are as follows (in millions):

	Twelve Months Ended December 31,	
	2023	2022
Confirmed obligations outstanding at the beginning of the year	$ 234	$ 226
Invoices confirmed during the year	569	656
Confirmed invoices paid during the year	(592)	(648)
Confirmed obligations outstanding at the end of the year	$ 211	$ 234

Income Taxes

The Company recognizes current tax liabilities and assets for the estimated taxes payable or refundable on the tax returns for the current year. Deferred tax balances reflect the impact of temporary differences between the carrying amount of assets and liabilities and their tax basis. Amounts are stated at enacted tax rates expected to be in effect when taxes are actually paid or recovered. In addition, realization of certain deferred tax assets is dependent upon our ability to generate future taxable income. The Company records a valuation allowance to reduce its deferred tax assets to the amount that it believes is more likely than not to be realized. In addition, the Company estimates tax reserves to cover potential taxing authority claims for income taxes and interest attributable to audits of open tax years. Please refer to Note 20 for additional disclosures related to Income Taxes.

Taxes Collected from Customers and Remitted to Government Authorities and Taxes Paid to Vendors

Taxes are assessed by various governmental authorities at different rates on many different types of transactions. The Company charges sales tax or value-added tax ("VAT") on sales to customers where applicable, as well as captures and claims back all available VAT that has been paid on purchases. VAT is recorded in separate payable or receivable accounts and does not affect Net Sales or Cost of Sales line items on the Consolidated Statement of Earnings. VAT receivable is recorded as a percentage of qualifying purchases at the time the vendor invoice is processed. VAT payable is recorded as a percentage of qualifying sales at the time an Owens Corning sale to a customer subject to VAT occurs. Amounts are paid to the taxing authority according to the method and collection prescribed by local regulations. Where applicable, VAT payable is netted against VAT receivable. The Company also pays sales tax to vendors who include a tax, required by government regulations, to the purchase price charged to the Company.

1. BUSINESS AND SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES (continued)

Pension and Other Postretirement Benefits

Accounting for pensions and other postretirement benefits involves estimating the cost of benefits to be provided well into the future and attributing that cost over the time period each employee works. To accomplish this, extensive use is made of assumptions about investment returns, discount rates, inflation, mortality, turnover and medical costs. Please refer to Notes 14 and 15 for additional disclosures related to Pension Plans and Other Postretirement Benefits, respectively.

Derivative Financial Instruments

The Company recognizes all derivative instruments as either assets or liabilities at fair value on the balance sheet. Please refer to Note 4 for further disclosure on derivatives.

The Company performs an analysis for effectiveness of its derivatives designated as hedging instruments at the end of each quarter based on the terms of the contracts and the underlying items being hedged. The change in the fair value of cash flow hedges is deferred in Accumulated other comprehensive income (deficit) ("AOCI") and is subsequently recognized in Cost of sales (for commodity and foreign currency cash flow hedges) on the Consolidated Statements of Earnings in order to mirror the location of the hedged items impacting earnings. Cash settlements for commodity and foreign currency hedges qualifying as cash flow hedges are included in Operating activities in the Consolidated Statements of Cash Flows.

The Company has translation exposure resulting from translating the financial statements of foreign subsidiaries into U.S. Dollars, which is recognized in Currency translation adjustment (a component of AOCI). The Company uses cross-currency forward contracts to hedge portions of the net investment in foreign subsidiaries against fluctuations in foreign exchange rates. The changes in fair values of these derivative instruments are recognized in Currency translation adjustment (a component of AOCI), with recognition of the excluded components amortized to Interest expense, net on the Consolidated Statements of Earnings. Cash settlements for derivatives qualifying as net investment hedges are included in Investing activities in the Consolidated Statements of Cash Flows.

The Company uses forward currency exchange contracts to manage existing exposures to foreign exchange rate risks related to assets and liabilities recorded on the Consolidated Balance Sheets. Gains and losses resulting from the changes in fair value of these instruments are recorded in Other expense (income), net on the Consolidated Statements of Earnings, and are substantially offset by net revaluation impacts on foreign currency denominated balance sheet exposures (which are also recorded in Other expense (income), net). Cash settlements for non-designated derivatives are included in the Consolidated Statements of Cash Flows in the category that is consistent with the nature of the derivative instrument, which is generally the same category as the underlying item being hedged.

Fair Value Measurements

The carrying value of cash and cash equivalents, accounts receivable and short-term debt approximate fair value because of the short-term maturity of the instruments.

Please refer to Notes 4, 13, 14 and 15 for fair value disclosures of derivative financial instruments, long-term debt, pension plans and postemployment and postretirement benefits other than pensions.

Foreign Currency

The functional currency of the Company's subsidiaries is generally the applicable local currency. Assets and liabilities of foreign subsidiaries are translated into United States Dollars at the period-end rate of exchange, and their Statements of Earnings (Loss) and Statements of Cash Flows are converted at the monthly average rate. The resulting translation adjustment is included in AOCI in the Consolidated Balance Sheets and Consolidated Statements of Stockholders' Equity. Transaction gains and losses that arise from exchange rate fluctuations on transactions denominated in a currency other than the functional currency are recorded in Other expense (income), net in the Consolidated Statements of Earnings as incurred. As discussed in the Derivative Financial Instruments section above, the Company uses non-designated foreign currency derivative financial instruments to mitigate this risk. The Company recorded foreign currency transactional (losses)/gains, net of associated derivative activity, of $(2) million, $(4) million and $1 million during the years ended December 31, 2023, 2022, and 2021, respectively. Please refer to Note 4 for additional disclosures related to non-designated derivatives.

1. BUSINESS AND SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES (continued)

Related Party Transactions

In the first quarter of 2021, a related party relationship was established as a result of a member of the Company's Board of Directors being named an executive officer of one of the Company's preexisting suppliers. The related party transactions with this supplier consist of the purchase of raw materials. Purchases from the related party supplier were $92 million for the year ended December 31, 2023. As of December 31, 2023, amounts due to the related party supplier were $5 million. Purchases from the related party supplier were $129 million and $87 million for the years ended December 31, 2022 and December 31, 2021, respectively. As of December 31, 2022 and December 31, 2021, amounts due to the related party supplier were $3 million and $1 million, respectively.

Failed Sale-Leaseback - Fort Smith, Arkansas Municipal Tax Incentive

In the fourth quarter of 2023, the Company signed a municipal tax incentive agreement, as part of which, the Company sold its Fort Smith, Arkansas plant and equipment (together, the "facility") to the municipality of Fort Smith for cash of $165 million. The Company then, on the same day, entered into an agreement to lease the facility from the municipality of Fort Smith over ten years for a total lease liability of $165 million and immediately purchased ten-year municipal bonds at 6.7% interest issued by the municipality of Fort Smith with cash of $165 million. In the Consolidated Statements of Cash Flows, the cash proceeds from the sale of the facility and the cash used for the bond purchase are presented on a net basis within the Net cash flows used by investing activities.

The monthly lease payments under the financing lease obligation and the semi-annual bond coupon payments associated with the bond investment are legally offset and, as such, the offset lease obligation and bond investment amounts are presented on a net basis on the Consolidated Balance Sheets. There will be no cash payments made by either party over the ten-year period. At the termination of the lease agreement, a non-cash exchange will occur where the municipality will call the bond and return title of the facility to the Company.

The transaction did not qualify for sale-leaseback treatment under ASC 842 and, therefore, the plant's net book value, as well as the net book value of the equipment sold, remains in Property, plant and equipment, net on the Consolidated Balance Sheets. Depreciation expense on the assets sold remains within Cost of Sales on the Consolidated Statements of Earnings.

Accounting Pronouncements

The following table summarizes recent Accounting Standard Updates ("ASU") issued by the Financial Accounting Standards Board ("FASB") that could have an impact on the Company's Consolidated Financial Statements:

Standard	Description	Effective Date for Company	Effect on the Consolidated Financial Statements
Recently issued standards:			
ASU 2023-06 "Disclosure Improvements"	The amendments in this update modify the disclosure or presentation requirements of a variety of Topics	The effective date for each topic is contingent on future SEC rule setting.	We are currently assessing the impact adopting this standard will have on our Consolidated Financial Statement disclosures. We do not believe the adoption of this guidance will have a material effect on the results of operations.
ASU 2023-07 "Segment Reporting (Topic 280): Improvements to Reportable Segment Disclosures"	The amendments in this update improve reportable segment disclosure requirements, primarily through enhanced disclosures about significant segment expenses.	January 1, 2024	We are currently assessing the impact adopting this standard will have on our Consolidated Financial Statement disclosures. We do not believe the adoption of this guidance will have a material effect on the results of operations.
ASU 2023-08 "Intangibles—Goodwill and Other— Crypto Assets (Subtopic 350-60): Accounting for and Disclosure of Crypto Assets"	This standard provides that certain crypto assets should be accounted for under fair value methodology rather than as indefinite-lived intangibles.	January 1, 2024	We do not believe the adoption of this guidance will have a material effect on our Consolidated Financial Statements.
ASU 2023-09 "Income Taxes (Topic 740): Improvements to Income Tax Disclosures"	This standard modifies the rate reconciliation and income taxes paid disclosures by requiring consistent categories and greater disaggregation of information in the rate reconciliation, as well as requiring income taxes paid to be disaggregated by jurisdiction.	January 1, 2025	We are currently assessing the impact adopting this standard will have on our Consolidated Financial Statements.

2. SEGMENT INFORMATION

The Company has three reportable segments: Roofing, Insulation and Composites. Accounting policies for the segments are the same as those for the Company. The Company's three reportable segments are defined as follows:

Roofing – Within our Roofing segment, the Company manufactures and sells residential roofing shingles, oxidized asphalt materials, and roofing components used in residential and commercial construction and specialty applications.

Insulation – Within our Insulation segment, the Company manufactures and sells thermal and acoustical batts, loosefill insulation, spray foam insulation, foam sheathing and accessories. It also manufactures and sells glass fiber pipe insulation, energy efficient flexible duct media, bonded and granulated stone wool insulation, cellular glass insulation, and foam insulation used in above- and below-grade construction applications.

Composites – Within our Composites segment, the Company manufactures, fabricates and sells glass reinforcements in the form of fiber. Glass reinforcement materials are also used by the Composites segment to manufacture and sell high value applications in the form of non-wovens, fabrics and composite lumber.

NET SALES

The following tables show a disaggregation of our net sales by segment and geographic region (in millions). Corporate eliminations (shown below) largely reflect intercompany sales from Composites to Roofing. External customer sales are attributed to geographic region based upon the location from which the product is sold to the external customer.

	Twelve Months Ended December 31, 2023				
Reportable Segments	**Roofing**	**Insulation**	**Composites**	**Eliminations**	**Consolidated**
Disaggregation Categories					
U.S. residential	$ 3,781	$ 1,407	$ 359	$ (296)	$ 5,251
U.S. commercial and industrial	137	869	845	(8)	$ 1,843
Total United States	3,918	2,276	1,204	(304)	$ 7,094
Europe	19	771	492	(3)	$ 1,279
Asia-Pacific	1	147	429	—	$ 577
Rest of world	92	474	161	—	$ 727
NET SALES	$ 4,030	$ 3,668	$ 2,286	$ (307)	$ 9,677

	Twelve Months Ended December 31, 2022				
Reportable Segments	**Roofing**	**Insulation**	**Composites**	**Eliminations**	**Consolidated**
Disaggregation Categories					
U.S. residential	$ 3,355	$ 1,523	$ 352	$ (262)	$ 4,968
U.S. commercial and industrial	160	796	872	(4)	$ 1,824
Total United States	3,515	2,319	1,224	(266)	$ 6,792
Europe	21	805	671	(5)	$ 1,492
Asia-Pacific	6	160	550	—	$ 716
Rest of world	116	430	215	—	$ 761
NET SALES	$ 3,658	$ 3,714	$ 2,660	$ (271)	$ 9,761

2. SEGMENT INFORMATION (continued)

	Twelve Months Ended December 31, 2021				
Reportable Segments	**Roofing**	**Insulation**	**Composites**	**Eliminations**	**Consolidated**
Disaggregation Categories					
U.S. residential	$ 2,958	$ 1,194	$ 312	$ (230)	$ 4,234
U.S. commercial and industrial	120	705	637	$ —	$ 1,462
Total United States	3,078	1,899	949	$ (230)	$ 5,696
Europe	19	718	653	$ (6)	$ 1,384
Asia-Pacific	7	187	552	$ —	$ 746
Rest of world	105	380	187	$ —	$ 672
NET SALES	$ 3,209	$ 3,184	$ 2,341	$ (236)	$ 8,498

Our contracts with customers are broadly similar in nature throughout our reportable segments, but the amount, timing and uncertainty of revenue and cash flows may vary in each reportable segment due to geographic and end-market economic factors.

Sales to major customer - One customer, which is a customer of both the Roofing and Insulation segments, accounted for $1 billion (11%) and $895 million (11%) of consolidated sales in 2023 and 2021, respectively. No individual customers accounted for 10% or more of consolidated sales in 2022.

In the United States, sales are primarily related to the residential housing market and commercial and industrial applications. Residential market demand is driven by housing starts and repair and remodeling activity (influenced by existing home sales, seasonal home improvement and damage from major storms). Significant portions of our residential products across our three reportable segments are used interchangeably in both new construction and repair and remodeling, and our customers typically distribute (or use) the products for both applications. U.S. commercial and industrial revenues are largely driven by U.S. industrial production growth, commercial construction activity and overall economic conditions in the U.S.

Outside of the United States (Europe, Asia-Pacific and Rest of world), sales are primarily related to commercial and industrial applications and, to a lesser extent, residential applications in certain countries. Throughout the international regions, demand is primarily driven by industrial production growth, commercial construction activity and overall economic conditions in each respective geographical region.

2. SEGMENT INFORMATION (continued)

EARNINGS BEFORE INTEREST AND TAXES

Earnings before interest and taxes (EBIT) by segment consists of net sales less related costs and expenses and are presented on a basis that is used internally for evaluating segment performance. Certain items, such as general corporate expenses or income and certain other expense or income items, are excluded from the internal evaluation of segment performance. Accordingly, these items are not reflected in EBIT for our reportable segments and are included within Corporate, Other and Eliminations.

The following table summarizes EBIT by segment (in millions):

	Twelve Months Ended December 31,		
	2023	2022	2021
Reportable Segments			
Roofing	$ 1,174	$ 831	$ 753
Insulation	619	612	446
Composites	242	498	376
Total reportable segments	2,035	1,941	1,575
Corporate, Other and Eliminations			
Restructuring costs	(169)	(48)	(34)
Gain on sale of Santa Clara, California site	189	—	—
Pension settlement losses	(145)	—	—
Gain on sale of land in India	—	—	15
Gain on sale of Shanghai, China facility	—	27	—
Gains on sale of certain precious metals	2	18	53
Intangible assets impairment charge	—	(96)	—
Acquisition and divestiture-related costs	—	(7)	—
Recognition of acquisition inventory fair value step-up	—	—	(1)
Paroc marine recall	(15)	—	—
Loss on sale of Chambery, France DUCS business	—	(30)	—
Loss on sale of Russian operations	—	(33)	—
Gain on remeasurement of Fiberteq equity investment	—	130	—
General corporate expense and other	(230)	(179)	(160)
Total Corporate, other and eliminations	(368)	(218)	(127)
EBIT	$ 1,667	$ 1,723	$ 1,448

2. SEGMENT INFORMATION (continued)

TOTAL ASSETS AND PROPERTY, PLANT AND EQUIPMENT

The following table summarizes total assets by segment and property, plant and equipment by geographic region (in millions):

	December 31,	
TOTAL ASSETS	**2023**	**2022**
Reportable Segments		
Roofing	$ 1,863	$ 1,897
Insulation	3,934	3,930
Composites	3,153	3,202
Total reportable segments	8,950	9,029
Cash and cash equivalents	1,615	1,099
Noncurrent deferred income taxes	24	16
Investments in affiliates	29	27
Assets held for sale	—	45
Corporate property, plant and equipment, other assets and eliminations	619	536
CONSOLIDATED TOTAL ASSETS	$ 11,237	$ 10,752

	December 31,	
PROPERTY, PLANT AND EQUIPMENT BY GEOGRAPHIC REGION	**2023**	**2022**
United States	$ 2,508	$ 2,383
Europe	606	541
Asia-Pacific	462	541
Rest of world	265	264
CONSOLIDATED PROPERTY, PLANT AND EQUIPMENT	$ 3,841	$ 3,729

PROVISION FOR DEPRECIATION AND AMORTIZATION

The following table summarizes the provision for depreciation and amortization by segment (in millions):

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
Reportable Segments			
Roofing	$ 64	$ 62	$ 59
Insulation	210	206	208
Composites	172	175	162
Total reportable segments	446	443	429
General corporate depreciation and amortization (a)	163	88	73
CONSOLIDATED PROVISION FOR DEPRECIATION AND AMORTIZATION	$ 609	$ 531	$ 502

(a) In 2023, 2022 and 2021, General corporate depreciation and amortization expense included $101 million, $26 million and $13 million, respectively, of accelerated depreciation and amortization related to restructuring actions further explained in Note 12 to the Consolidated Financial Statements.

2. SEGMENT INFORMATION (continued)

ADDITIONS TO PROPERTY, PLANT AND EQUIPMENT

The following table summarizes cash paid for property, plant and equipment by segment (in millions):

	Twelve Months Ended December 31,		
	2023	2022	2021
Reportable Segments			
Roofing	$ 65	$ 50	$ 54
Insulation	182	196	144
Composites	212	151	153
Total reportable segments	459	397	351
General corporate additions	67	49	65
CONSOLIDATED ADDITIONS TO PROPERTY, PLANT AND EQUIPMENT	$ 526	$ 446	$ 416

3. INVENTORIES

Inventories consist of the following (in millions):

	December 31,	
	2023	2022
Finished goods	$ 742	$ 843
Materials and supplies	456	491
Total inventories	$ 1,198	$ 1,334

4. DERIVATIVE FINANCIAL INSTRUMENTS

The Company is exposed to, among other risks, the impact of changes in commodity prices, foreign currency exchange rates, and interest rates in the normal course of business. The Company's risk management program is designed to manage the exposure and volatility arising from these risks, and utilizes derivative financial instruments to offset a portion of these risks. The Company uses derivative financial instruments only to the extent necessary to hedge identified business risks, and does not enter into such transactions for trading purposes.

The Company generally does not require collateral or other security with counterparties to these financial instruments and is therefore subject to credit risk in the event of nonperformance; however, the Company monitors credit risk and currently does not anticipate nonperformance by other parties. Contracts with counterparties generally contain right of offset provisions. These provisions effectively reduce the Company's exposure to credit risk in situations where the Company has gain and loss positions outstanding with a single counterparty. It is the Company's policy to offset on the Consolidated Balance Sheets the amounts recognized for derivative instruments with any cash collateral arising from derivative instruments executed with the same counterparty under a master netting agreement. As of December 31, 2023 and 2022, the Company did not have any amounts on deposit with any of its counterparties, nor did any of its counterparties have any amounts on deposit with the Company.

Derivative Fair Values

Our derivatives consist of natural gas forward swaps, cross-currency swaps, foreign exchange forward contracts and U.S. treasury rate lock agreements, all of which are over-the-counter and not traded through an exchange. The Company uses widely accepted valuation tools to determine fair value, such as discounting cash flows to calculate a present value for the derivatives. The models use Level 2 inputs, which are observable market-based inputs or unobservable inputs that are corroborated by market data. Examples include forward curves and other commonly quoted observable transactions and prices. The fair value of our derivatives and hedging instruments are all classified as Level 2 investments within the three-tier hierarchy.

4. DERIVATIVE FINANCIAL INSTRUMENTS (continued)

The following table presents the fair value of derivatives and hedging instruments and the respective location on the Consolidated Balance Sheets (in millions):

	Location	Fair Value at December 31, 2023	Fair Value at December 31, 2022
Derivative assets designated as hedging instruments:			
Cash flow hedges:			
Natural gas forward swaps	Other current assets	$ —	$ 2
Derivative liabilities designated as hedging instruments:			
Cash flow hedges:			
Natural gas forward swaps	Other current liabilities	$ 15	$ 32
Derivative assets not designated as hedging instruments:			
Foreign exchange forward contracts	Other current assets	$ 1	$ 1
Derivative liabilities not designated as hedging instruments:			
Foreign exchange forward contracts	Other current liabilities	$ 1	$ 2

Consolidated Statements of Earnings Activity

The following table presents the impact and respective location of derivative activities on the Consolidated Statements of Earnings (in millions):

	Location	Twelve Months Ended December 31, 2023	2022	2021
Derivative activity designated as hedging instruments:				
Natural gas cash flow hedges:				
Amount of loss (gain) reclassified from AOCI (as defined below) into earnings (a)	Cost of sales	$ 52	$ (52)	$ (15)
Cross-currency swap net investment hedges:				
Amount of gain recognized in earnings on derivative amounts excluded from effectiveness testing	Interest expense, net	$ —	$ (1)	$ (5)
Derivative activity not designated as hedging instruments:				
Foreign currency:				
Amount of loss (gain) recognized in earnings (b)	Other expense (income), net	$ 10	$ (29)	$ (41)
Treasury interest rate lock:				
Amount of gain recognized in earnings	Other expense (income), net	$ —	$ (6)	$ —

(a) Accumulated Other Comprehensive Earnings (Deficit) (“AOCI”)

(b) Losses (gains) related to foreign currency derivatives were substantially offset by net revaluation impacts on foreign currency denominated balance sheet exposures, which were also recorded in Other expense (income), net. Please refer to the “Other Derivatives” section below for additional detail.

4. DERIVATIVE FINANCIAL INSTRUMENTS (continued)

Consolidated Statements of Comprehensive Earnings Activity

The following table presents the impact of derivative activities on the Consolidated Statements of Comprehensive Earnings (in millions):

		Amount of Gain (Loss) Recognized in Comprehensive Earnings	
		Twelve Months Ended December 31,	
Hedging Type	**Derivative Financial Instrument**	**2023**	**2022**
Net investment hedge	Cross-currency swaps	$ —	$ 5
Cash flow hedge	Natural gas forward swaps	$ 15	$ (40)
Cash flow hedge	Treasury interest rate lock	$ —	$ 20

Cash Flow Hedges

The Company uses a combination of derivative financial instruments, which qualify as cash flow hedges, and physical contracts to manage forecasted exposure to electricity and natural gas prices. As of December 31, 2023, the notional amounts of these natural gas forward swaps were 7 million MMBTu (or MMBTu equivalent based on U.S. and European indices), compared with the notional amounts of 8 million MMBTu at December 31, 2022. The Company has designated these natural gas forward swaps as cash flow hedges, with the last hedge maturing no later than March 2025. A net unrecognized loss of $15 million related to these natural gas forward swaps was included in AOCI as of December 31, 2023, $15 million of which is expected to be reclassified into earnings in the next 12 months.

In 2020, the Company entered into a $175 million forward U.S. Treasury rate lock agreement to manage the U.S. Treasury portion of its interest rate risk associated with the anticipated issuance of certain 10-year fixed rate senior notes. The Company designated this outstanding forward U.S. Treasury rate lock agreement, which expired on December 15, 2022, as a cash flow hedge. The locked fixed rate of this agreement was 0.994%. In September 2022, a gain of $6 million was recognized as a result of a change in the forecasted issuance of certain senior notes. In December 2022, the Company received cash of $37 million upon the settlement of the rate lock agreement, of which $31 million will be amortized as a component of interest expense upon the future issuance of senior notes. This unrecognized gain of $31 million was included in AOCI as of December 31, 2023.

Net Investment Hedges

The Company has translation exposure resulting from translating the financial statements of foreign subsidiaries into U.S. Dollars, which is recognized in Currency translation adjustment (a component of AOCI). In the second quarter of 2022, the Company terminated the remaining cross-currency forward contracts related to the hedged portions of the net investment in foreign subsidiaries, resulting in cash proceeds of $11 million.

Other Derivatives

The Company uses forward currency exchange contracts to manage existing exposures to foreign exchange risk related to assets and liabilities recorded on the Consolidated Balance Sheets. As of December 31, 2023, the Company had notional amounts of $183 million for non-designated derivative financial instruments related to foreign currency exposures in U.S. Dollars primarily related to the Brazilian Real, Indian Rupee, Chinese Yuan, Hong Kong Dollar, South Korean Won, and the European Euro. In addition, the Company had notional amounts of $45 million for non-designated derivative financial instruments related to foreign currency exposures in European Euro primarily related to the Polish Złoty, British Pound Sterling, and the U.S. Dollar.

5. GOODWILL AND OTHER INTANGIBLE ASSETS

The Company tests goodwill and indefinite-lived intangible assets for impairment as of October 1st each year, or more frequently should circumstances change or events occur that would more likely than not reduce the fair value of a reporting unit below its carrying amount.

Goodwill

The changes in the net carrying amount of goodwill by segment are as follows (in millions):

	Roofing	Insulation	Composites	Total
Gross carrying amount at December 31, 2022	$ 394	$ 1,499	$ 425	$ 2,318
Acquisitions (see Note 7)	—	—	(1)	(1)
Foreign currency translation	1	21	1	23
Gross carrying amount at December 31, 2023	395	1,520	425	2,340
Accumulated impairment losses at December 31, 2022	—	(935)	—	(935)
Foreign currency translation	—	(13)	—	(13)
Accumulated impairment losses at December 31, 2023	—	(948)	—	(948)
Balance, net of impairment at December 31, 2023	$ 395	$ 572	$ 425	$ 1,392

	Roofing	Insulation	Composites	Total
Gross carrying amount at December 31, 2021	$ 397	$ 1,481	$ 75	$ 1,953
Acquisitions (see Note 7)	—	62	353	415
Foreign currency translation	(3)	(44)	(3)	(50)
Gross carrying amount at December 31, 2022	394	1,499	425	2,318
Accumulated impairment losses at December 31, 2021	—	(963)	—	(963)
Foreign currency translation	—	28	—	28
Accumulated impairment losses at December 31, 2022	—	(935)	—	(935)
Balance, net of impairment at December 31, 2022	$ 394	$ 564	$ 425	$ 1,383

The annual tests performed in the fourth quarter of 2023 and 2022 resulted inno impairment of goodwill. The annual 2023 testing indicated that the business enterprise values for the Roofing and Insulation reporting units substantially exceeded their respective carrying values. The business enterprise value of the Composites reporting unit exceeded its carrying value by approximately 5%. There is uncertainty surrounding the macroeconomic factors that impact this reporting unit and a sustained downturn in these factors or a change in the long-term revenue growth or profitability for this reporting unit could increase the likelihood of a future impairment.

5. GOODWILL AND OTHER INTANGIBLE ASSETS (continued)

Other Intangible Assets

Other intangible assets consist of the following (in millions):

	December 31, 2023			December 31, 2022		
	Gross Carrying Amount	Accumulated Amortization	Net Carrying Amount	Gross Carrying Amount	Accumulated Amortization	Net Carrying Amount
Indefinite-lived trademarks and trade names	$ 992	$ —	$ 992	$ 989	$ —	$ 989
Amortizable intangible assets						
Customer relationships	614	(283)	331	638	(243)	395
Technology	326	(203)	123	330	(187)	143
Trademarks	12	(2)	10	12	—	12
Other (a)	74	(2)	72	66	(3)	63
Total other intangible assets	$ 2,018	$ (490)	$ 1,528	$ 2,035	$ (433)	$ 1,602

(a) Other primarily includes emissions rights.

Indefinite-Lived Intangible Assets

Fair values used in testing for potential impairment of our trademarks and trade names are calculated using the relief-from-royalty method by applying an estimated market value royalty rate to the forecasted revenues of the businesses that utilize those assets. The assumed cash flows from this calculation are discounted at a rate based on a market participant discount rate.

The annual test performed in the fourth quarter of 2023 resulted inno impairment of indefinite-lived intangible assets.

There are two indefinite-lived intangible assets that are at an increased risk of impairment, both of which are used by our Insulation segment and were partially impaired in the fourth quarter of 2022. A change in the estimated long-term revenue growth rate or discount rate for the segment could increase the likelihood of a future impairment.

The following table presents the carrying values of these assets as of December 31, 2023 (in millions):

Trade names and trademarks	December 31, 2023
European building and technical insulation trade name	$ 90
Global cellular glass insulation trademark	$ 80

All remaining indefinite-lived intangible assets substantially exceeded their carrying values as of December 31, 2023.

The annual tests performed in the fourth quarter of 2022 resulted in impairment offive of the Company's indefinite-lived intangible assets. Based on the results of this testing, the Company recorded pre-tax non-cash impairment charges totaling $96 million in the fourth quarter of 2022. These charges were recorded in Other expense (income), net on the Consolidated Statements of Earnings, and were included in the Corporate, Other and Eliminations reporting category.

These charges included the following within the Insulation segment: a pre-tax impairment charge of $63 million for a trade name used by our European building and technical insulation business due to the effect of a higher discount rate, associated with rising interest rates, and general economic and geopolitical uncertainty within the European markets resulting in a slightly lower profitability outlook; a pre-tax impairment charge of $12 million related to a trademark used on global cellular glass insulation products due to the effect of a higher discount rate, associated with rising interest rates, and general economic and geopolitical uncertainty within the European markets; a pre-tax impairment charge of $8 million for a trademark used on stone wool insulation products sold in the United States due to forecasted profitability of the product line.

5. GOODWILL AND OTHER INTANGIBLE ASSETS (continued)

The remaining $13 million pre-tax impairment charge for trademarks used within the components business in our Roofing segment was due to the effect of a higher discount rate, associated with rising interest rates, and forecasted profitability of a specific product line.

Definite-Lived Intangible Assets

The Company amortizes the cost of other intangible assets over their estimated useful lives which, individually, range up to 45 years. The Company's future cash flows are not materially impacted by its ability to extend or renew agreements related to its amortizable intangible assets.

Amortization expense for intangible assets for the years ended December 31, 2023, 2022 and 2021 was $94 million, $55 million, and $49 million, respectively. In 2023, amortization expense included $25 million of accelerated amortization related to restructuring actions further explained in Note 12 to the Consolidated Financial Statements.

The estimated amortization expense for intangible assets for the next five years is as follows (in millions):

Period	Amortization
2024	$ 65
2025	$ 58
2026	$ 44
2027	$ 35
2028	$ 34

6. PROPERTY, PLANT AND EQUIPMENT

Property, plant and equipment consist of the following (in millions):

	December 31,	
	2023	2022
Land	$ 168	$ 166
Buildings and leasehold improvements	1,263	1,221
Machinery and equipment	5,402	5,220
Construction in progress	665	522
	7,498	7,129
Accumulated depreciation	(3,657)	(3,400)
Property, plant and equipment, net	$ 3,841	$ 3,729

Machinery and equipment includes certain precious metals used in our production tooling, which comprise approximately 10% of total machinery and equipment as of December 31, 2023 and December 31, 2022.

6. PROPERTY, PLANT AND EQUIPMENT (continued)

Our production tooling needs in our Composites segment are changing in response to economic and technological factors. As a result, we exchanged certain precious metals used in production tooling for certain other precious metals to be used in production tooling. There were no non-cash exchanges during the twelve months ended December 31, 2023. During the twelve months ended December 31, 2022, these non-cash exchanges resulted in a net increase to Machinery and equipment of $18 million and gains totaling $18 million. These gains are included in Other expense (income), net on the Consolidated Statements of Earnings and reflected in the Corporate, Other and Eliminations reporting category. These non-cash investing activities are not included in Net cash flow used by investing activities in the Consolidated Statements of Cash Flows. We do not expect these non-cash exchanges to materially impact our current or future capital expenditure requirements or rate of depletion.

For the years ended December 31, 2023, 2022 and 2021, depreciation expense was $481 million, $447 million and $429 million, respectively, which includes depletion expense related to precious metals used in our production tooling. In 2023, 2022 and 2021, depreciation expense included $76 million, $26 million and $13 million, respectively, of accelerated depreciation related to restructuring actions further explained in Note 12 to the Consolidated Financial Statements.

7. ACQUISITIONS

On September 1, 2022, the Company acquired the remaining 50% interest in Fiberteq, LLC ("Fiberteq"), the joint venture between Owens Corning and IKO Industries, Ltd, which produces high-quality wet-formed fiberglass mat for roofing applications for $140 million, net of cash acquired. During the third quarter of 2023, an additional $6 million of consideration was paid as a result of final working capital adjustments. The acquisition advances the Composites strategy to focus on high-value material solutions and expands Owens Corning's capacity to produce non-woven mat. The Company's 50% interest in Fiberteq was accounted for as an equity-method investment and had a carrying value of $17 million at the acquisition date. The Company used the discounted cash flow method to remeasure the previously held equity method investment to its fair value of $147 million, resulting in the recognition of a gain of $130 million, which is recorded in Gain on equity method investment on the 2022 Consolidated Statements of Earnings. The operating results for Fiberteq have been included in the Composites segment within the Consolidated Financial Statements since the date of the acquisition. The purchase price allocation included $58 million in intangible assets, which primarily consists of customer relationships with an estimated weighted average life of 3 years, a $62 million unfavorable contract liability and $242 million in goodwill, of which 50% is tax deductible. The factors contributing to the recognition of the amount of goodwill are based on several strategic and synergistic benefits that are expected to be realized from the acquisition. The pro-forma effect of this acquisition on revenues and earnings was not material.

On August 1, 2022, the Company acquired Natural Polymers, LLC ("Natural Polymers"), an innovative manufacturer of spray polyurethane foam insulation for building and construction applications for $111 million, net of cash acquired. The acquisition advances the Owens Corning strategy to strengthen the Company's core building and construction products and expand its addressable markets into higher-growth segments. The operating results for Natural Polymers have been included in the Insulation segment within the Consolidated Financial Statements since the date of the acquisition. The purchase price allocation included $44 million in intangible assets and $62 million in goodwill, of which all is tax deductible. The intangible assets consist of definite-lived trademarks of $5 million with an estimated weighted average life of 10 years, technology of $12 million with an estimated weighted average life of 6 years and customer relationships of $27 million with an estimated weighted average life of 17 years. The factors contributing to the recognition of the amount of goodwill are based on several strategic and synergistic benefits that are expected to be realized from the acquisition. The pro-forma effect of this acquisition on revenues and earnings was not material.

On June 1, 2022, the Company acquired all of the outstanding assets of WearDeck®, a premium producer of composite weather-resistant decking for commercial and residential applications, for $133 million, net of cash acquired. The acquisition advances the Composites business growth strategy to focus on high-value material solutions within the building and construction industry. The operating results for WearDeck® have been included in the Composites segment within the Consolidated Financial Statements since the date of the acquisition. The purchase price allocation included $38 million in intangible assets and $68 million in goodwill, of which $61 million is tax deductible. The intangible assets consist of definite-lived trademarks of $7 million with an estimated average life of 10 years, technology of $10 million with an estimated weighted average life of 11 years and customer relationships of $21 million with an estimated weighted average life of 15 years. The factors contributing to the recognition of the amount of goodwill are based on several strategic and synergistic benefits that are expected to be realized from the acquisition. The pro-forma effect of this acquisition on revenues and earnings was not material.

7. ACQUISITIONS (continued)

On May 23, 2022, Owens Corning and Pultron Composites ("Pultron") formed a joint venture (the "JV") to manufacture and sell fiberglass rebar. The Company contributed approximately $47 million to acquire a 65.5% controlling interest and has established a redeemable noncontrolling interest of $25 million related to Pultron, the minority holder. The JV expands Owens Corning's capability to produce high-value material solutions by combining the Company's glass-fiber material technology, channel access and extensive industry experience with Pultron's manufacturing expertise and process efficiency. The fully consolidated operating results for the JV have been included in the Company's Composites segment within the Consolidated Financial Statements since the date of the formation of the JV. Subsequent to the JV formation, the JV acquired assets and technology from Pultron for approximately $65 million. The purchase price allocation included $15 million in intangible assets, consisting of technology, with an estimated weighted average life of 15 years and $42 million in goodwill, of which $37 million is tax deductible. The factors contributing to the recognition of the amount of goodwill are based on several strategic and synergistic benefits that are expected to be realized from the acquisition. The pro-forma effect of this acquisition on revenues and earnings was not material.

On July 13, 2021, the Company acquired vliepa GmbH ("vliepa"), which specializes in the coating, printing, and finishing of non-wovens, paper, and film for the building materials industry in Europe, for $42 million, net of cash acquired. The acquisition broadens the Company's global non-wovens portfolio to better serve European customers and accelerate growth of building and construction market applications in the region. The operating results for vliepa have been included in the Company's Composites segment within the Consolidated Financial Statements since the date of the acquisition. The acquisition resulted in the recognition of $13 million in intangible assets, primarily consisting of customer relationships with an estimated weighted average life of 12 years, and $16 million in goodwill. The pro-forma effect of this acquisition on revenues and earnings was not material.

8. DIVESTITURES

During the third quarter of 2021, the Company entered into a purchase and sale agreement for the Company's Insulation site in Santa Clara, California and ceased operations in the fourth quarter of 2022. Assets held for sale as of December 31, 2022 were $45 million and primarily consisted of land. In March 2023, the Company finalized the sale for total proceeds of $234 million, net of transaction fees. Total proceeds included a non-refundable deposit of $50 million received in the third quarter of 2021. As a result, the Company recognized a pre-tax gain of $189 million in the first quarter of 2023, which is recorded in Gain on sale of site on the Consolidated Statements of Earnings.

On November 24, 2022, the Company finalized the sale of its Russian operations within the Composites and Insulation segments. As a result of this sale, the Company received $104 million, net of cash sold, in consideration and recorded a pre-tax loss of $33 million in Other expense (income), net on the Consolidated Statements of Earnings.

On July 1, 2022, the Company finalized the sale of the European portion of the dry-use chopped strands ("DUCS") product line located in Chambéry, France, within the Composites segment. As a result of this sale, the Company received $80 million, net of cash sold, in consideration and recorded a pre-tax loss of $30 million in Other expense (income), net on the Consolidated Statements of Earnings.

9. LEASES

The Company leases certain equipment and facilities under both operating and finance leases expiring on various dates through 2050. The nature of these leases generally fall into the following five categories: real estate, material handling equipment, fleet vehicles, office equipment and energy equipment.

For leases with initial terms greater than 12 months, we consider these our right-of-use assets and record the related asset and obligation at the present value of lease payments over the term. For leases with initial terms equal to or less than 12 months, we do not consider them as right-of-use assets and instead consider them short-term lease costs that are recognized on a straight-line basis over the lease term.

Our leases may include escalation clauses, renewal options and/or termination options that are factored into our determination of lease payments when reasonably certain. These options to extend or terminate a lease are at our discretion. We have elected to take the practical expedient and not separate lease and non-lease components of contracts. We estimate our incremental borrowing rate to discount the lease payments based on information available at lease commencement. Our lease agreements do not contain any material residual value guarantees.

Balance Sheet Classification

The table below presents the lease-related assets and liabilities recorded on the balance sheet (in millions):

		December 31,	
Leases	**Classification on Balance Sheet**	**2023**	**2022**
Assets			
Operating lease assets	Operating lease right-of-use assets	$ 222	$ 204
Finance lease assets	Other non-current assets	149	127
Total lease assets		$ 371	$ 331
Liabilities			
Current			
Operating	Current operating lease liabilities	$ 62	$ 52
Finance	Long-term debt – current portion	32	28
Non-current			
Operating	Non-current operating lease liabilities	165	152
Finance	Long-term debt, net of current portion	122	103
Total lease liabilities		$ 381	$ 335

9. LEASES (continued)

Lease Costs

The table below presents lease-related costs (in millions):

	Twelve Months Ended December 31,		
	2023	2022	2021
Operating lease cost	$ 81	$ 67	$ 69
Finance lease cost			
Amortization of right-of-use asset	$ 35	$ 30	$ 23
Interest on lease liabilities	$ 6	$ 4	$ 4
Short-term lease cost	$ 6	$ 12	$ 7
Variable lease cost	$ 8	$ 6	$ 5

Cash paid for operating leases included $76 million, $68 million and $70 million in the years ended December 31, 2023, 2022 and 2021, respectively. Cash paid for finance leases included $33 million for financing activities and $6 million for operating activities for the year ended December 31, 2023. Cash paid for finance leases included $30 million for financing activities and $4 million for operating activities for the year ended December 31, 2022. Cash paid for finance leases included $23 million for financing activities and $4 million for operating activities for the year ended December 31, 2021.

We added $104 million, $119 million and $81 million of operating lease liabilities as a result of obtaining operating lease right-of-use assets in the years ended December 31, 2023, 2022 and 2021 respectively. We added $61 million, $70 million and $51 million of finance lease liabilities as a result of obtaining finance lease right-of-use assets in the years ended December 31, 2023, 2022 and 2021 respectively.

Other Information

The tables below present supplemental information related to leases:

	December 31,		
Weighted-average remaining lease term (years)	**2023**	**2022**	**2021**
Operating leases	4.6	4.8	4.2
Finance leases	9.5	9.6	6.9

	December 31,		
Weighted-average discount rate	**2023**	**2022**	**2021**
Operating leases	3.79 %	3.32 %	3.27 %
Finance leases	4.02 %	3.46 %	3.67 %

9. LEASES (continued)

Maturities of Lease Liabilities

The table below reconciles the undiscounted cash flows for each of the first five years and the total of the remaining years to the finance lease liabilities and operating lease liabilities recorded on the balance sheet as of December 31, 2023 (in millions):

Period	Operating Leases	Finance Leases
2024	$ 70	$ 37
2025	58	29
2026	43	20
2027	30	13
2028	17	10
2029 and beyond	30	87
Total minimum lease payments	248	196
Less: implied interest	21	42
Present value of future minimum lease payments	227	154
Less: current lease obligations	62	32
Long-term lease obligations	$ 165	$ 122

10. OTHER CURRENT LIABILITIES

Other current liabilities consist of the following current portions of these liabilities (in millions):

	December 31, 2023	December 31, 2022
Payroll, vacation pay and incentive compensation	$ 224	$ 233
Income, property, and other non-payroll taxes	91	108
Other	300	338
Total other current liabilities	$ 615	$ 679

11. WARRANTIES

The Company records a liability for warranty obligations at the date the related products are sold. Adjustments are made as new information becomes available. Please refer to Note 1 for information about our separately-priced extended warranty contracts. A reconciliation of the warranty liability is as follows (in millions):

	December 31, 2023	December 31, 2022
Beginning balance	$ 88	$ 81
Amounts accrued for current year	26	20
Settlements of warranty claims	(17)	(13)
Ending balance	$ 97	$ 88

12. RESTRUCTURING, ACQUISITION AND DIVESTITURE-RELATED COSTS

The Company may incur restructuring, transaction and integration costs related to acquisitions and divestitures, and may incur restructuring and other exit costs in connection with its global cost reduction, product line and productivity initiatives and the Company's growth strategy.

Global Composites Restructuring

In December 2023, the Company took actions to reduce costs throughout its global Composites segment given current market conditions, primarily through global workforce reductions, as well as streamlining manufacturing and supply chain operations. These actions primarily include salaried workforce reductions and the relocation of the Changzhou, China operations to Hangzhou, China.

In connection with these actions, the Company estimates it will incur cash charges in the range of $20 million to $30 million, primarily related to severance and other exit costs, including termination costs, and non-cash charges in the range of $15 million to $20 million, primarily related to accelerated depreciation.

During 2023, the Company recorded $16 million of charges, of which $4 million were non-cash charges, primarily related to accelerated depreciation and $12 million of cash charges, primarily related to severance.

Building Materials Asia-Pacific Optimization

In December 2023, the Company took actions to further its ongoing cost optimization strategy for the Insulation segment by permanently closing the Xi'an, China facility, which had previously ceased operations, and permanently closing one idled production line at the Guangde, China facility. These actions are expected to result in cumulative costs of approximately $20 million, primarily related to accelerated depreciation.

During 2023, the Company recorded $18 million of charges primarily related to accelerated depreciation. The Company does not expect to recognize significant incremental costs related to these actions.

Protective Packaging Exit

In May 2023, the Company made the decision to exit the Protective Packaging business within the Roofing segment, including the production and sale of wood packaging, metal packaging and custom products. Exiting Protective Packaging will allow the Company to focus resources on the growth of its building materials products, which supports the future growth aspirations of the enterprise. With the exit of the Protective Packaging business, the Company closed its plants in Dorval, Quebec and Mission, British Columbia, Canada. The Company also ceased operations at its Qingdao, China facility.

In connection with the exit of the Protective Packaging business, the Company estimates that it will incur cash charges of approximately $15 million, primarily related to severance and other exit costs. Additionally, the Company will incur total non-cash charges in the range of $70 to $75 million, primarily related to accelerated depreciation of property, plant and equipment and accelerated amortization of definite-lived intangibles.

During 2023, the Company recorded $78 million of charges, of which $69 million were non-cash charges, primarily related to accelerated depreciation, amortization and inventory write-offs and $9 million of cash charges, primarily related to severance.

Wabash Facility Closure

In April 2023, the Company took actions to support its strategy to operate a flexible and cost-efficient manufacturing network through decisions to relocate the Wabash, Indiana mineral wool operations to Joplin, Missouri, and to exit the U.S. granulated mineral wool market. These actions are expected to result in cumulative costs of approximately $30 million, primarily related to severance and accelerated depreciation.

During 2023, the Company recorded $33 million of charges, primarily related to accelerated depreciation, severance and other exit costs. The Company does not expect to recognize significant incremental costs related to these actions.

12. RESTRUCTURING, ACQUISITION AND DIVESTITURE-RELATED COSTS (continued)

European Operating Structure Optimization

In March 2023, the Company took actions to optimize the operating structure of its segments across Europe to increase its competitiveness. These actions are expected to result in cumulative costs of approximately $20 million, primarily related to severance and other exit costs.

During 2023, the Company recorded $12 million of charges primarily related to severance costs.

Composites Strategic Realignment Actions

On July 1, 2022, the Company finalized the sale of the European portion of the DUCS product line located in Chambéry, France, within the Company's Composites segment. The Company recorded a pre-tax charge of $30 million in Other expense (income), net on the Consolidated Statements of Earnings in 2022 to reflect the fair value less cost to sell the assets. The Company also took actions to convert the DUCS manufacturing facilities located in Anderson, South Carolina and Kimchon, Korea to produce other glass fiber products needed to support our growth strategy in building and construction applications.

As a result, during 2023, the Company recorded $2 million of charges primarily related to other exit costs. The Company does not expect to recognize significant incremental costs related to these actions.

Roofing Restructuring Actions

In December 2021, the Company took actions to restructure operations within the Roofing segment's components product line by relocating production assets from China to India, which allowed the business to optimize its manufacturing network and support a tariff mitigation strategy.

During 2023, the Company recorded $2 million of charges primarily related to other exit costs. The Company does not expect to recognize significant incremental charges related to these actions.

Santa Clara Insulation Site

During the third quarter of 2021, the Company entered into a sales agreement for the Company's Insulation site in Santa Clara, California as part of the Company's on-going strategy to operate a flexible, cost-efficient manufacturing network and geographically locate its assets to better serve its customers. On March 3, 2023, the Company finalized the sale of this site for total proceeds of $234 million, net of transaction fees. Total proceeds included a non-refundable deposit of $50 million received in the third quarter of 2021.

During 2023, the Company recorded $6 million of charges, primarily related to other exit costs. The Company does not expect to recognize significant incremental costs related to this action.

12. RESTRUCTURING, ACQUISITION AND DIVESTITURE-RELATED COSTS (continued)

Consolidated Statements of Earnings Classification

The following table presents the impact and respective location of total restructuring, acquisition and divestiture-related costs on the Consolidated Statements of Earnings, which are included within Corporate, Other and Eliminations (in millions):

		Twelve Months Ended December 31,		
Type of Cost	**Location**	**2023**	**2022**	**2021**
Accelerated depreciation	Cost of sales	$ (76)	$ (26)	$ (13)
Other exit (costs) gains	Cost of sales	(26)	(16)	(1)
Recognition of acquisition inventory fair value step-up	Cost of sales	—	—	(1)
Other exit (costs) gains	Marketing and administrative expenses	(2)	—	(2)
Acquisition and divestiture-related (costs) gains	Marketing and administrative expenses	—	(7)	—
Severance	Other expense (income), net	(34)	(1)	(11)
Other exit (costs) gains	Other expense (income), net	(6)	(41)	10
Accelerated amortization	Other expense (income), net	(25)	—	—
Acquisition and divestiture-related (costs) gains	Gain on equity method investment	—	130	—
Gain on sale of Santa Clara, California site	Gain on sale of site	189	—	—
Other exit (costs) gains	Non-operating expense (income), net	—	—	(2)
Total restructuring, acquisition and divestiture-related gains (costs)		$ 20	$ 39	$ (20)

12. RESTRUCTURING, ACQUISITION AND DIVESTITURE-RELATED COSTS (continued)

Summary of Unpaid Liabilities

The following tables summarize the status of the unpaid liabilities from the Company's restructuring activities (in millions):

	December 31, 2023			
	Global Composites Restructuring	Building Materials Asia-Pacific Optimization	Protective Packaging Exit	Wabash Facility Closure
Balance at December 31, 2022	$ —	$ —	$ —	$ —
Restructuring costs	16	18	78	33
Payments	—	(1)	(8)	(4)
Accelerated depreciation and other non-cash items	(4)	(17)	(69)	(26)
Balance at December 31, 2023	$ 12	$ —	$ 1	$ 3
Cumulative charges incurred	$ 16	$ 18	$ 78	$ 33

	December 31, 2023 (continued)			
	European Operating Structure Optimization	Composites Strategic Realignment Actions	Roofing Restructuring Actions	Santa Clara Insulation Site
Balance at December 31, 2022	$ —	$ 1	$ —	$ 7
Restructuring costs	12	2	2	6
Payments	(6)	(3)	(2)	(12)
Accelerated depreciation and other non-cash items	—	—	—	(1)
Balance at December 31, 2023	$ 6	$ —	$ —	$ —
Cumulative charges incurred	$ 12	$ 11	$ 10	$ 66

As of December 31, 2023, the remaining liability balance is comprised of $22 million of severance, which the Company expects to pay over the next twelve months.

	December 31, 2022				
	Composites Strategic Realignment Actions	Roofing Restructuring Actions	Santa Clara Insulation Site	2020 Insulation Restructuring Actions	Acquisition-Related Restructuring
Balance at December 31, 2021	$ —	$ 1	$ 13	$ 1	$ 5
Restructuring costs (gains)	9	3	35	2	(1)
Payments	(2)	—	(19)	(1)	(3)
Accelerated depreciation and other non-cash items	(6)	(4)	(22)	(2)	1
Balance at December 31, 2022	$ 1	$ —	$ 7	$ —	$ 2
Cumulative charges incurred	$ 9	$ 8	$ 60	$ 29	$ 26

As of December 31, 2022, the remaining liability balance is comprised of $11 million of severance, which the Company expects to pay over the next twelve months.

13. DEBT

Details of the Company's outstanding long-term debt, as well as the fair values, are as follows (in millions):

	December 31, 2023		December 31, 2022	
	Carrying Value	Fair Value	Carrying Value	Fair Value
4.200% senior notes, net of discount and financing fees, due 2024	$ 399	99 %	$ 398	99 %
3.400% senior notes, net of discount and financing fees, due 2026	398	96 %	398	94 %
3.950% senior notes, net of discount and financing fees, due 2029	447	95 %	446	90 %
3.875% senior notes, net of discount and financing fees, due 2030	298	94 %	298	89 %
7.000% senior notes, net of discount and financing fees, due 2036	369	116 %	368	107 %
4.300% senior notes, net of discount and financing fees, due 2047	589	88 %	589	78 %
4.400% senior notes, net of discount and financing fees, due 2048	391	87 %	390	78 %
Various finance leases, due through 2050 (a)	154	100 %	131	100 %
Other	1	N/A	2	N/A
Total long-term debt	3,046	N/A	3,020	N/A
Less – current portion of senior notes	399	99 %	—	— %
Less – current portion of other long-term debt (a)	32	100 %	28	100 %
Long-term debt, net of current portion	$ 2,615	N/A	$ 2,992	N/A

(a) The Company determined that the book value of the above noted long-term debt instruments approximates fair value.

The fair values of the Company's outstanding long-term debt instruments were estimated using market observable inputs, including quoted prices in active markets, market indices and interest rate measurements. Within the hierarchy of fair value measurements, these are Level 2 fair values.

Senior Notes

The Company issued $300 million of 2030 senior notes on May 12, 2020. Interest on the notes is payable semiannually in arrears on June 1 and December 1 each year, beginning on December 1, 2020. The proceeds from these notes were used for general corporate purposes.

The Company issued $450 million of 2029 senior notes on August 12, 2019. Interest on the notes is payable semiannually in arrears on February 15 and August 15 each year, beginning on February 15, 2020. The proceeds from these notes were used to repay $416 million of our 2022 senior notes and $34 million of our 2036 senior notes.

The Company issued $400 million of 2048 senior notes on January 25, 2018. Interest on the notes is payable semiannually in arrears on January 30 and July 30 each year, beginning on July 30, 2018. The proceeds from these notes were used, along with borrowings on a $600 million term loan commitment and borrowings on the Receivables Securitization Facility (as defined below), to fund the purchase of Paroc in the first quarter of 2018.

The Company issued $600 million of 2047 senior notes on June 26, 2017. Interest on the notes is payable semiannually in arrears on January 15 and July 15 each year, beginning on January 15, 2018. A portion of the proceeds from these notes was used to fund the purchase of Pittsburgh Corning in 2017 and for general corporate purposes. The remaining proceeds were used to repay $144 million of our 2019 senior notes and $140 million of our 2036 senior notes.

The Company issued $400 million of 2026 senior notes on August 8, 2016. Interest on the notes is payable semiannually in arrears on February 15 and August 15 each year, beginning on February 15, 2017. A portion of the proceeds from these notes was used to redeem $158 million of our 2016 senior notes. The remaining proceeds were used to pay down portions of our Receivables Securitization Facility and for general corporate purposes.

13. DEBT (continued)

The Company issued $400 million of 2024 senior notes on November 12, 2014. Interest on the notes is payable semiannually in arrears on June 1 and December 1 each year, beginning on June 1, 2015. A portion of the proceeds from these notes was used to repay $ 242 million of our 2016 senior notes and $105 million of our 2019 senior notes. The remaining proceeds were used to pay down our Senior Revolving Credit Facility (as defined below), finance general working capital needs, and for general corporate purposes. As of December 31, 2023, the $399 million outstanding principal related to the senior notes was recorded in Long-term debt – current portion.

The Company issued $600 million of 2022 senior notes on October 17, 2012. Interest on the notes was payable semiannually in arrears on June 15 and December 15 each year, beginning on June 15, 2013. The proceeds of these notes were used to repay $ 250 million of our 2016 senior notes and $100 million of our 2019 senior notes and pay down our Senior Revolving Credit Facility. On August 19, 2021, the Company issued a make-whole call to repay the remaining portion of its outstanding 2022 senior notes, and the redemption was completed in the third quarter of 2021. The Company recognized approximately $9 million of loss on extinguishment of debt in the third quarter of 2021 associated with these actions.

The Company issued $550 million of 2036 senior notes on October 31, 2006. Interest on the notes is payable semiannually in arrears on June 1 and December 1 each year, beginning on June 1, 2007. The proceeds of these notes were used to pay certain unsecured and administrative claims, finance general working capital needs and for general corporate purposes.

Collectively, the senior notes above are referred to as the “Senior Notes.” The Senior Notes are general unsecured obligations of the Company and rank*pari passu* with all existing and future senior unsecured indebtedness of the Company.

The Company has the option to redeem all or part of the Senior Notes at any time at a “make-whole” redemption price. The Company is subject to certain covenants in connection with the issuance of the Senior Notes that it believes are usual and customary. The Company was in compliance with these covenants as of December 31, 2023.

Senior Revolving Credit Facility

The Company has an $800 million senior revolving credit facility (the “Senior Revolving Credit Facility”) with a maturity date in July 2026 that includes both borrowings and letters of credit. Borrowings under the Senior Revolving Credit Facility may be used for general corporate purposes and working capital. The Company has the discretion to borrow under multiple options, which provide for varying terms and interest rates including the United States prime rate, federal funds rate plus a spread or the Secured Overnight Financing Rate (“Term SOFR”) plus a spread.

The Senior Revolving Credit Facility contains various covenants, including a maximum allowed leverage ratio, that the Company believes are usual and customary for a senior unsecured credit agreement. The Company was in compliance with these covenants as of December 31, 2023. Please refer to the Credit Facility Utilization section below for liquidity information as of December 31, 2023.

In May 2023, the Senior Revolving Credit Facility was amended to formally adopt Term SOFR plus a spread as the benchmark reference rate in anticipation of the June 30, 2023 discontinuation of the London Interbank Offered Rate (“LIBOR”).

Receivables Securitization Facility

The Company has a Receivables Purchase Agreement (“RPA”) that is accounted for as secured borrowings in accordance with ASC 860, “Accounting for Transfers and Servicing.” Owens Corning Sales, LLC and Owens Corning Receivables LLC, each a subsidiary of the Company, have a $280 million RPA with certain financial institutions. The Company has the ability to borrow at the lenders' cost of funds, which approximates A-1/P-1 commercial paper rates vs. Term SOFR plus a spread. As of the June 30, 2023 discontinuation of LIBOR, fallback language in the RPA took effect to transition the facility to Term SOFR plus a spread. The RPA has been amended from time to time, with a maturity date in April 2024.

The RPA contains various covenants, including a maximum allowed leverage ratio, that the Company believes are usual and customary for a securitization facility. The Company was in compliance with these covenants as of December 31, 2023. Please refer to the Credit Facility Utilization section below for liquidity information as of December 31, 2023.

13. DEBT (continued)

Owens Corning Receivables LLC's sole business consists of the purchase or acceptance through capital contributions of trade receivables and related rights from Owens Corning Sales, LLC and the subsequent retransfer of or granting of a security interest in such trade receivables and related rights to certain purchasers who are party to the RPA. Owens Corning Receivables LLC is a separate legal entity with its own separate creditors who will be entitled, upon its liquidation, to be satisfied out of Owens Corning Receivables LLC's assets prior to any assets or value in Owens Corning Receivables LLC becoming available to Owens Corning Receivables LLC's equity holders. The assets of Owens Corning Receivables LLC are not available to pay creditors of the Company or any other affiliates of the Company or Owens Corning Sales, LLC.

Credit Facility Utilization

The following table shows how the Company utilized its primary sources of liquidity (in millions):

	Balance at December 31, 2023	
	Senior Revolving Credit Facility	Receivables Securitization Facility
Facility size	$ 800	$ 280
Collateral capacity limitation on availability	N/A	—
Outstanding borrowings	—	—
Outstanding letters of credit	4	1
Availability on facility	$ 796	$ 279

Debt Maturities

The aggregate maturities for all outstanding long-term debt borrowings for each of the five years following December 31, 2023 and thereafter are presented in the table below (in millions). The maturities below are the aggregate par amounts of the outstanding senior notes and finance lease liabilities:

Period	Maturities
2024	$ 432
2025	24
2026	416
2027	10
2028	7
2029 and beyond	2,191
Total	$ 3,080

Short-Term Debt

At December 31, 2023 and December 31, 2022, short-term borrowings were $1 million. The short-term borrowings for both periods consisted of various operating lines of credit. The weighted average interest rate on all short-term borrowings was approximately 5.1% and 2.8% for December 31, 2023 and December 31, 2022, respectively.

14. PENSION PLANS

Pension Plans

The Company sponsors defined benefit pension plans. Under the plans, pension benefits are based on an employee's years of service and, for certain categories of employees, qualifying compensation. Company contributions to these pension plans are determined by an independent actuary to meet or exceed minimum funding requirements. In our U.S. plan, the unrecognized cost of any retroactive amendments and actuarial gains and losses are amortized over the average remaining life expectancy of inactive participants. In all of our Non-U.S. plans, the unrecognized cost of any retroactive amendments and actuarial gains and losses are amortized over the average future service period of plan participants expected to receive benefits.

During the fourth quarter of 2023, the Company entered into two agreements to purchase non-participating annuity contracts from insurance companies to transfer $291 million of the Company's outstanding pension projected benefit obligation ("PBO") related to certain U.S. and non-U.S. pension plans. These transactions were funded with pension plan assets of $268 million. As a result of these transactions, the Company recognized a pre-tax settlement charge of $145 million in the fourth quarter of 2023 from the accelerated recognition of a pro rata portion of plan actuarial losses. This charge was recorded in Non-operating expense (income), net on the Consolidated Statements of Earnings. These transactions did not have a material effect on the plans' funded statuses.

The following tables provide a reconciliation of the change in the projected benefit obligation, the change in plan assets and the net amount recognized in the Consolidated Balance Sheets (in millions):

	December 31, 2023			December 31, 2022		
	U.S.	Non-U.S.	Total	U.S.	Non-U.S.	Total
Change in Projected Benefit Obligation						
Benefit obligation at beginning of period	$ 638	$ 326	$ 964	$ 841	$ 476	$ 1,317
Service cost	3	3	6	5	4	9
Interest cost	29	16	45	23	11	34
Actuarial (gain) loss	(21)	9	(12)	(174)	(103)	(277)
Currency loss (gain)	—	11	11	—	(38)	(38)
Benefits paid	(36)	(20)	(56)	(58)	(18)	(76)
Settlements/curtailments	(247)	(42)	(289)	—	(4)	(4)
Other	—	1	1	1	(2)	(1)
Benefit obligation at end of period	$ 366	$ 304	$ 670	$ 638	$ 326	$ 964

	December 31, 2023			December 31, 2022		
	U.S.	Non-U.S.	Total	U.S.	Non-U.S.	Total
Change in Plan Assets						
Fair value of assets at beginning of period	$ 602	$ 302	$ 904	$ 816	$ 443	$ 1,259
Actual return on plan assets	22	14	36	(156)	(92)	(248)
Currency gain (loss)	—	9	9	—	(35)	(35)
Company contributions	—	18	18	—	8	8
Benefits paid	(36)	(20)	(56)	(58)	(18)	(76)
Settlements/curtailments	(247)	(42)	(289)	—	(4)	(4)
Fair value of assets at end of period	341	281	622	602	302	904
Funded status	$ (25)	$ (23)	$ (48)	$ (36)	$ (24)	$ (60)

The following table presents the amount recorded and respective location in the Consolidated Balance sheets (in millions):

14. PENSION PLANS (continued)

		December 31, 2023			December 31, 2022		
	Location	U.S.	Non-U.S.	Total	U.S.	Non-U.S.	Total
Prepaid pension cost	Other non-current assets	$ —	$ 23	$ 23	$ —	$ 21	$ 21
Accrued pension cost – current	Other current liabilities	—	(2)	(2)	(1)	(2)	(3)
Accrued pension cost – non-current	Pension plan liability	(25)	(44)	(69)	(35)	(43)	(78)
Total amount recorded		$ (25)	$ (23)	$ (48)	$ (36)	$ (24)	$ (60)

The following table represents the amounts recorded to Accumulated Other Comprehensive Income (in millions):

	December 31, 2023			December 31, 2022		
Amounts Recorded in AOCI	U.S.	Non-U.S.	Total	U.S.	Non-U.S.	Total
Net actuarial loss	$ (191)	$ (74)	$ (265)	$ (340)	$ (72)	$ (412)
Net prior service cost	—	(5)	(5)	—	(5)	(5)
Total amount recorded	$ (191)	$ (79)	$ (270)	$ (340)	$ (77)	$ (417)

For the year ended December 31, 2023, the actuarial gain of $12 million was largely the result of the impacts related to the annuity purchase settlement offset by losses due to decreased discount rates. In the U.S. plan, the actuarial gain was primarily driven by the settling of the annuity purchase at a lower value relative to the PBO held at the time. The gain was slightly offset by an actuarial loss due to the decrease in the discount rate. In the Non-U.S. plans, the actuarial loss was primarily driven by a decrease in the discount rates across all the plans.

For the year ended December 31, 2022, the actuarial gain of $277 million was largely the result of increases in discount rates across all plans. In the U.S. plan, the actuarial gain was primarily driven by the increase in the discount rate. The gain was slightly offset by the unfavorable impact of differences between expected and actual pension experience. In the Non-U.S. plans, the actuarial gain was driven by an increase in the discount rate of the U.K. and Canada plans, partially offset by the unfavorable impact of differences between expected and actual pension experience.

The following table presents information about the projected benefit obligation, accumulated benefit obligation ("ABO") and plan assets of the Company's pension plans (in millions):

	December 31, 2023			December 31, 2022		
	U.S.	Non-U.S.	Total	U.S.	Non-U.S.	Total
Plans with PBO in excess of fair value of plan assets:						
Projected benefit obligation	$ 366	$ 221	$ 587	$ 638	$ 208	$ 846
Fair value of plan assets	$ 341	$ 174	$ 515	$ 602	$ 163	$ 765
Plans with ABO in excess of fair value of plan assets:						
Accumulated benefit obligation	$ 366	$ 203	$ 569	$ 638	$ 192	$ 830
Fair value of plan assets	$ 341	$ 164	$ 505	$ 602	$ 153	$ 755

14. PENSION PLANS (continued)

Weighted-Average Assumptions Used to Determine Benefit Obligation

The following table presents weighted average assumptions used to determine benefit obligations at the measurement dates:

	December 31,	
	2023	2022
United States Plans		
Discount rate	5.00 %	5.15 %
Cash balance interest crediting rate	4.63 %	3.77 %
Non-United States Plans		
Discount rate	4.62 %	5.02 %
Rate of compensation increase	3.25 %	3.31 %

Components of Net Periodic Pension Cost

The following table presents the components of net periodic pension cost (in millions):

	Twelve Months Ended December 31,		
	2023	2022	2021
United States Plans			
Service cost	$ 3	$ 5	$ 5
Interest cost	29	23	22
Expected return on plan assets	(37)	(36)	(36)
Amortization of actuarial loss	5	11	12
Settlement/curtailment	138	—	—
Other	—	—	2
Net periodic pension cost	$ 138	$ 3	$ 5
Non-United States Plans			
Service cost	$ 3	$ 4	$ 6
Interest cost	16	11	9
Expected return on plan assets	(15)	(16)	(18)
Amortization of actuarial loss	3	2	4
Settlement/curtailment	7	(1)	—
Other	1	—	—
Net periodic pension cost	$ 15	$ —	$ 1
Total			
Service cost	$ 6	$ 9	$ 11
Interest cost	45	34	31
Expected return on plan assets	(52)	(52)	(54)
Amortization of actuarial loss	8	13	16
Settlement/curtailment	145	(1)	—
Other	1	—	2
Net periodic pension cost	$ 153	$ 3	$ 6

14. PENSION PLANS (continued)

Weighted-Average Assumptions Used to Determine Net Periodic Pension Cost

The following table presents weighted-average assumptions used to determine net periodic pension costs for the periods noted:

	Twelve Months Ended December 31,		
	2023	2022	2021
United States Plans			
Discount rate	5.15 %	2.85 %	2.50 %
Expected return on plan assets	5.75 %	4.75 %	4.75 %
Cash balance interest crediting rate	3.76 %	1.26 %	0.79 %
Rate of compensation increase	N/A (a)	N/A (a)	N/A (a)
Non-United States Plans			
Discount rate	5.02 %	2.35 %	1.73 %
Expected return on plan assets	5.00 %	3.93 %	4.08 %
Rate of compensation increase	3.31 %	3.31 %	3.00 %

(a) Not applicable due to changes in plan made on August 1, 2009 that were effective beginning January 1, 2010.

The expected return on plan assets assumption is derived by taking into consideration the target plan asset allocation, historical rates of return on those assets, projected future asset class returns and net outperformance of the market by active investment managers. An asset return model is used to develop an expected range of returns on plan investments over a 30 year period, with the expected rate of return selected from a best estimate range within the total range of projected results. The result is then rounded down to the nearest 25 basis points.

Items Measured at Fair Value

The Company classifies and discloses pension plan assets in one of the following three categories:

Level 1: Quoted market prices in active markets for identical assets.

Level 2: Observable market-based inputs or unobservable inputs that are corroborated by market data.

Level 3: Unobservable inputs that are not corroborated by market data.

Plan Assets

The tables in this section show pension plan asset fair values and fair value leveling information. The assets are categorized into one of the three levels of the fair value hierarchy or are not subject to leveling, in the case of investments that are valued using the net asset value per share (or its equivalent) practical expedient (“NAV”).

14. PENSION PLANS (continued)

The following tables summarize the fair values and applicable fair value hierarchy levels of United States pension plan assets (in millions):

	December 31, 2023			
Asset Category	**Level 1**	**Level 2**	**Level 3**	**Total**
Equities:				
Domestic	$ —	$ —	$ —	$ —
Fixed income and cash equivalents:				
Corporate bonds	12	82	—	94
Government debt	—	20	—	20
Total United States plan assets subject to leveling	$ 12	$ 102	$ —	114
Plan assets measured at NAV:				
Equities				69
Real assets				11
Fixed income and cash equivalents				110
Absolute return strategies				37
Total United States plan assets				$ 341

	December 31, 2022			
Asset Category	**Level 1**	**Level 2**	**Level 3**	**Total**
Equities:				
Domestic	$ 39	$ —	$ —	$ 39
Fixed income and cash equivalents:				
Corporate bonds	26	230	—	256
Government debt	—	58	—	58
Total United States plan assets subject to leveling	$ 65	$ 288	$ —	353
Plan assets measured at NAV:				
Equities				95
Real assets				25
Fixed income and cash equivalents				87
Absolute return strategies				42
Total United States plan assets				$ 602

14. PENSION PLANS (continued)

The following tables summarize the fair values and applicable fair value hierarchy levels of non-United States pension plan assets (in millions):

	December 31, 2023			
Asset Category	**Level 1**	**Level 2**	**Level 3**	**Total**
Equities	$ —	$ 1	$ —	$ 1
Fixed income and cash equivalents:				
Cash and cash equivalents	—	57	—	57
Fixed income	—	11	—	11
Total non-United States plan assets subject to leveling	$ —	$ 69	$ —	69
Plan assets measured at NAV:				
Equities				28
Fixed income and cash equivalents				122
Absolute return strategies and other				62
Total non-United States plan assets				$ 281

	December 31, 2022			
Asset Category	**Level 1**	**Level 2**	**Level 3**	**Total**
Equities	$ —	$ 1	$ —	$ 1
Fixed income and cash equivalents:				
Cash and cash equivalents	—	56	—	56
Corporate bonds	—	11	—	11
Total non-United States plan assets subject to leveling	$ —	$ 68	$ —	68
Plan assets measured at NAV:				
Equities				22
Fixed income and cash equivalents				127
Absolute return strategies				85
Total non-United States plan assets				$ 302

Investment Strategy

The current targeted asset allocation for the United States pension plan is to have23% of assets invested in equities, 70% in intermediate and long-term fixed income securities, high yield and cash and 7% in other strategies. Assets are rebalanced at least quarterly to conform to policy tolerances. The Company actively evaluates the reasonableness of its asset mix given changes in the projected benefit obligation and market dynamics. Our investment policy and asset mix for the non-United States pension plans varies by location and is based on projected benefit obligation and market dynamics.

14. PENSION PLANS (continued)

Estimated Future Benefit Payments

The following table shows estimated future benefit payments from the Company's pension plans (in millions):

Year	Estimated Benefit Payments
2024	$ 47
2025	$ 47
2026	$ 49
2027	$ 46
2028	$ 48
2029-2033	$ 246

Contributions

The Company expects to contribute $20 million in cash to its defined benefit pension plans during 2024. Actual contributions to the plans may change as a result of a variety of factors, including changes in laws that impact funding requirements.

Defined Contribution Plans

The Company sponsors two defined contribution plans which are available to substantially all United States employees. The Company matches a percentage of employee contributions up to a maximum level and contributes up to 2% of an employee's wages regardless of employee contributions. The Company recognized expense of $65 million, $57 million and $52 million during the years ended December 31, 2023, 2022 and 2021, respectively, related to these plans.

15. POSTEMPLOYMENT AND POSTRETIREMENT BENEFITS OTHER THAN PENSIONS

The Company maintains health care and life insurance benefit plans for certain retired employees and their dependents. The health care plans in the United States are non-funded and pay either (1) stated percentages of covered medically necessary expenses, after subtracting payments by Medicare or other providers and after stated deductibles have been met, or (2) fixed amounts of medical expense reimbursement.

Employees hired on or before December 31, 2005 become eligible to participate in the United States health care plans upon retirement if they have accumulated 10 years of service after age 45, 48 or 50, depending on the category of employee. For employees hired after December 31, 2005, the Company does not provide subsidized retiree health care. Some of the plans are contributory, with some retiree contributions adjusted annually. The Company has reserved the right to change or eliminate these benefit plans subject to the terms of collective bargaining agreements.

The following table provides a reconciliation of the change in the projected benefit obligation and the net amount recognized in the Consolidated Balance Sheets for the years ended December 31, 2023 and 2022 (in millions):

	December 31, 2023			December 31, 2022		
	U.S.	Non-U.S.	Total	U.S.	Non-U.S.	Total
Change in Projected Benefit Obligation						
Benefit obligation at beginning of period	$ 115	$ 10	$ 125	$ 151	$ 13	$ 164
Service cost	1	—	1	1	—	1
Interest cost	5	1	6	4	—	4
Actuarial gain	(2)	—	(2)	(30)	(2)	(32)
Currency gain	—	—	—	—	(1)	(1)
Benefits paid	(11)	—	(11)	(11)	—	(11)
Benefit obligation at end of period	$ 108	$ 11	$ 119	$ 115	$ 10	$ 125
Funded status	$ (108)	$ (11)	$ (119)	$ (115)	$ (10)	$ (125)

The following table presents the amount recorded and respective location in the Consolidated Balance sheet (in millions):

		December 31, 2023			December 31, 2022		
		U.S.	Non-U.S.	Total	U.S.	Non-U.S.	Total
	Location						
Accrued benefit obligation – current	Other current liabilities	$ (11)	$ (1)	$ (12)	$ (11)	$ (1)	$ (12)
Accrued benefit obligation – non-current	Other employee benefits liability	(97)	(10)	(107)	(104)	(9)	(113)
Net amount recorded		$ (108)	$ (11)	$ (119)	$ (115)	$ (10)	$ (125)

15. POSTEMPLOYMENT AND POSTRETIREMENT BENEFITS OTHER THAN PENSIONS (continued)

The following table represents the amounts recorded to Accumulated Other Comprehensive Income (in millions):

	December 31, 2023			December 31, 2022		
Amounts Recorded in AOCI	**U.S.**	**Non-U.S.**	**Total**	**U.S.**	**Non-U.S.**	**Total**
Net actuarial (loss) gain	$ (54)	$ (4)	$ (58)	$ 60	$ 5	$ 65
Net prior service credit	—	—	—	—	—	—
Total amount recorded	$ (54)	$ (4)	$ (58)	$ 60	$ 5	$ 65

Weighted-Average Assumptions Used to Determine Benefit Obligations

The following table presents weighted average assumptions used to determine benefit obligations at the measurement dates:

	December 31,	
	2023	**2022**
United States plans		
Discount rate	4.90 %	5.10 %
Rate of compensation increase	N/A	N/A
Non-United States plans		
Discount rate	5.64 %	5.93 %
Rate of compensation increase	3.00 %	3.00 %

Components of Net Periodic Postretirement Benefit Income

The following table presents the components of net periodic postretirement benefit income (in millions):

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
United States plans			
Service cost	$ 1	$ 1	$ 1
Interest cost	5	4	4
Amortization of prior service credit	—	—	(1)
Amortization of actuarial loss	(8)	(7)	(8)
Net periodic postretirement benefit income	$ (2)	$ (2)	$ (4)

There was no significant net periodic postretirement income attributable to non-U.S. plans.

15. POSTEMPLOYMENT AND POSTRETIREMENT BENEFITS OTHER THAN PENSIONS (continued)

Weighted-Average Assumptions Used to Determine Net Periodic Postretirement Benefit Income

The following table presents the discount rates used to determine net periodic postretirement benefit income:

	Twelve Months Ended December 31,		
	2023	2022	2021
United States plans	5.10 %	2.70 %	2.25 %
Non-United States plans	5.93 %	3.63 %	3.04 %

The following table presents health care cost trend rates used to determine net periodic postretirement benefit income, as well as information regarding the ultimate rate and the year in which the ultimate rate is reached:

	Twelve Months Ended December 31,		
	2023	2022	2021
United States plans:			
Initial rate at end of year	11.27 %	8.86 %	8.10 %
Ultimate rate	4.50 %	4.50 %	4.50 %
Year in which ultimate rate is reached	2033	2031	2029
Non-United States plans:			
Initial rate at end of year	4.97 %	4.65 %	4.25 %
Ultimate rate	4.31 %	4.13 %	3.87 %
Year in which ultimate rate is reached	2037	2040	2040

Estimated Future Benefit Payments

The following table shows estimated future benefit payments from the Company's postretirement benefit plans (in millions):

Year	Estimated Benefit Payments
2024	$ 12
2025	$ 12
2026	$ 11
2027	$ 11
2028	$ 11
2029-2033	$ 46

Postemployment Benefits

The Company may also provide benefits to former or inactive employees after employment but before retirement under certain conditions. These benefits include continuation of benefits such as health care and life insurance coverage. The accrued postemployment benefits liability was $5 million at December 31, 2023 and 2022. The net periodic postemployment benefit expense/(income) for the years ended December 31, 2023, 2022 and 2021 were less than $1 million, less than $1 million and less than $(1) million, respectively.

16. CONTINGENT LIABILITIES AND OTHER MATTERS

The Company may be involved in various legal and regulatory proceedings relating to employment, antitrust, tax, product liability, environmental, contracts, intellectual property and other matters (collectively, "Proceedings"). The Company regularly reviews the status of such Proceedings along with legal counsel. Liabilities for such Proceedings are recorded when it is probable that the liability has been incurred and when the amount of the liability can be reasonably estimated. Liabilities are adjusted when additional information becomes available. Management believes that the amount of any reasonably possible losses in excess of any amounts accrued, if any, with respect to such Proceedings or any other known claim, including the matters described below under the caption Environmental Matters (the "Environmental Matters"), are not material to the Company's financial statements. Management believes that the ultimate disposition of the Proceedings and the Environmental Matters will not have a material adverse effect on the Company's financial condition. While the likelihood is remote, the disposition of the Proceedings and Environmental Matters could have a material impact on the results of operations, cash flows or liquidity in any given reporting period.

Litigation and Regulatory Proceedings

The Company is involved in litigation and regulatory proceedings from time to time in the regular course of its business. The Company believes that adequate provisions for resolution of all contingencies, claims and pending matters have been made for probable losses that are reasonably estimable.

During the second quarter of 2023, the Company's subsidiary, Paroc Group OY ("Paroc"), which the Company acquired in 2018, notified the appropriate European maritime regulatory authorities that specific products in its marine insulation product line may not meet certain fire safety requirements in accordance with their certifications. Paroc voluntarily withdrew these specific products from the market, issued recalls, and suspended distribution and sales of these products. Paroc continues to cooperate with the applicable regulatory and government authorities and work with its customers and end-users to assist with remediation. During 2023, the Company established an estimated liability for expected future costs related to the marine recall on our Consolidated Balance Sheet as of December 31, 2023. The estimated liability is primarily based on assumptions related to the estimated costs of the remedy for the recall. We will reevaluate these assumptions each period, and the related liability may be adjusted when factors indicate that the liability is either not sufficient to cover or exceeds the estimated product recall costs. Based on the factors currently known, we believe the appropriate liability has been established at this time. It is reasonably possible that additional product recall costs could be incurred that exceed the estimated liability by amounts that could be material to our consolidated financial statements.

As part of its review of the Paroc insulation product portfolio, the Company discovered potential nonconformances relating to certain ventilation duct insulation products. In January 2024, Paroc suspended sales of the affected insulation products as a precautionary measure while it reviews the potential nonconformances. The Company is continuing its review.

Environmental Matters

The Company has established policies and procedures designed to ensure that its operations are conducted in compliance with all relevant laws and regulations and that enable the Company to meet its high standards for corporate sustainability and environmental stewardship. Our manufacturing facilities are subject to numerous foreign, federal, state and local laws and regulations relating to the presence of hazardous materials, pollution and protection of the environment, including emissions to air, reductions of greenhouse gases, discharges to water, management of hazardous materials, handling and disposal of solid wastes, use of chemicals in our manufacturing processes, and remediation of contaminated sites. All Company manufacturing facilities are either ISO 14001 certified or deploy environmental management systems based on ISO 14001 principles. The Company's 2030 Sustainability Goals include targets related to significant global reductions in energy use, water consumption, waste to landfill, and emissions of greenhouse gases, fine particulate matter, and volatile organic air emissions, and protection of biodiversity.

Owens Corning is involved in remedial response activities and is responsible for environmental remediation at a number of sites, including certain of its currently owned or formerly owned plants. These responsibilities arise under a number of laws, including, but not limited to, the Federal Resource Conservation and Recovery Act, and similar state or local laws pertaining to the management and remediation of hazardous materials and petroleum. The Company has also been named a potentially responsible party under the U.S. Federal Superfund law, or state equivalents, at a number of disposal sites. The Company

16. CONTINGENT LIABILITIES AND OTHER MATTERS (continued)

became involved in these sites as a result of government action or in connection with business acquisitions. As of December 31, 2023, the Company was involved with a total of 22 sites worldwide, including 10 Superfund and state or country equivalent sites and 12 owned or formerly owned sites. None of the liabilities for these sites are individually significant to the Company.

Remediation activities generally involve a potential range of activities and costs related to soil, groundwater, and sediment contamination. This can include pre-cleanup activities such as fact-finding and investigation, risk assessment, feasibility studies, remedial action design and implementation (where actions may range from monitoring to removal of contaminants, to installation of longer-term remediation systems). A number of factors affect the cost of environmental remediation, including the number of parties involved in a particular site, the determination of the extent of contamination, the length of time the remediation may require, the complexity of environmental regulations, variability in clean-up standards, the need for legal action, and changes in remediation technology. Taking these factors into account, Owens Corning has predicted the costs of remediation reasonably estimated to be paid over a period of years. The Company accrues an amount on an undiscounted basis, consistent with the reasonable estimates of these costs when it is probable that a liability has been incurred. Actual cost may differ from these estimates for the reasons mentioned above. At December 31, 2023, the Company had an accrual totaling $4 million for these costs, of which the current portion is $1 million. Changes in required remediation procedures or timing of those procedures, or discovery of contamination at additional sites, could result in material increases to the Company's environmental obligations.

17. STOCK COMPENSATION

Description of the Plan

On April 20, 2023, the Company's stockholders approved the Owens Corning 2023 Stock Plan (the "2023 Stock Plan") which authorizes grants of stock options, stock appreciation rights, stock awards (including restricted stock awards, restricted stock units and bonus stock awards), performance share awards and performance share units. At December 31, 2023, the number of shares remaining available under the 2023 Stock Plan for all stock awards was 3.4 million.

Prior to the 2023 Stock Plan, employees were eligible to receive stock awards under the Owens Corning 2019 Stock Plan.

Total Stock-Based Compensation Expense

Stock-based compensation expense included in Marketing and administrative expenses in the accompanying Consolidated Statements of Earnings is as follows (in millions):

	Twelve Months Ended December 31,		
	2023	2022	2021
Total stock-based compensation expense	$ 51	$ 51	$ 50
Income tax benefit recognized on stock-based compensation expense	$ 14	$ 11	$ 14

Stock Options

The Company has granted stock options under its stockholder approved stock plans. The Company calculates a weighted-average grant-date fair value using a Black-Scholes valuation model for options granted. Compensation expense for options is measured based on the fair market value of the option on the date of grant, and is recognized on a straight-line basis over a four-year vesting period. In general, the exercise price of each option awarded was equal to the closing market price of the Company's common stock on the date of grant and an option's maximum term is 10 years. The volatility assumption was based on a benchmark study of our peers prior to 2014. Starting with the options granted in 2014, the volatility was based on the Company's historic volatility.

The Company has not granted stock options since the year ended December 31, 2014. As of December 31, 2023, there was no unrecognized compensation cost related to stock options and the exercise prices on outstanding stock options was $37.65.

17. STOCK COMPENSATION (continued)

The following table summarizes the Company's stock option activity in 2023:

	Number of Options	Weighted-Average Exercise Price	Weighted-Average Remaining Contractual Life (in years)	Intrinsic Value (in millions)
Outstanding, December 31, 2022	27,000	$ 37.65	1.10	$ 1
Exercised	(17,900)	37.65		
Outstanding, December 31, 2023	9,100	$ 37.65	0.10	$ 1
Exercisable, December 31, 2023	9,100	$ 37.65	0.10	$ 1

The total cash received from the exercise of stock options and the resulting tax benefits received were as follows (in millions):

	Twelve Months Ended December 31,		
	2023	2022	2021
Cash received upon exercise of stock option awards	$ 1	$ 1	$ 11
Income tax benefit received for stock option awards exercised	$ —	$ —	$ 4

Restricted Stock Units

The Company has granted restricted stock units ("RSUs") under its stockholder-approved stock plans. Generally, all outstanding RSUs will fully settle in stock. Compensation expense for RSUs is measured based on the closing market price of the stock at date of grant and is recognized on a straight-line basis over the vesting period, which is typically three or four years. The Stock Plan allows alternate vesting schedules for death, disability and retirement.

The weighted average grant date fair value of RSUs granted in 2023, 2022 and 2021 was $104.27, $90.41 and $84.03, respectively.

The following table shows a summary of the Company's RSUs:

	Number of RSUs	Weighted-Average Fair Value
Balance at January 1, 2023	1,276,160	$ 69.16
Granted	408,890	104.27
Vested	(391,096)	68.32
Forfeited	(68,286)	87.10
Balance at December 31, 2023	1,225,668	$ 79.72

As of December 31, 2023, there was $40 million of total unrecognized compensation cost related to RSUs. That cost is expected to be recognized over a weighted-average period of 2.42 years. The total grant date fair value of stock vested during the years ended December 31, 2023, 2022 and 2021 was $27 million, $22 million and $26 million, respectively.

Performance Share Units

The Company has granted performance share units ("PSUs") as a part of its long-term incentive plan. All outstanding PSUs will fully settle in stock. The amount of shares ultimately distributed from the 2023, 2022 and 2021 grants is contingent on meeting internal company-based metrics or an external-based stock performance metric.

In 2023, 2022 and 2021, the Company granted both internal company-based and external-based metric PSUs.

17. STOCK COMPENSATION (continued)

Internal Company-based metrics

The internal Company-based metric PSUs are based on various Company metrics and typically vest after athree-year period. The amount of stock distributed will vary from 0% to 200% of PSUs awarded depending on each award's design and performance versus the Company-based metrics.

The initial fair value for all internal company-based metric PSUs assumes that the performance goals will be achieved and is based on the grant date stock price. This assumption is monitored quarterly and if it becomes probable that such goals will not be achieved or will be exceeded, compensation expense recognized will be adjusted and previous surplus compensation expense recognized will be reversed or additional expense will be recognized. The expected term represents the period from the grant date to the end of the three-year performance period. Pro-rata vesting may be utilized in the case of death, disability or retirement, and awards, if earned, will be paid at the end of the three-year period.

The following table provides a summary of the grant date fair values of the internal Company-based metric PSUs:

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
Grant date fair value of units granted	$92.97	$87.06	$78.01

External based metrics

The external-based metric PSUs vest after a three-year period. Outstanding grants issued in or after 2018 until 2022 were based on the Company's total stockholder return relative to the performance of the Dow Jones U.S. Construction & Materials Index. Outstanding grants issued in 2023 are based on the Company's total stockholder return relative to a peer group. The amount of stock distributed will vary from 0% to 200% of PSUs awarded depending on the relative stockholder return performance. The fair value of external-based metric PSUs has been estimated at the grant date using a Monte Carlo simulation that uses various assumptions.

The following table provides a summary of the assumptions for PSUs granted in 2023, 2022 and 2021:

	Twelve Months Ended December 31,		
	2023	**2022**	**2021**
Expected volatility	44.66%	41.65%	42.74% — 43.67%
Risk free interest rate	3.75%	1.36%	0.18% — 0.24%
Expected term (in years)	2.91	2.91	2.56 — 2.90
Grant date fair value of units granted	$119.33	$122.69	$99.19 — $127.37

The risk-free interest rate was based on zero-coupon United States Treasury STRIPS at the grant date. The expected term represents the period from the grant date to the end of the three-year performance period.

PSU Summary

As of December 31, 2023, there was $17 million total unrecognized compensation cost related to PSUs. That cost is expected to be recognized over a weighted-average period of 1.64 years. The total grant date fair value of shares vested during the years ended December 31, 2023, 2022 and 2021, was $21 million, $9 million and $8 million, respectively.

17. STOCK COMPENSATION (continued)

The following table shows a summary of the Company's PSUs:

	Number of PSUs	Weighted-Average Grant Date Fair Value
Balance as of January 1, 2023	303,716	$ 91.47
Granted	266,149	95.01
Vested	(245,697)	84.74
Forfeited	(55,491)	94.98
Balance as of December 31, 2023	268,677	$ 100.57

Employee Stock Purchase Plan

The Owens Corning Employee Stock Purchase Plan ("ESPP") is a tax-qualified plan under Section 423 of the Internal Revenue Code. The purchase price of shares purchased under the ESPP is equal to 85% of the lower of the fair market value of shares of Owens Corning common stock at the beginning or ending of the offering period, which is a six month period ending on May 31 and November 30 of each year. On April 16, 2020, the Company's stockholders approved the Amended and Restated Owens Corning Employee Stock Purchase Plan which increased the number of shares available for issuance under the plan by 4.2 million shares. As of December 31, 2023, 3.2 million shares remain available for purchase.

Included in total stock-based compensation expense is $7 million, $6 million and $6 million of expense related to the Company's ESPP for the years ended December 31, 2023, 2022 and 2021, respectively. As of December 31, 2023, the Company had $3 million of total unrecognized compensation costs related to the ESPP. Under the outstanding ESPP as of February 9, 2024, employees have contributed $5 million to purchase shares for the current purchase period ending May 31, 2024.

The following table shows a summary of employee purchase activity under the ESPP:

	Twelve Months Ended December 31,		
	2023	2022	2021
Total shares purchased by employees	287,732	293,364	289,945
Average purchase price	$ 83.51	$ 74.19	$ 66.68

18. CHANGES IN ACCUMULATED OTHER COMPREHENSIVE DEFICIT

The following table summarizes the changes in accumulated other comprehensive income (deficit) (in millions):

	Twelve Months Ended December 31,	
	2023	2022
Currency Translation Adjustment		
Beginning balance	$ (380)	$ (279)
Net investment hedge amounts classified into AOCI, net of tax	—	4
Gain (loss) on foreign currency translation	62	(105)
Other comprehensive income (loss), net of tax	62	(101)
Ending balance	$ (318)	$ (380)
Pension and Other Postretirement Adjustment		
Beginning balance	$ (301)	$ (318)
Amounts reclassified from AOCI to net earnings, net of tax (a)	—	5
Pension annuity settlement charge reclassified from AOCI, net of tax (b)	109	—
Amounts classified into AOCI, net of tax	(4)	12
Other comprehensive income, net of tax	105	17
Ending balance	$ (196)	$ (301)
Hedging Adjustment		
Beginning balance	$ —	$ 16
Amounts reclassified from AOCI to net earnings, net of tax (c)	39	(44)
Amounts classified into AOCI, net of tax	(28)	28
Other comprehensive income (loss), net of tax	11	(16)
Ending balance	$ 11	$ —
Total AOCI ending balance	$ (503)	$ (681)

(a) These AOCI components are included in the computation of total Pension and Other Postretirement cost and are recorded in Non-operating expense (income), net. See Notes 14 and 15 for additional information.

(b) These amounts reclassified from AOCI relate to a pension annuity settlement which occurred in the fourth quarter of 2023. See Note 14 for additional information.

(c) Amounts reclassified from (loss) gain on cash flow hedges are reclassified from AOCI to income when the hedged item affects earnings and is recognized in Cost of sales or Interest expense, net depending on the hedged item. See Note 4 for additional information.

19. EARNINGS PER SHARE

The following table is a reconciliation of weighted-average shares for calculating basic and diluted earnings per-share (in millions, except per share amounts):

	Twelve Months Ended December 31,		
	2023	2022	2021
Net earnings attributable to Owens Corning	$ 1,196	$ 1,241	$ 995
Weighted-average number of shares outstanding used for basic earnings per share	90.1	96.6	103.5
Non-vested restricted and performance shares	0.9	1.1	0.8
Weighted-average number of shares outstanding and common equivalent shares used for diluted earnings per share	91.0	97.7	104.3
Earnings per common share attributable to Owens Corning common stockholders:			
Basic	$ 13.27	$ 12.85	$ 9.61
Diluted	$ 13.14	$ 12.70	$ 9.54

Basic earnings per share is calculated by dividing earnings attributable to Owens Corning by the weighted-average number of shares of the Company's common stock outstanding during the period. Outstanding shares consist of issued shares less treasury stock.

The Board of Directors approved two share repurchase programs in 2022 under which the Company is authorized to repurchase up to an aggregate of 20 million shares of the Company's outstanding common stock (the "Repurchase Authorization"). The Repurchase Authorization enables the Company to repurchase shares through the open market, privately negotiated, or other transactions. The actual number of shares repurchased will depend on timing, market conditions and other factors and will be at the Company's discretion. The Company repurchased 5.4 million shares of its common stock for $629 million, inclusive of applicable taxes, during the year ended December 31, 2023 under the Repurchase Authorization. As of December 31, 2023, 8.9 million shares remained available for repurchase under the Repurchase Authorization.

For the year ended December 31, 2023, the Company did not have any non-vested restricted stock units or non-vested performance share units that had an anti-dilutive effect on earnings per share. For the year ended December 31, 2022, the Company did not have any non-vested restricted stock units or non-vested performance share units that had an anti-dilutive effect on earnings per share. For the year ended December 31, 2021, the number of shares used in the calculation of diluted earnings per share did not include 0.1 million shares of Company stock underlying unvested performance share units due to their anti-dilutive effect.

20. INCOME TAXES

The following table summarizes our Earnings before taxes and Income tax expense (in millions):

	Twelve Months Ended December 31,		
	2023	2022	2021
Earnings before taxes:			
United States	$ 1,360	$ 1,286	$ 868
Foreign	231	328	445
Total	$ 1,591	$ 1,614	$ 1,313
Income tax expense:			
Current			
United States	$ 286	$ 180	$ 139
State and local	62	38	27
Foreign	65	125	90
Total current	413	343	256
Deferred			
United States	(4)	50	53
State and local	5	(6)	(3)
Foreign	(13)	(14)	13
Total deferred	(12)	30	63
Total income tax expense	$ 401	$ 373	$ 319

The reconciliation between the United States federal statutory rate and the Company's effective income tax rate from continuing operations is:

	Twelve Months Ended December 31,		
	2023	2022	2021
United States federal statutory rate	21 %	21 %	21 %
State and local income taxes, net of federal tax benefit	3	2	3
Foreign tax credits	—	—	(1)
R&D Credits	—	(1)	—
Other, net	1	1	1
Effective tax rate	25 %	23 %	24 %

The Company continues to assert indefinite reinvestment in accordance with ASC 740 based on the laws as of enactment of the Tax Act. As of December 31, 2023, the Company has not provided for withholding or income taxes on approximately $1.5 billion of undistributed reserves of its foreign subsidiaries and affiliates as they are considered by management to be permanently reinvested. Quantification of the deferred tax liability associated with these undistributed reserves is not practicable.

20. INCOME TAXES (continued)

The cumulative temporary differences giving rise to the deferred tax assets and liabilities are as follows (in millions):

	December 31, 2023		December 31, 2022	
	Deferred Tax Assets	Deferred Tax Liabilities	Deferred Tax Assets	Deferred Tax Liabilities
Other employee benefits	$ 57	$ —	$ 59	$ —
Pension plans	13	—	16	—
Operating loss and tax credit carryforwards	92	—	108	—
Depreciation	—	363	—	334
Capitalized R&D	53	—	24	—
Leases - right of use assets	—	70	—	70
Leases - liabilities	62	—	56	—
Amortization	—	293	—	298
Inventory	30	—	24	—
Foreign tax credit carryforwards	53	—	52	—
Other	103	—	120	—
Subtotal	463	726	459	702
Valuation allowances	(140)	—	(129)	—
Total deferred taxes	$ 323	$ 726	$ 330	$ 702

The following table summarizes the amount and expiration dates of our deferred tax assets related to operating loss and tax credit carryforwards and foreign tax credit carryforwards at December 31, 2023 (in millions) (a):

	Expiration Dates	Amounts
Domestic loss and tax credit carryforwards	2024-2036	$ 85
Foreign loss and tax credit carryforwards (b)	2024 - Indefinite	60
Total operating loss and tax credit carryforwards		$ 145

(a) The use of certain of the Company's domestic loss carryforwards is limited pursuant to Internal Revenue Code (IRC) Section 382. IRC Section 382 imposes an annual limitation on a corporation's ability to use loss carryforwards that arose before a change in control. A change in control is generally defined as a cumulative change of more than 50% in the ownership positions of certain stockholders during a rolling three-year period. The Company believes that these limitations will not result in the loss of any of the loss carryforwards.

(b) The foreign net operating losses are related to various jurisdictions that provide for both indefinite carryforward periods and others with carryforward periods that range from the tax years 2024 to 2033.

Deferred income taxes are provided for temporary differences between amounts of assets and liabilities for financial reporting purposes and the basis of such assets and liabilities as measured under enacted tax laws and regulations, as well as NOLs, tax credits and other tax carryforwards. We have a variety of deferred tax assets in numerous tax jurisdictions. These deferred tax assets are subject to periodic assessment as to recoverability and if it is determined that it is more likely than not that the benefits will not be realized, valuation allowances are recognized. In evaluating whether it is more likely than not that we would recover these deferred tax assets, future taxable income, the reversal of existing temporary differences and tax planning strategies are considered.

We believe that our estimates for the valuation allowances recorded against deferred tax assets are appropriate based on current facts and circumstances. As of December 31, 2023 we had $140 million of valuation allowances on deferred tax assets, on a

20. INCOME TAXES (continued)

tax-effected basis, primarily related to U.S. federal foreign tax credit carryforwards and certain foreign deferred tax attributes as it is more likely than not that some portion or all of these tax attributes will not be realized.

We file a consolidated federal income tax return in the United States as well as tax returns in multiple state, local and foreign jurisdictions. In the normal course of business, the Company is subject to examination by the taxing authorities in each of the jurisdictions where we file tax returns. The Company is no longer subject to U.S. federal tax examinations for years before 2020 or state and foreign examinations for years before 2014.

We have on-going audits in various stages of completion in several state and foreign jurisdictions, one or more of which may conclude within the next 12 months. Such settlements could involve some or all of the following: the payment of additional taxes, the adjustment of certain deferred taxes and/or the recognition of unrecognized tax benefits. The resolution of these matters, in combination with the expiration of certain statutes of limitations in various jurisdictions, make it reasonably possible that our unrecognized tax benefits may decrease as a result of either payment or recognition by up to $44 million in the next 12 months, excluding interest.

A reconciliation of the beginning and ending amounts of unrecognized tax benefits is as follows (in millions):

	Twelve Months Ended December 31,		
	2023	2022	2021
Balance at beginning of period	$ 71	$ 74	$ 76
Tax positions related to the current year			
Gross additions	—	—	—
Tax positions related to prior years			
Gross additions	—	2	1
Gross reductions	—	—	(1)
Settlements	—	(1)	(1)
Expiration of statute of limitations	(1)	(3)	(1)
Impact of currency changes	—	(1)	—
Balance at end of period	$ 70	$ 71	$ 74

If these unrecognized tax benefits were to be recognized as of December 31, 2023, the Company's income tax expense would decrease by about $8 million.

The Company recognizes all interest and penalties related to unrecognized tax benefits as a component of income tax expense. Accrued interest and penalties, which are not presented in the rollforward table above, were $9 million, $7 million and $7 million as of December 31, 2023, 2022 and 2021, respectively. Related to interest and penalties, we recognized an income tax expense of $2 million, $1 million and $2 million, as of December 31, 2023, 2022 and 2021, respectively.

21. SUBSEQUENT EVENTS

On February 8, 2024, the Company entered into a definitive agreement to purchase all of the outstanding shares of Masonite. The purchase price for the acquisition of Masonite is approximately $3.9 billion in cash, which we expect to fund with cash on hand and new committed financing. Masonite is a leading global designer, manufacturer, marketer and distributor of interior and exterior doors and door systems for the new construction and repair, renovation and remodeling sectors of the residential and non-residential building construction markets. The transaction was unanimously approved by the board of directors of both companies and is expected to close mid-2024, subject to regulatory and other customary closing conditions, including the approval of Masonite shareholders.

On February 9, 2024, the Company announced the decision to review strategic alternatives for its global glass reinforcements (“GR”) business, consistent with our strategy to focus on building and construction materials. The GR business, which operates within our Composites segment, supplies a wide variety of glass fiber products for applications in wind energy, infrastructure, industrial, transportation, and consumer markets. The GR business generates annual revenues of approximately $1.3 billion and has operations in 11 countries, with 18 manufacturing facilities. While a range of options are under consideration, including a potential sale, spin-off or other strategic option, there can be no assurance that the strategic review will result in any transaction or other outcome.

OWENS CORNING AND SUBSIDIARIES

INDEX TO CONDENSED FINANCIAL STATEMENTS SCHEDULE

Number	Description	Page
II	Valuation and Qualifying Accounts and Reserves – for the years ended December 31, 2023, 2022 and 2021	117

OWENS CORNING AND SUBSIDIARIES

SCHEDULE II – VALUATION AND QUALIFYING ACCOUNTS AND RESERVES FOR THE YEARS ENDED DECEMBER 31, 2023, 2022 AND 2021
(in millions)

	Balance at Beginning of Period	Charged to Costs and Expenses	Charged to Other Accounts	Deductions	Balance at End of Period
FOR THE YEAR ENDED DECEMBER 31, 2023					
Allowance for doubtful accounts	$ 11	$ 1	$ —	$ (1) (a)	$ 11
Tax valuation allowance	$ 129	$ 11	$ 1	$ (1)	$ 140
FOR THE YEAR ENDED DECEMBER 31, 2022					
Allowance for doubtful accounts	$ 9	$ 3	$ —	$ (1) (a)	$ 11
Tax valuation allowance	$ 132	$ 5	$ (2)	$ (6)	$ 129
FOR THE YEAR ENDED DECEMBER 31, 2021					
Allowance for doubtful accounts	$ 10	$ 1	$ —	$ (2) (a)	$ 9
Tax valuation allowance	$ 133	$ 11	$ 8	$ (20)	$ 132

(a) Uncollectible accounts written off, net of recoveries.